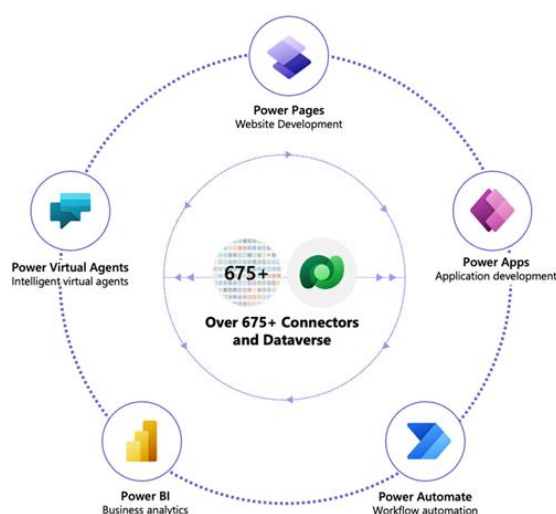


Power Pages

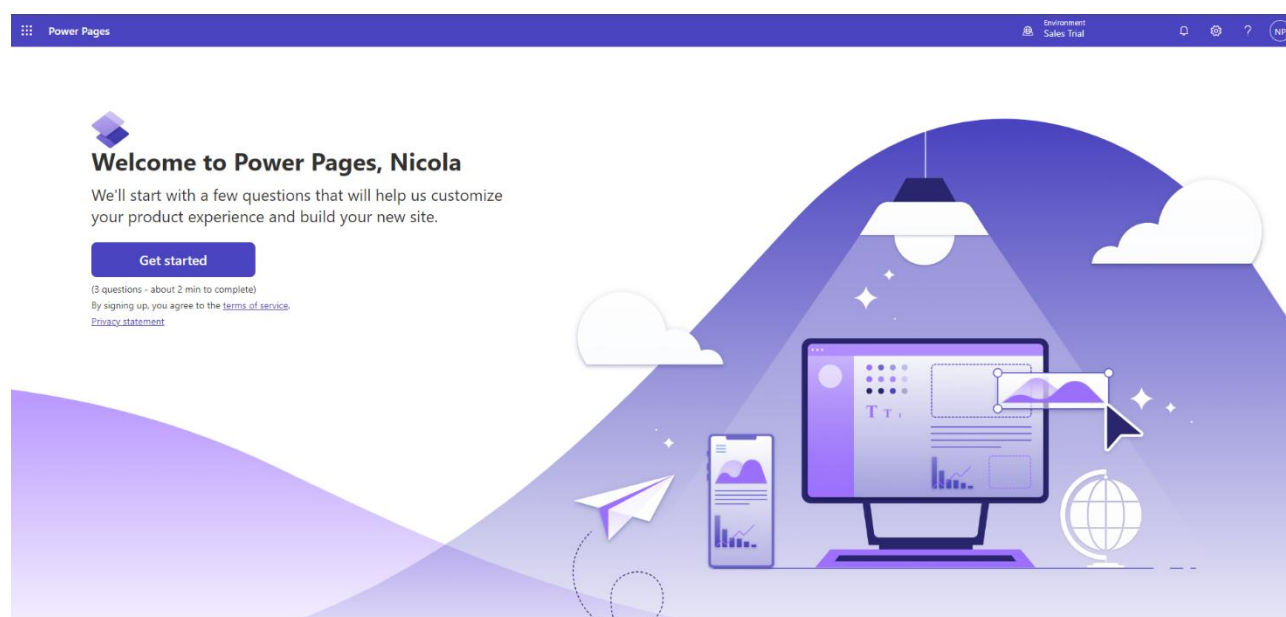
Microsoft **Power Pages** is a secure, enterprise-grade, low-code software as a service (SaaS) platform for creating, hosting, and administering modern external-facing business websites. Whether we are low-code makers or professional developers, Power Pages empowers us to rapidly design, configure, and publish websites that work across web browsers and devices. Power Pages provides rich customizable templates, a fluid visual experience through the design studio, and an integrated learning hub to quickly build sites for our unique business needs. Power Pages is the newest member of the Microsoft Power Platform family. With Power Pages, we can build sites by using the same shared business data stored in Microsoft Dataverse that we use for building apps, workflows, intelligent virtual agents, reports, and analytics with other Microsoft Power Platform components in our organization.



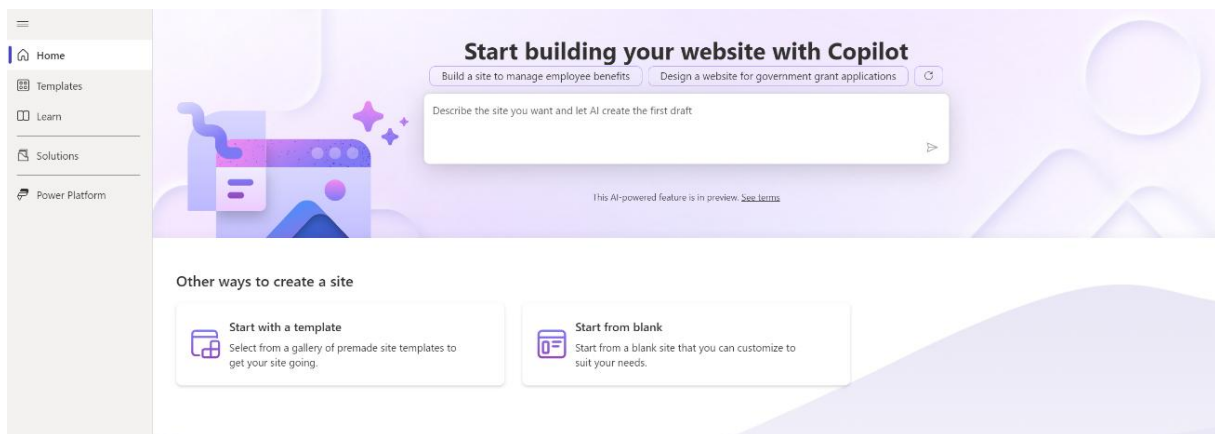
By means of this service, we can expose selected data to the public.

1. Create a Power Pages website

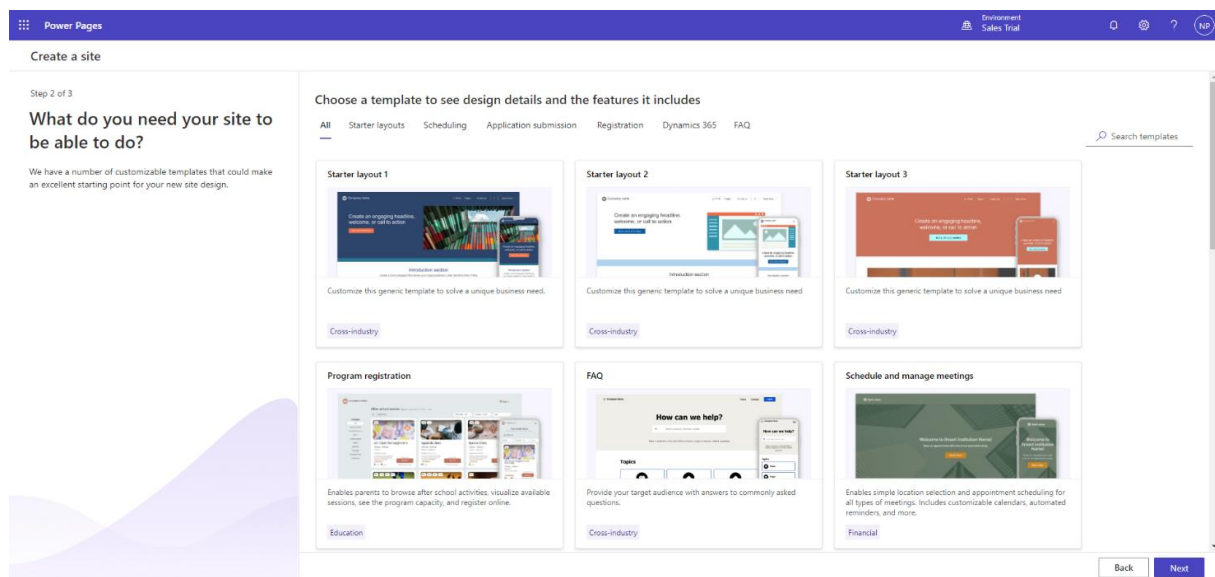
Let us navigate to <https://make.powerpages.microsoft.com> and choose the right environment.



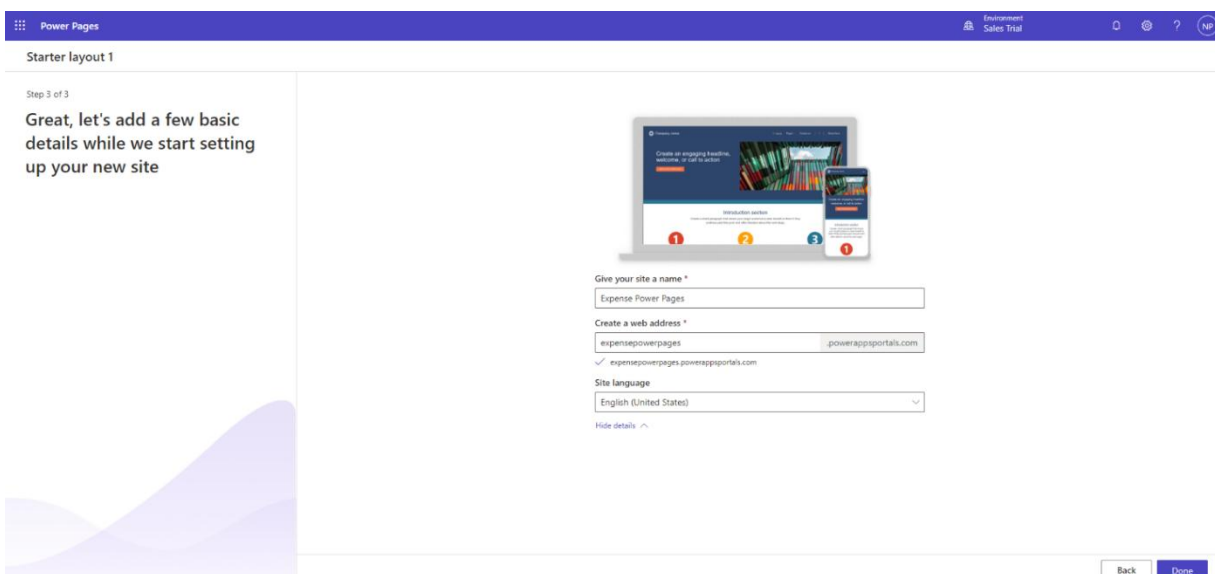
We can start building a new site using already existing templates or a blank one.



Some of the default templates which we can explore and use are displayed in the following figure.

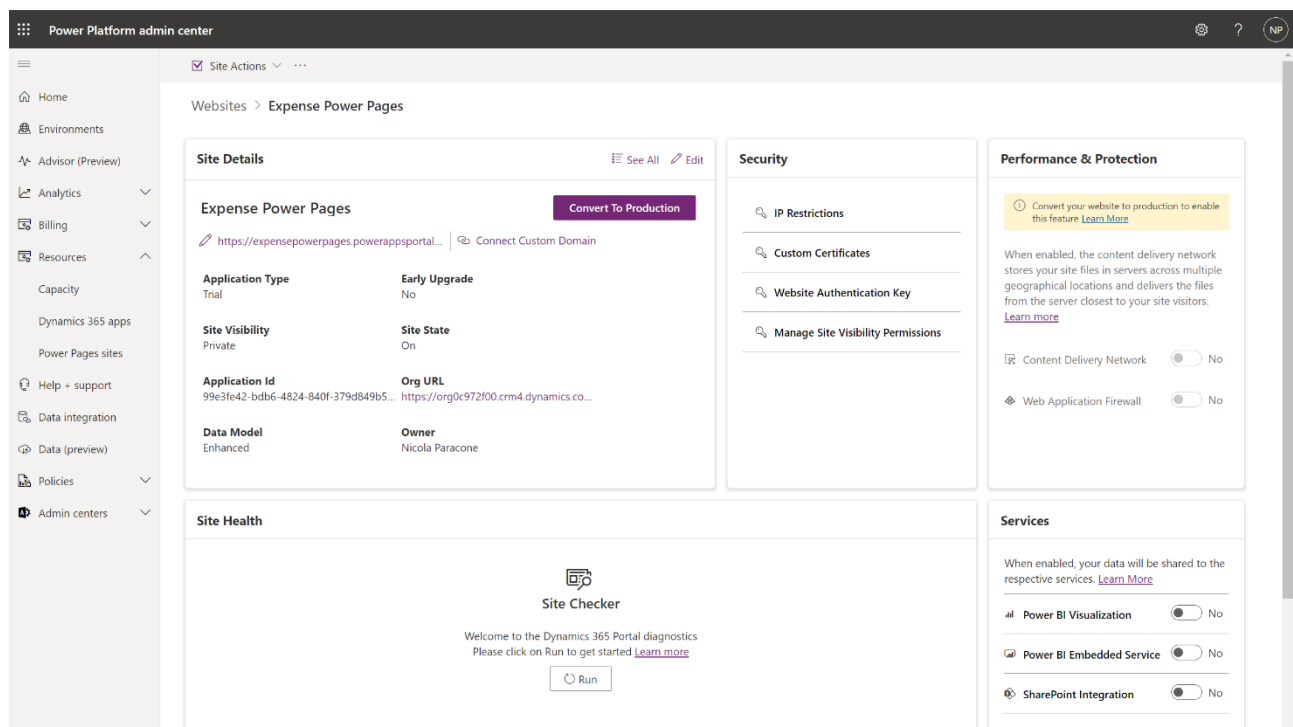
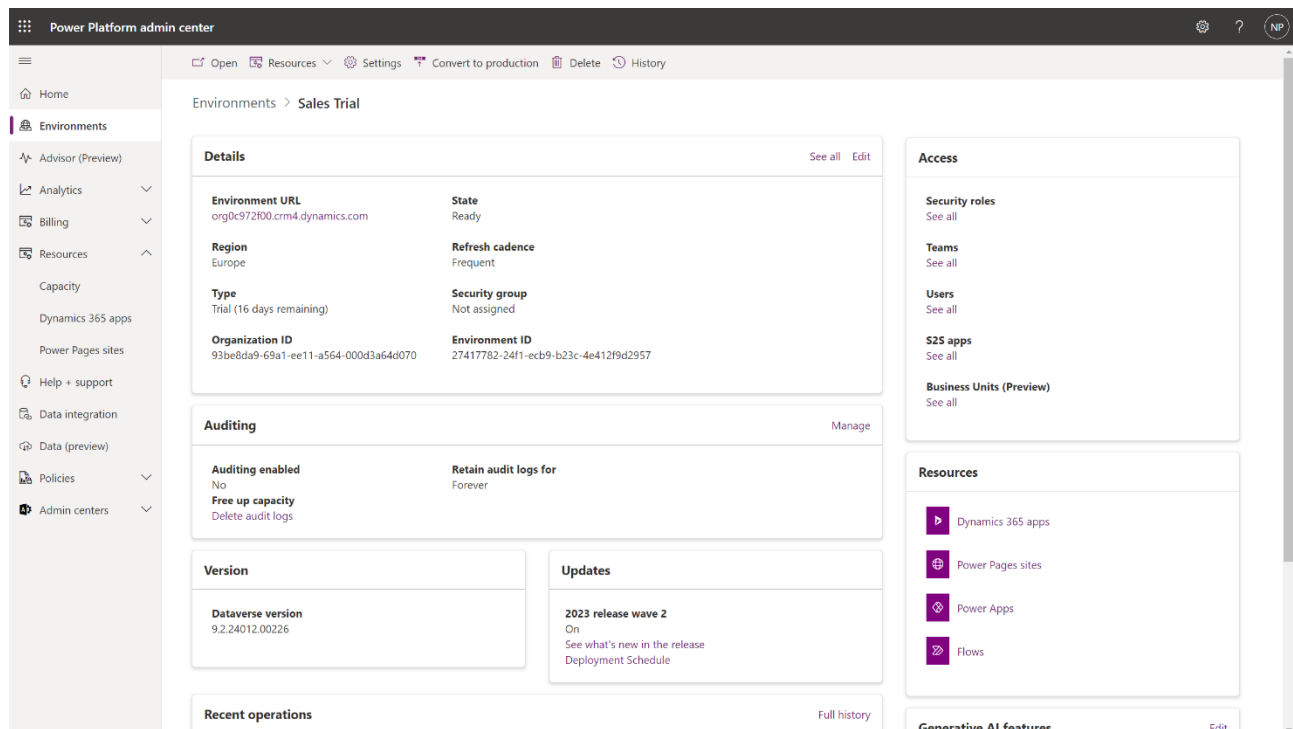


If we like the Starter layout 1 template, for example, we then proceed by clicking on Choose this template. Since we are trying to build a public-facing website for customers or clients, we need to choose an appropriate website name and address.



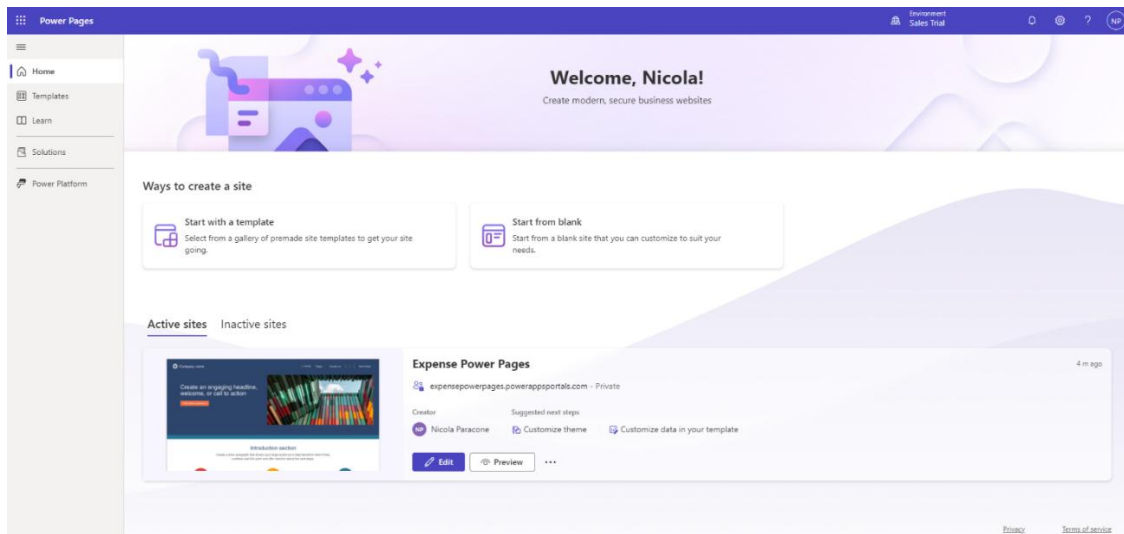
Click on Done and wait for a couple of minutes. By default we have two models: **standard** and **enanched**. It is advisable not to choose the standard model, since it uses features that were implemented for Portals rather than Power Pages. On the contrary, the enanched model uses solutions designed for Power Pages. Moreover, using the standard model complicates migration of Power Pages from one environment to another.

In the Power Platform Admin Center, under Resources, we have Power Pages sites.

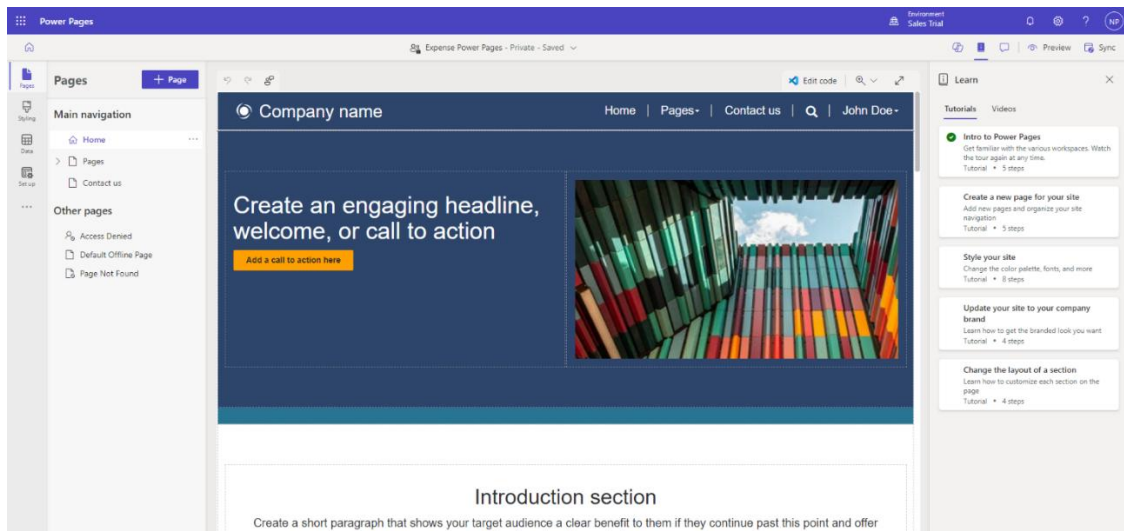


In this case, the data model of the website we are setting up is already enhanced.

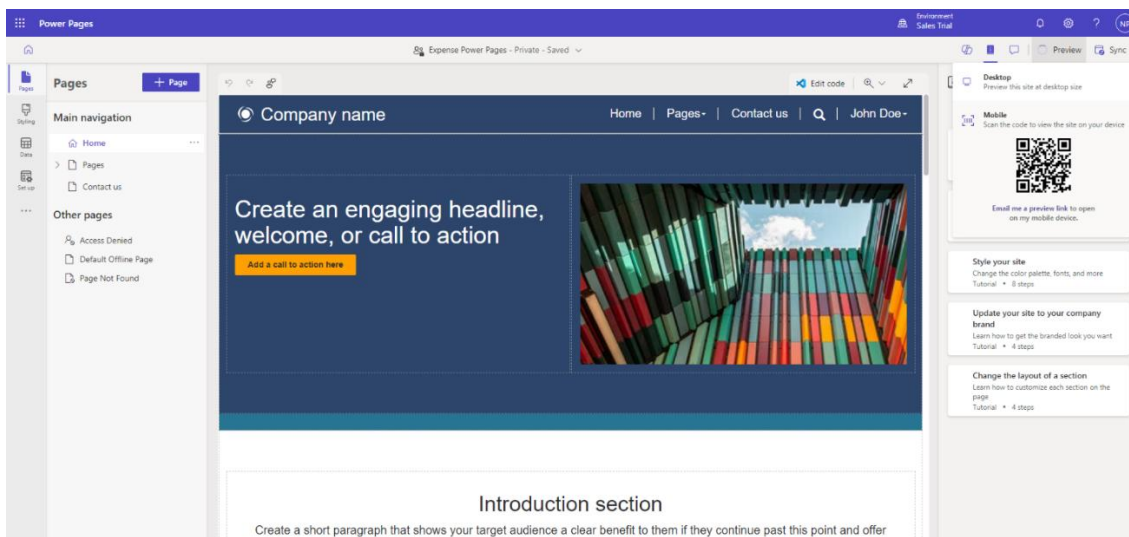
This is what we will see in Power Pages once our website is ready.



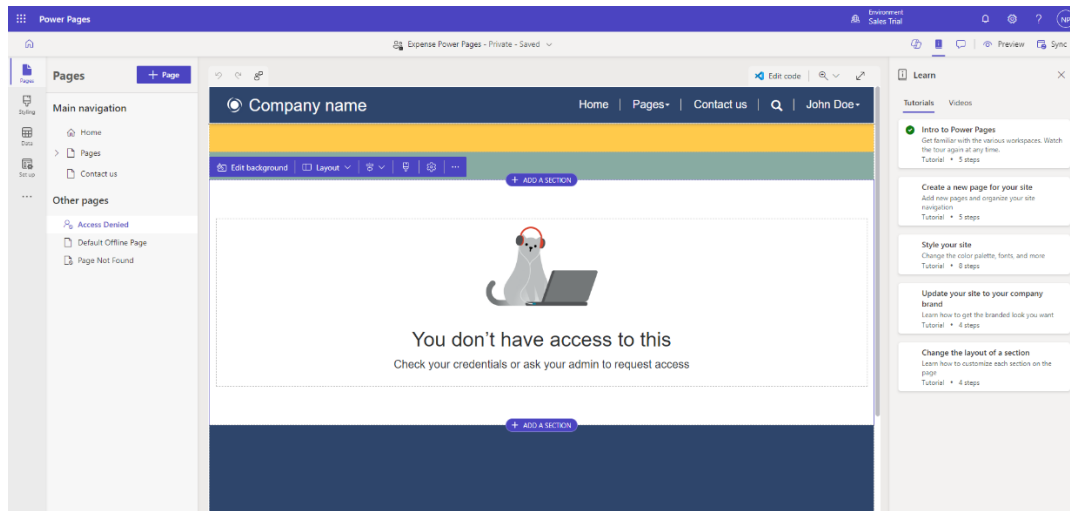
Select Expense Power Pages and click on Edit.



Whatever changes we will perform, we can always click the Preview button on the top right for obtaining an anticipation of how our website will look like both on desktop and mobile.

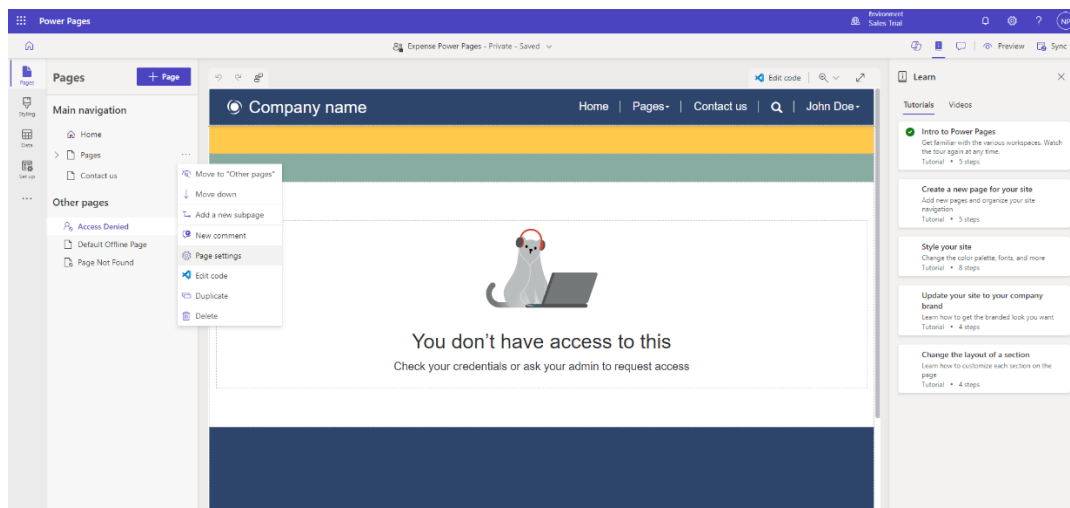


In the menu on the left we have all the pages of our website. The Access Denied page, for example, is the page which will be displayed to a user which is not authorized to see a particular content.

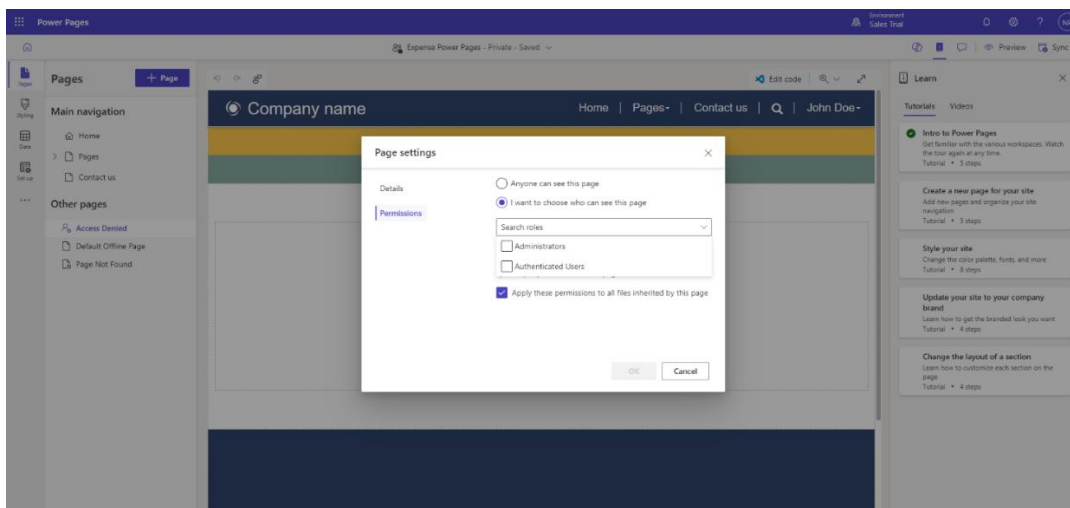


Similarly to Dynamics 365, in which we have security roles, in Power Pages we have **web roles**. According to the web role assigned to a user, a page will or will not be accessible to them.

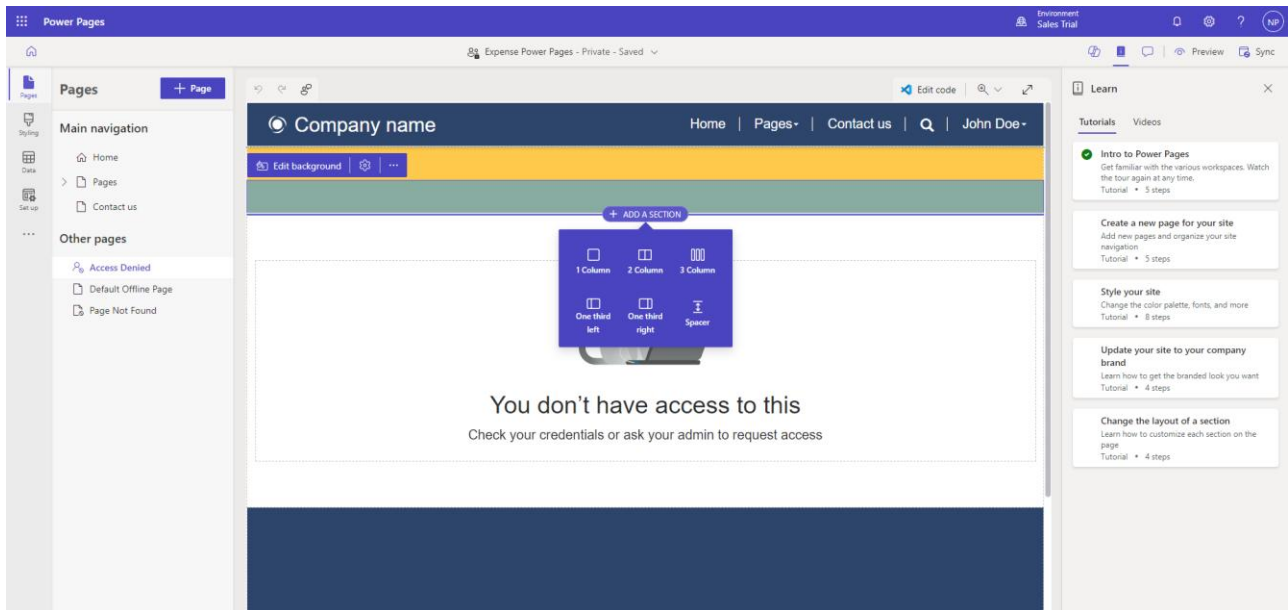
Click on Page settings.



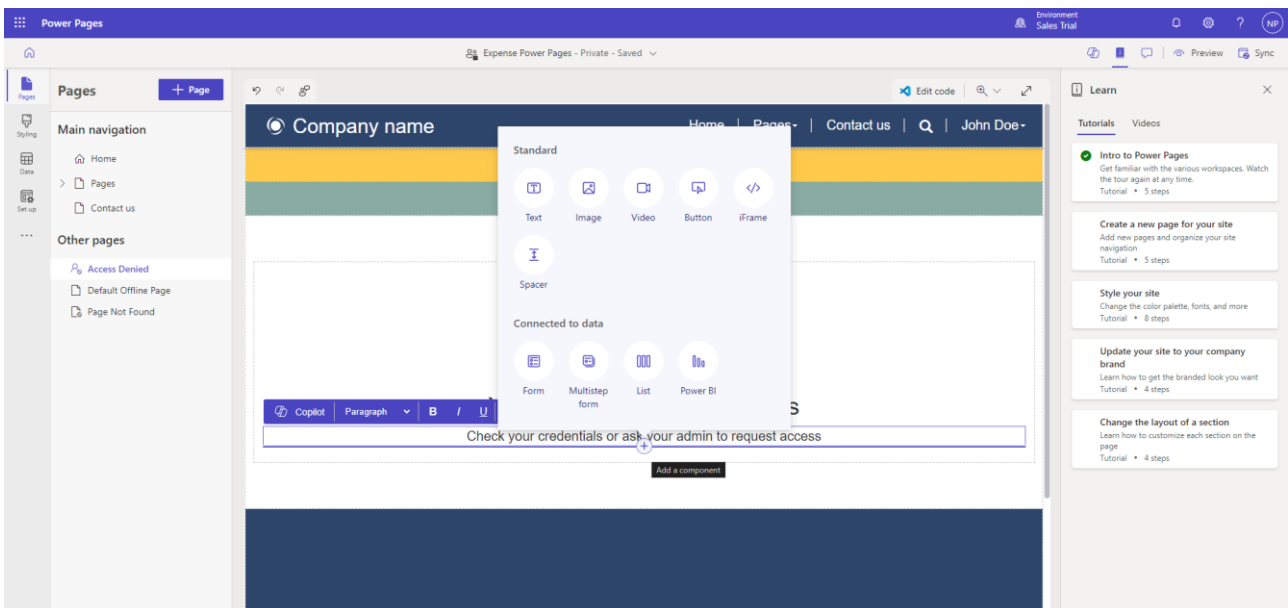
There we can set up permissions according to our will in an easy and intuitive way.



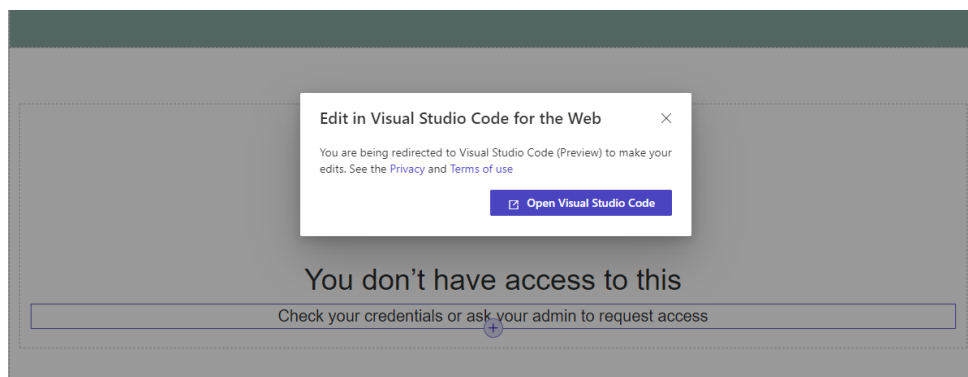
There are two default web roles, but we can also create other custom roles. Quite intuitively, we can add new sections to our page:

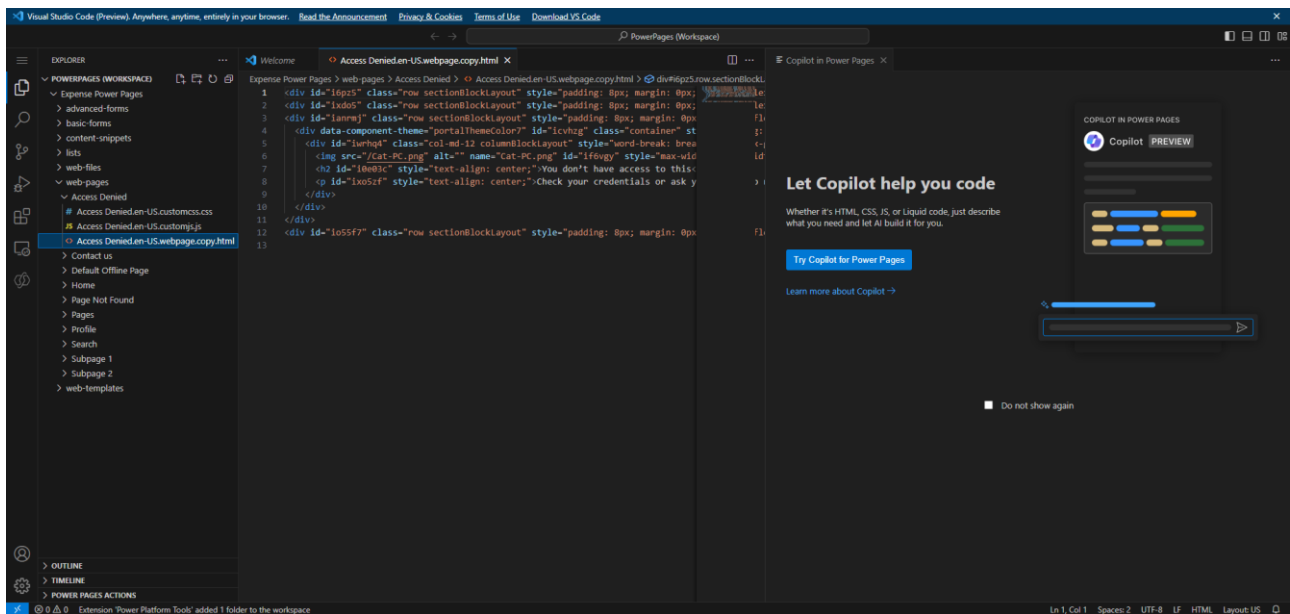


and, inside these sections, add components of various types.

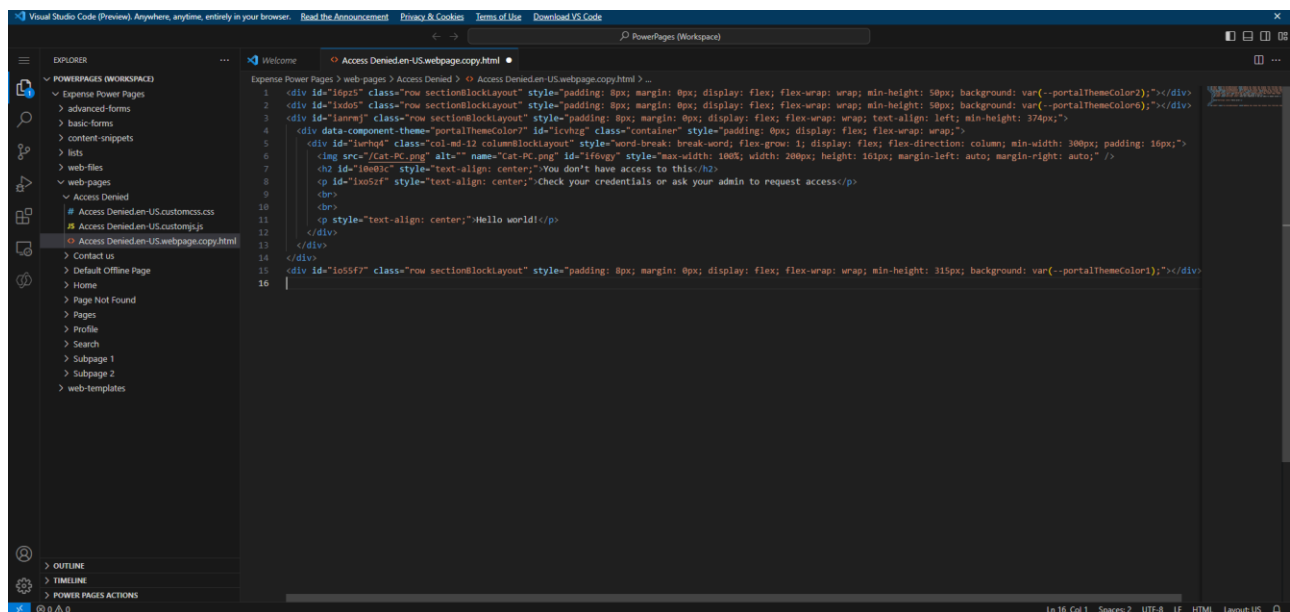


Everything is very user friendly and requires no code knowledge. If we want, however, we can edit code running on the webpage by clicking on  Edit code on the top right of the page.

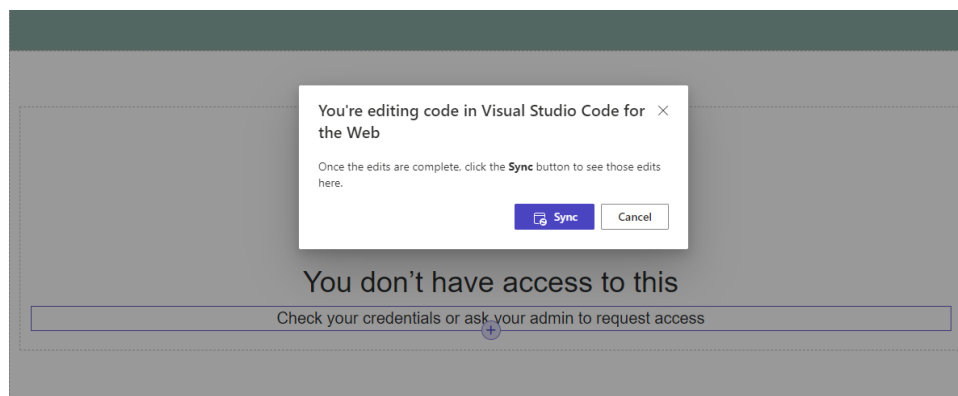




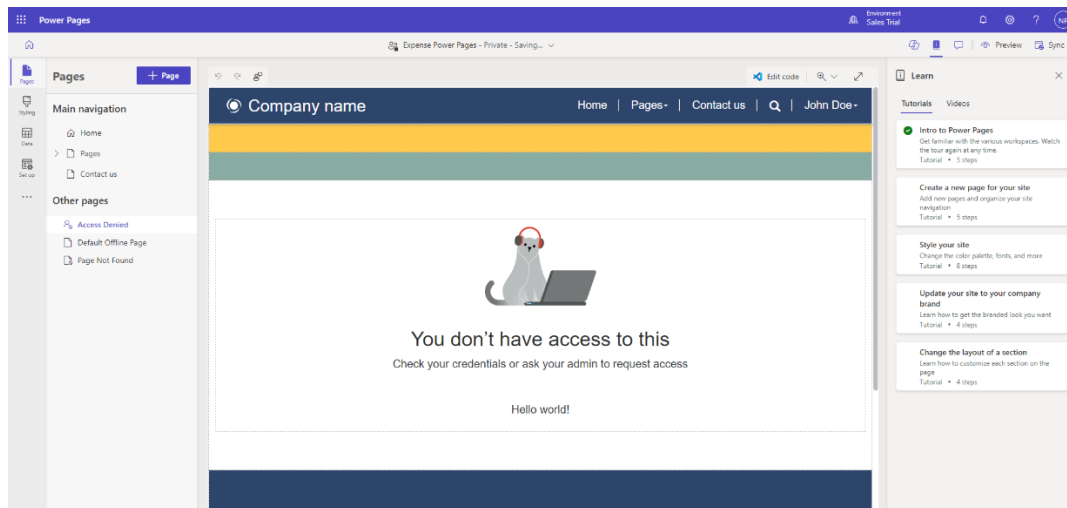
Let us add a simple line of code for displaying "Hello world!" (clicking Ctrl+S is the only way of saving the changes).



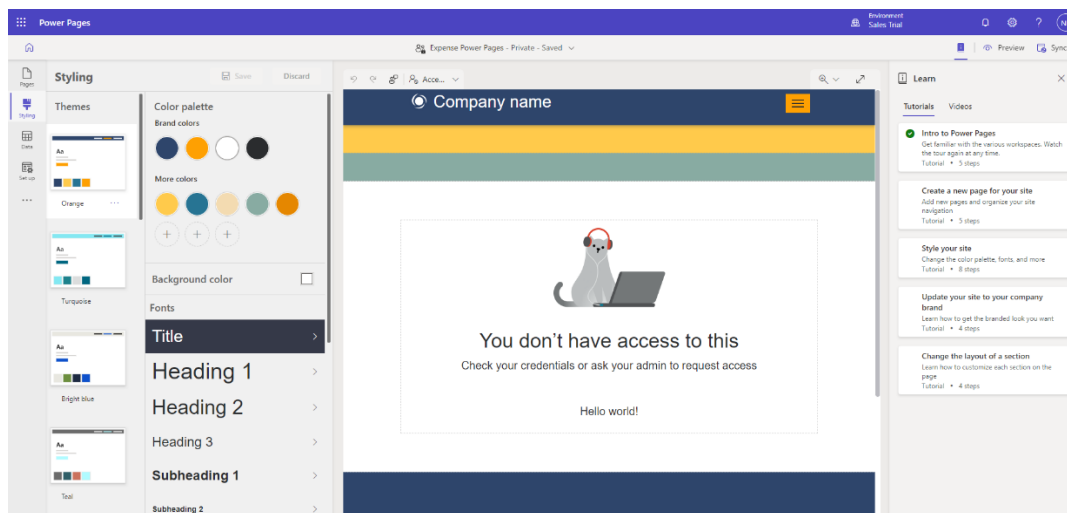
To visualize the modification in our site we always need to click on Sync.



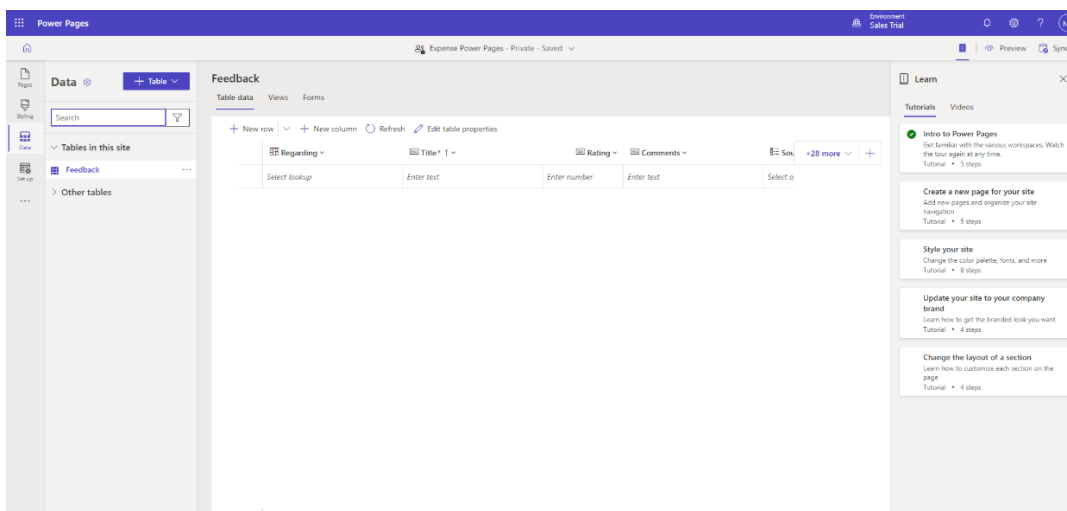
Our text will now appear on the page.



Let us click on Styling, in the left navigation panel.

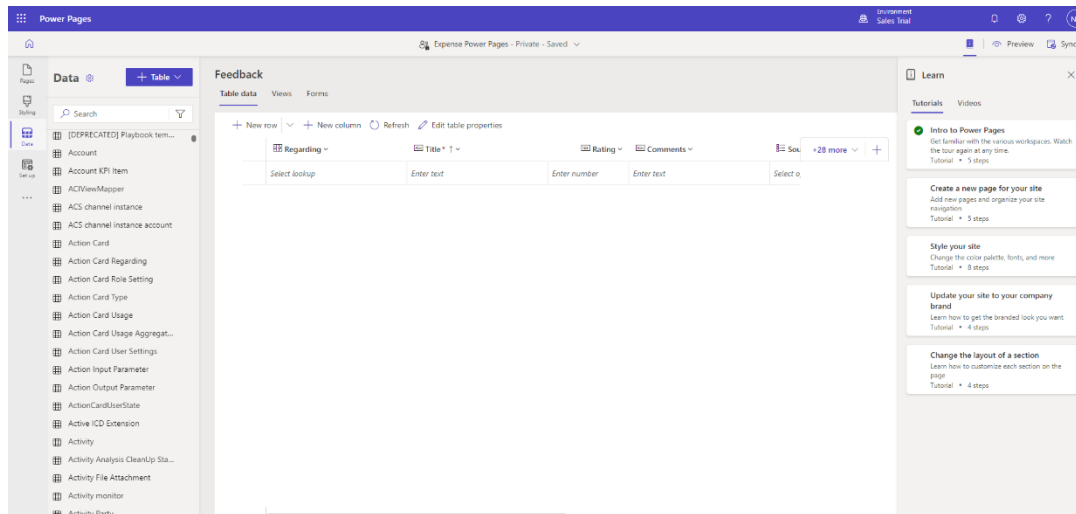


Here we find many different themes, for which we can customize colors and fonts (remember to save). Below Styling we have Data.

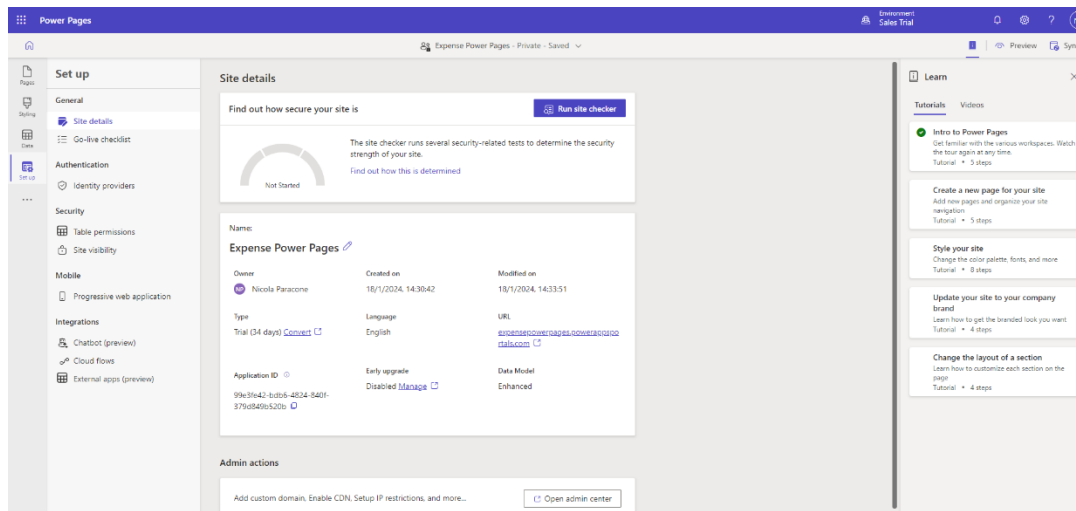


Here we find the tables which are already in the site, but also tables we have in Dataverse.

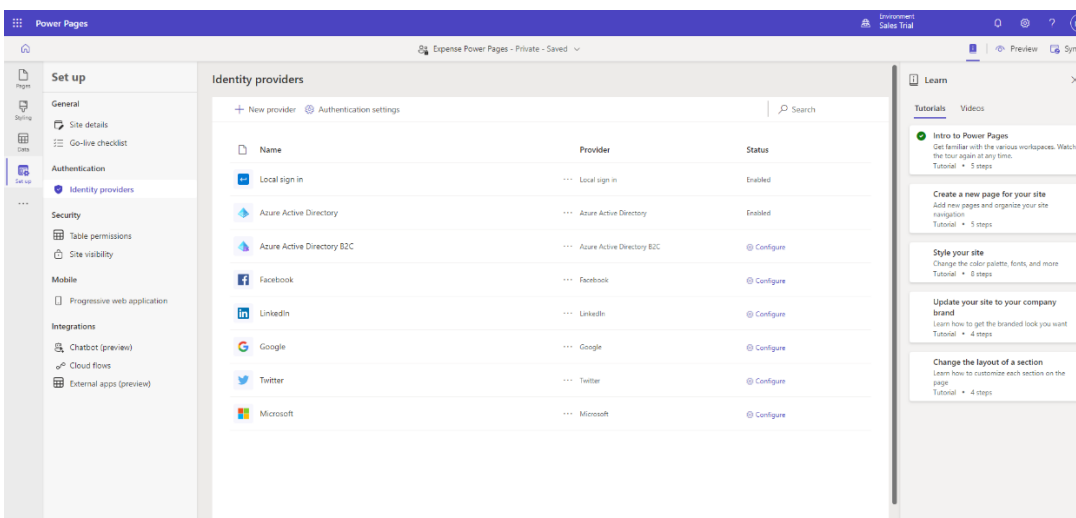
If we create a new table, however, the storage used by Power Pages won't be Dataverse. The newly created table, therefore, will not be available in Power Apps.



In Setup, instead, we have administrative operations and options.



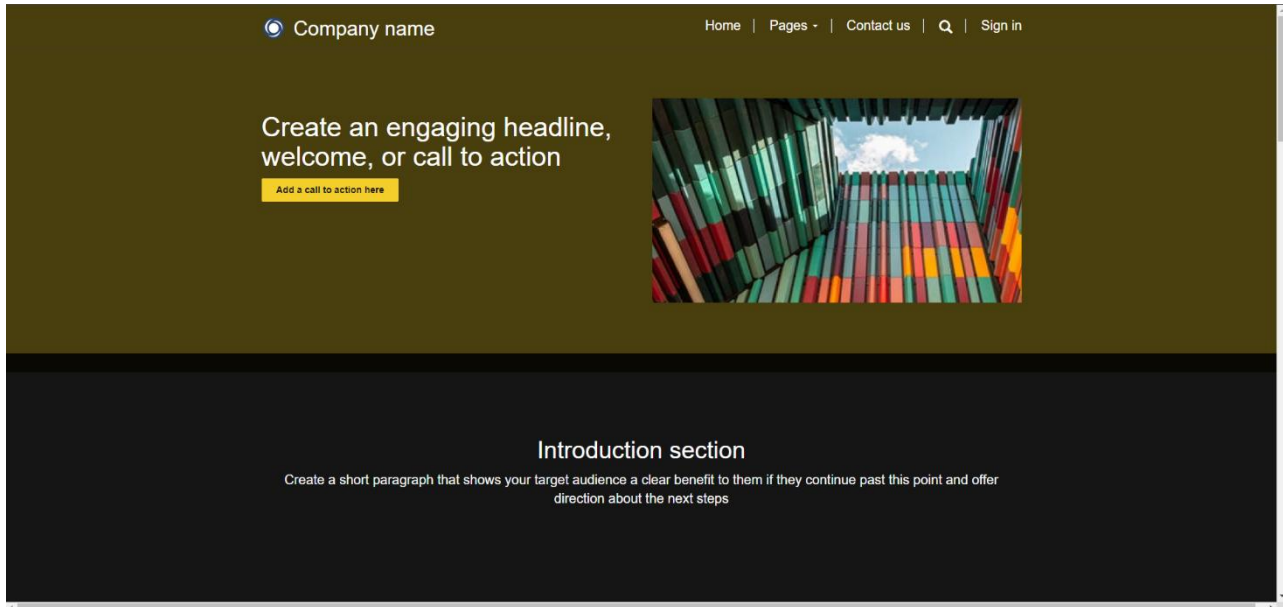
Click, for example, on Identity providers.



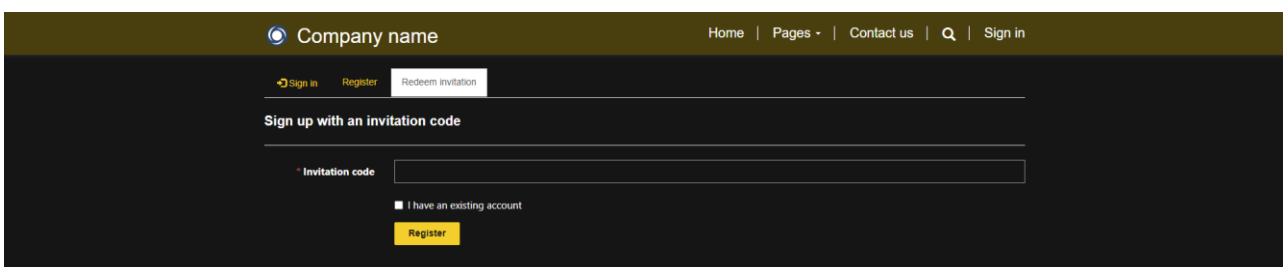
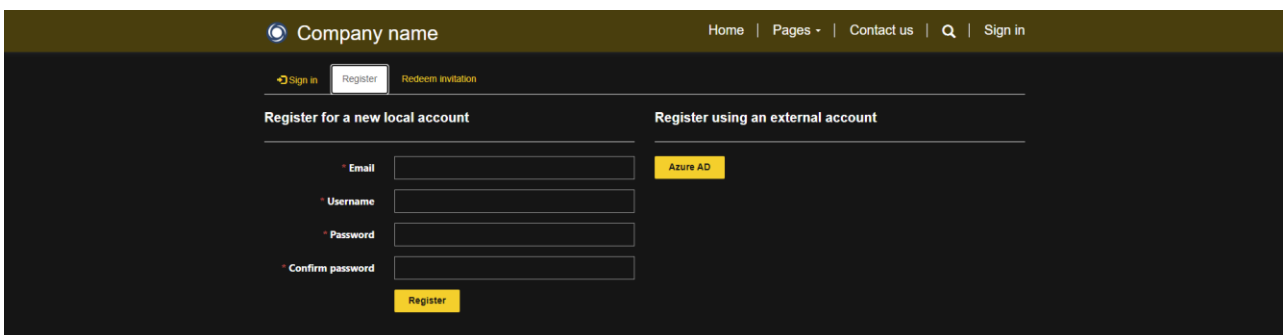
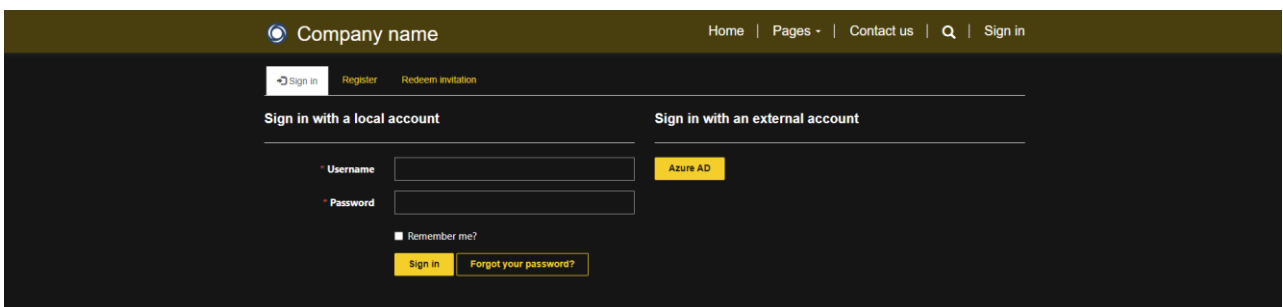
Here we find the providers which we can configure for signing-in to our website.

If you want to configure the Google provider, for example, navigate to <https://learn.microsoft.com/en-us/power-pages/security/authentication/oauth2-google> and simply follow the instructions step by step.

Let's have a preview of the website and click on Sign in.



A user can sign-in, register to the website, or redeem an invitation.



In CRM we have a concept called **Invitation**. From the account of a user, we can generate an invitation which will provide an invitation code. Using that code, the user can register to our website.

Notice that we also have Sign in with an external account. Azure AD is available only for users in our tenant.

The screenshot shows a user profile page for 'Nicola Paracone'. The page has a dark header with the company name and navigation links. The main content area is divided into a left sidebar with 'Profile' and 'Security' sections, and a main form area. The 'Profile' section includes a user card and a list of links: 'Profile', 'Set password', 'Change email', and 'Manage external authentication'. The 'Security' section is currently selected. The main form area contains instructions for providing information and a 'Your information' section with fields for First Name, Last Name, E-mail, Business Phone, Organization Name, Title, Nickname, and Web Site. A confirmation message states: 'Your email requires confirmation. Confirm Email'.

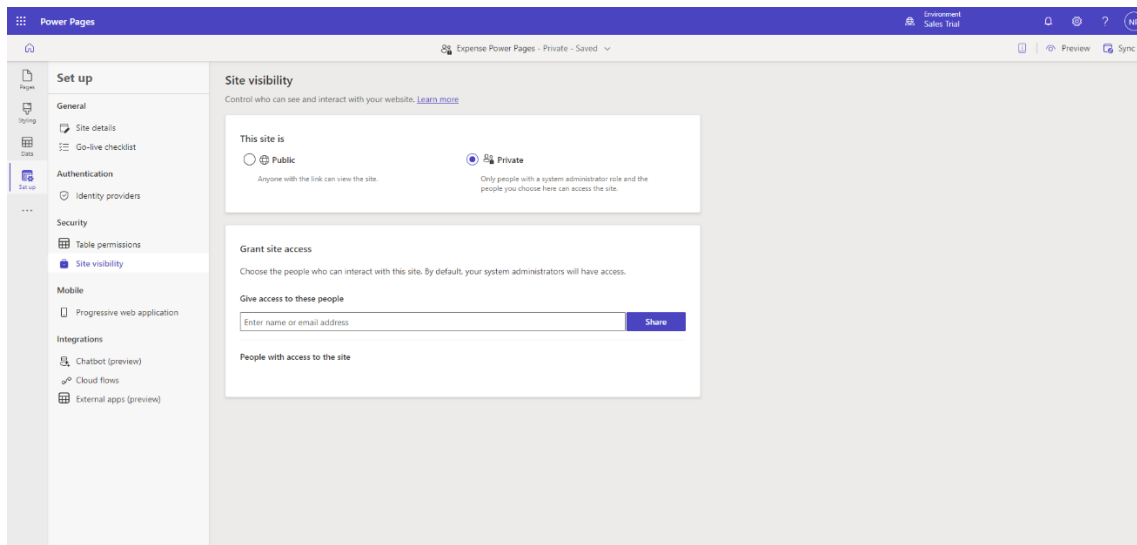
Another thing we have is Table permissions.

The screenshot shows the 'Table permissions' configuration page in Power Pages. The page has a blue header with 'Power Pages' and 'Expense Power Pages - Private - Saved'. The left sidebar shows 'Set up' with options for General, Authentication, Security, Mobile, and Integrations. The main content area is titled 'Table permissions' and contains a table with columns: Name, Status, Table, Access Type, Roles, Relationship, Read, Update, Create, Delete, Append, and Append to. The table lists a 'Feedback' table with 'Global access' and 'Anonymous Users +2 more' roles. The 'Create' checkbox is checked, while others are unchecked.

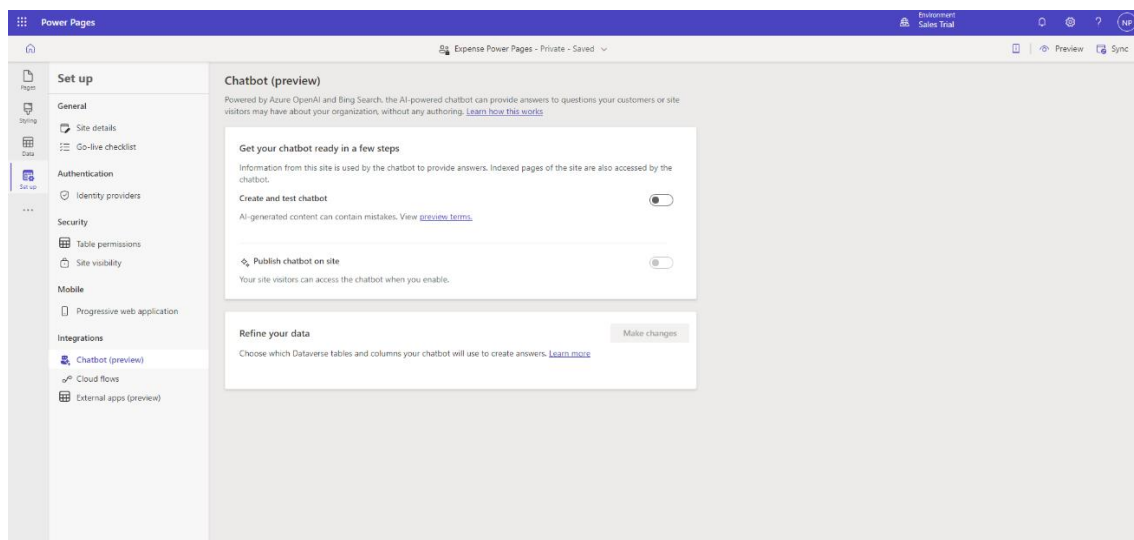
Name	Status	Table	Access Type	Roles	Relationship	Read	Update	Create	Delete	Append	Append to
Feedback	Active	Feedback	Global access	Anonymous Users +2 more	--	☐	☐	☒	☐	☐	☐

Assume that we have ten tables. Here we can configure which tables will be accessible to a user based on their web role. In the view above, for example, the Feedback table is accessible even for users which are not registered.

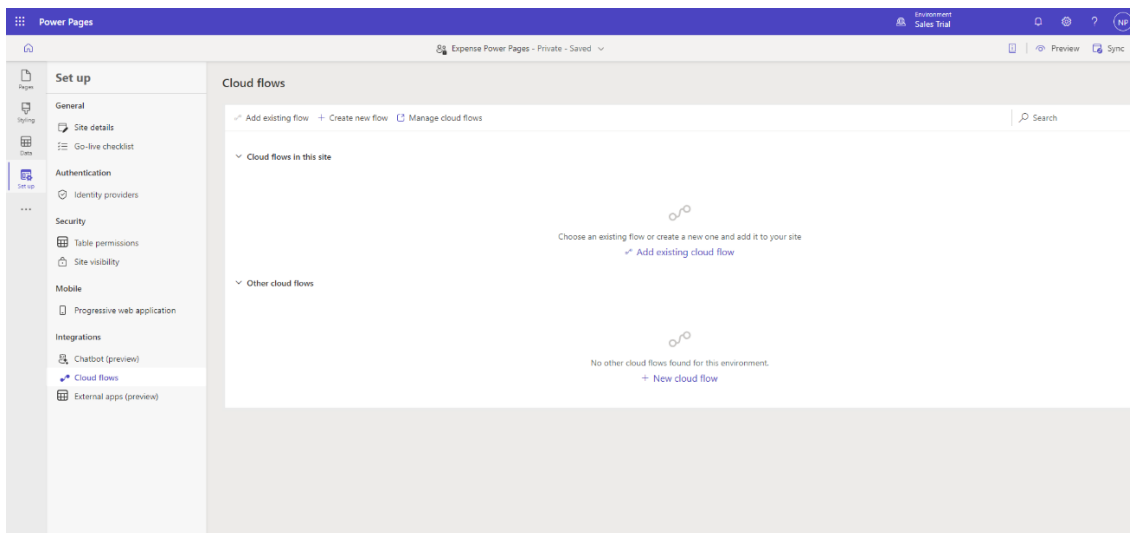
By default, our website will be private.



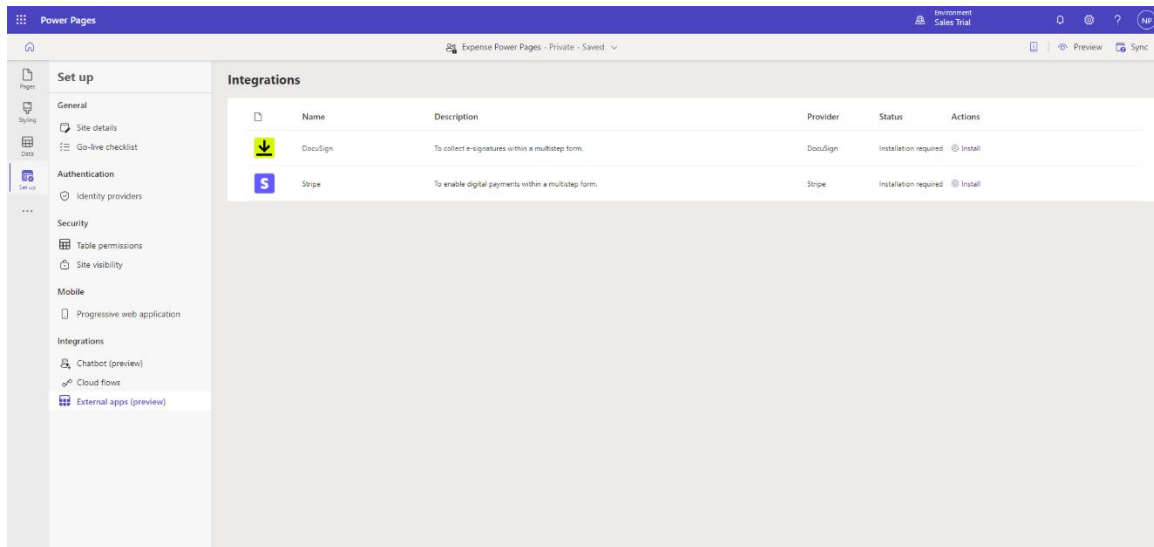
Under Site visibility, however, we can decide who can access the site. In addition, the integration of a chatbot is useful for helping users with predefined answers and interacting with the CRM.



Then we have Cloud flows, which are essentially Power Automate flows.

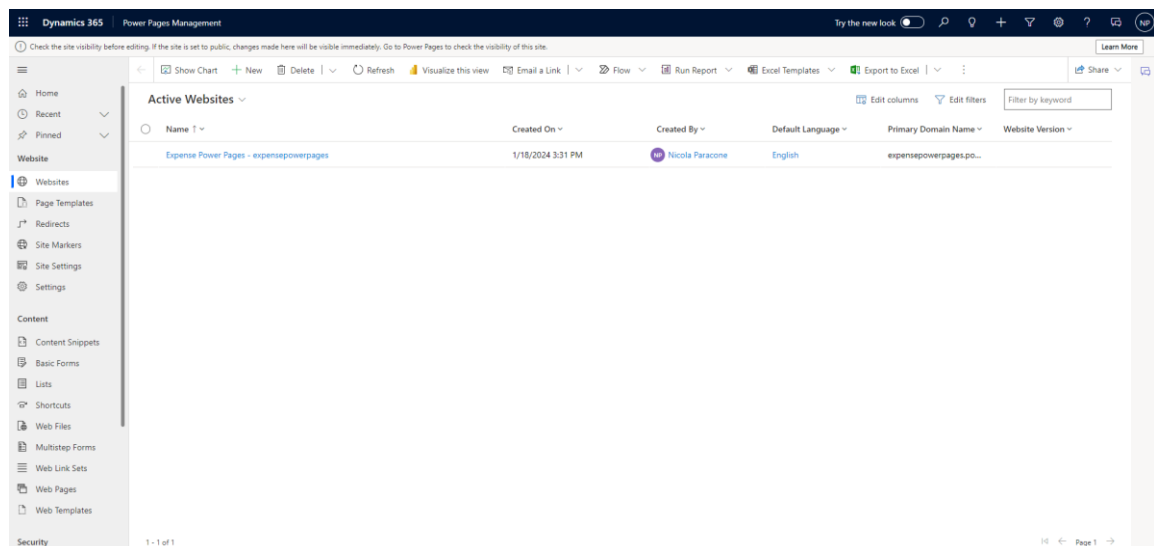
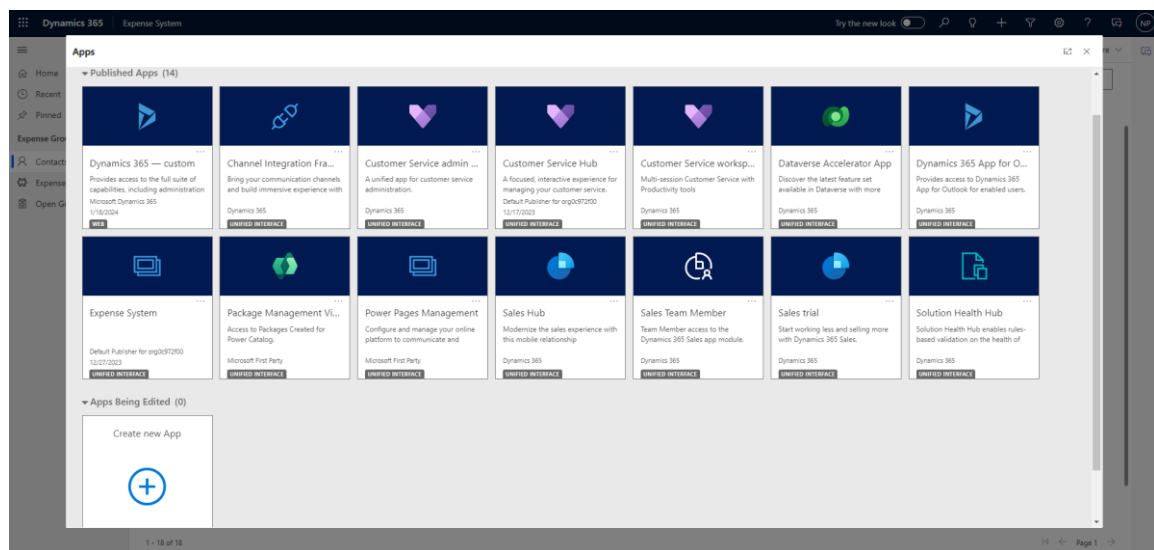


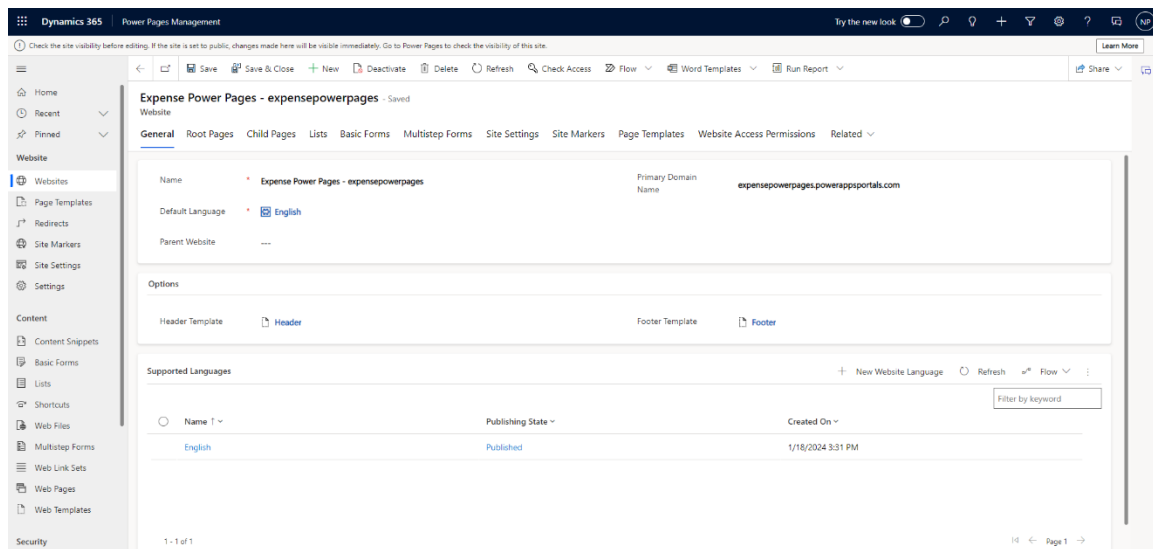
In External apps we find applications that we can integrate in our website.



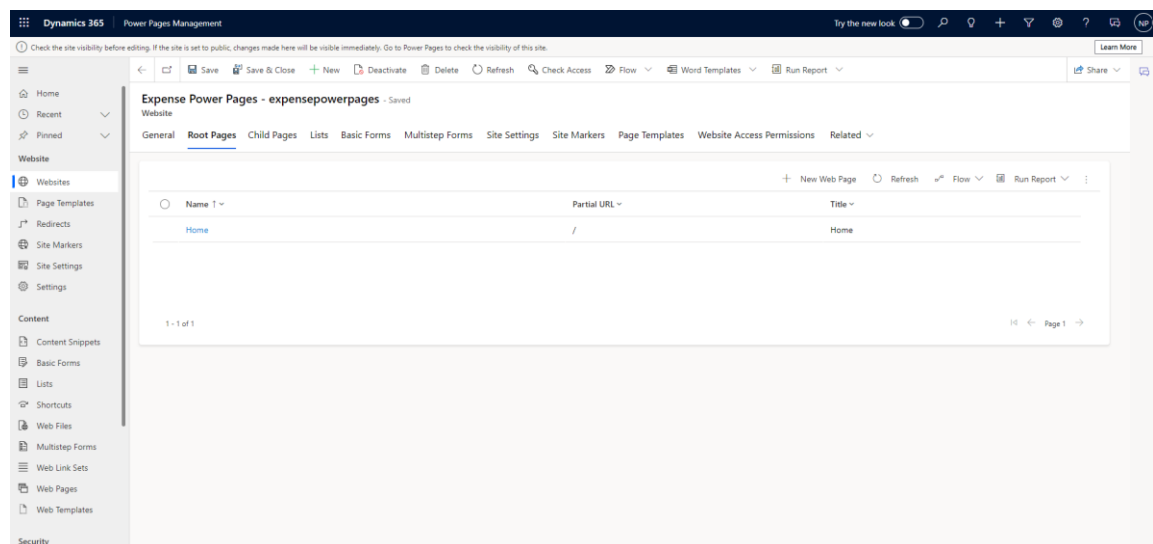
2. Power Pages management in Dynamics 365

If we now move to Dynamics 365, we can open the Power Pages Management app.

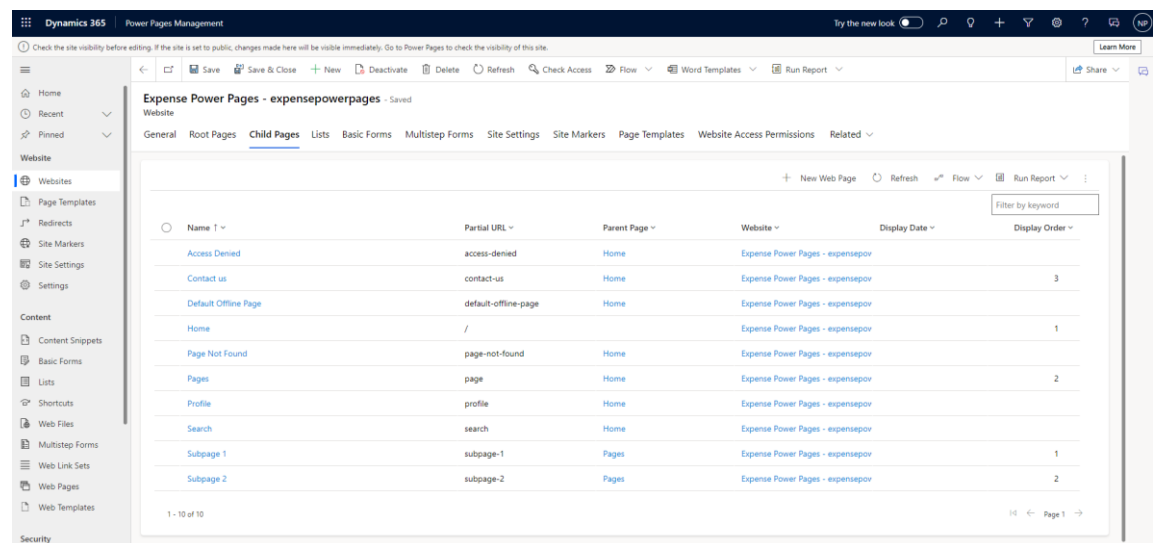




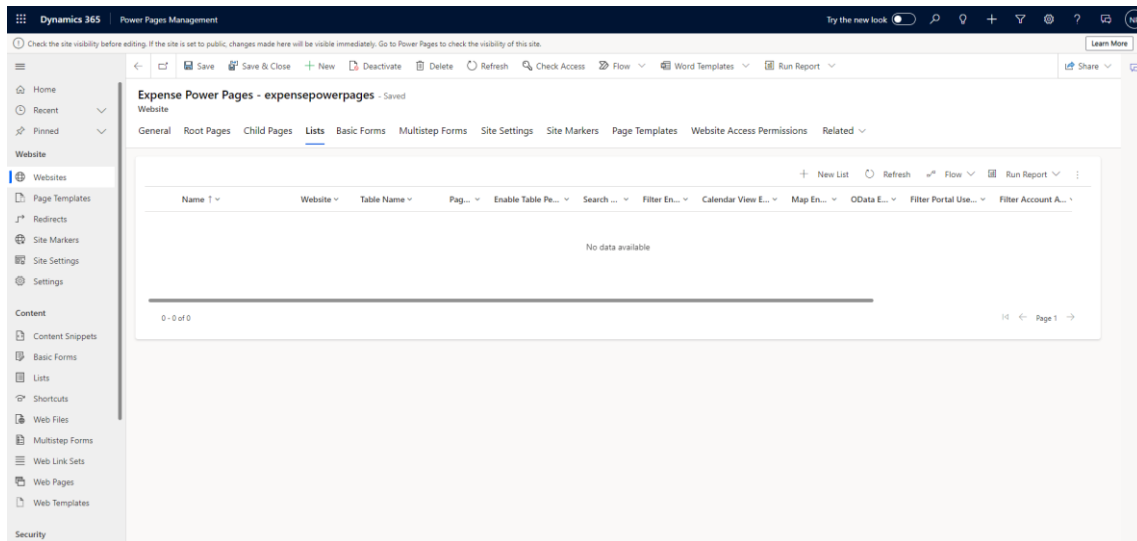
In Root Pages we find the pages which are shown by default when we are logging to our website.



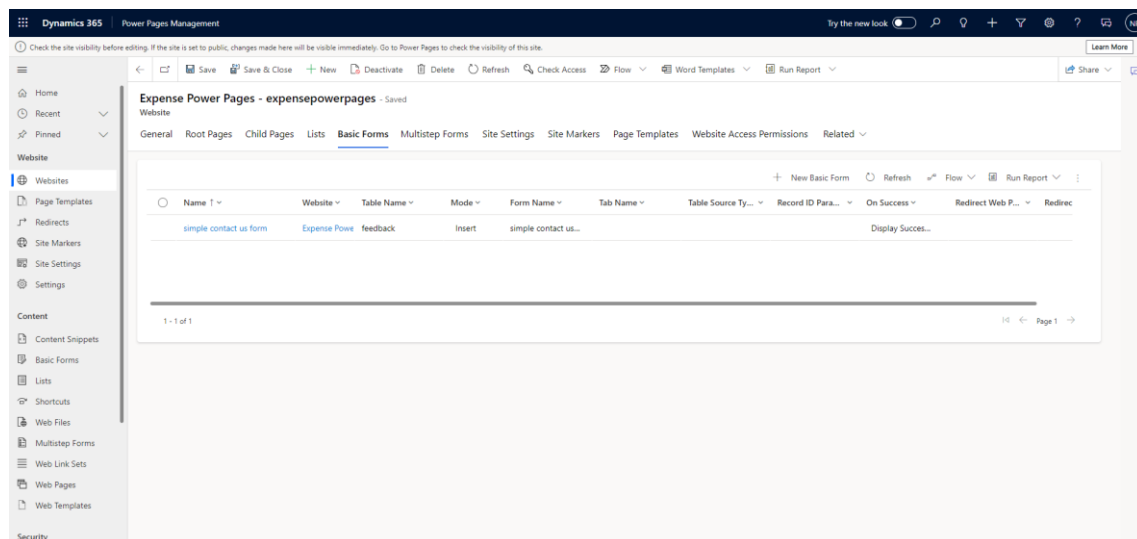
Under Child Pages we see all the pages under a parent page.



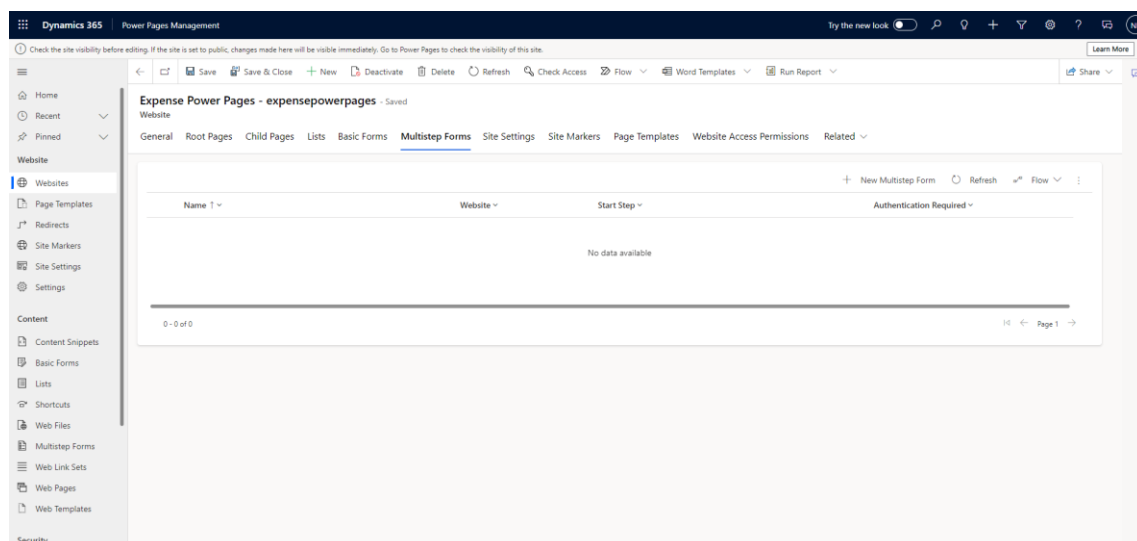
A list is a collection of records we have in the CRM that we want to display on our website.



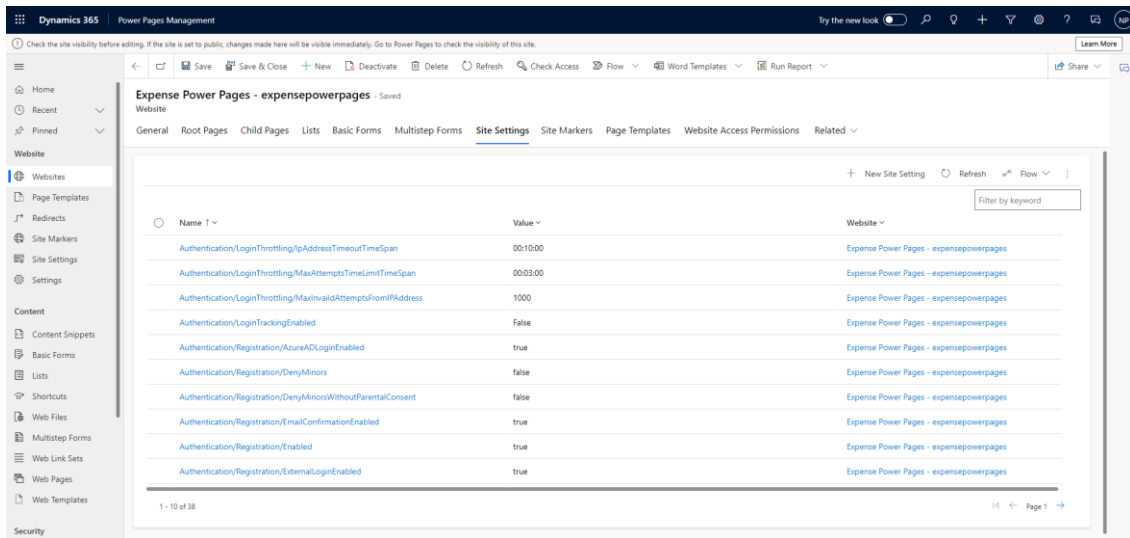
Then we have Basic Forms.



Under Multistep Forms we find the forms that a user has to follow step by step.



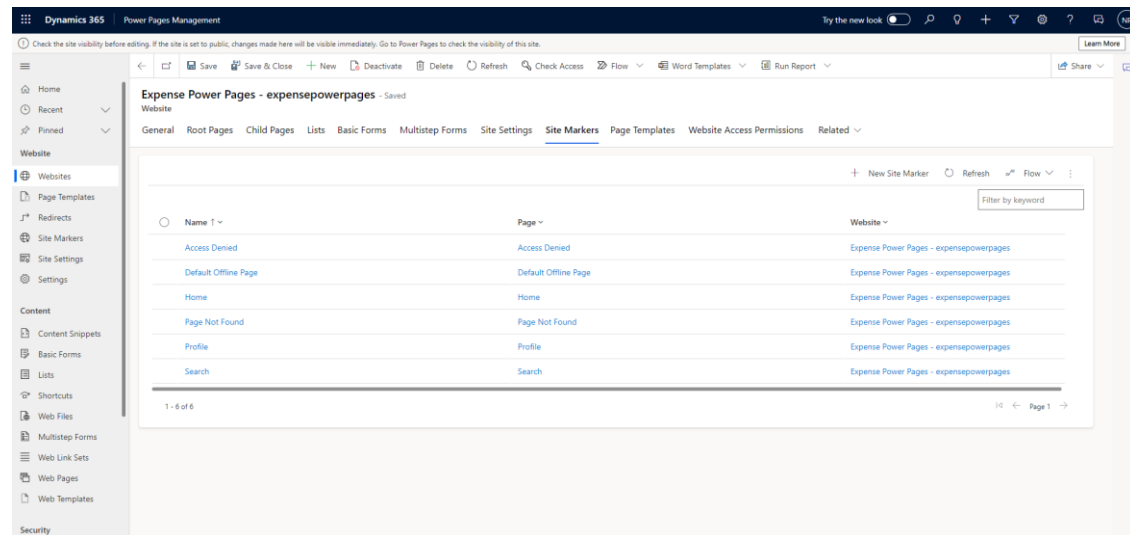
Site Settings defines a collection of settings which can be enabled, disabled, or edited.



The screenshot shows the 'Site Settings' tab for a website named 'Expense Power Pages - expensepowerpages'. The left sidebar contains navigation options: Home, Recent, Pinned, Website, Websites, Page Templates, Redirects, Site Markers, Site Settings (selected), Settings, Content, Content Snippets, Basic Forms, Lists, Shortcuts, Web Files, Multistep Forms, Web Link Sets, Web Pages, and Web Templates. The main area displays a table of settings for the website.

Name	Value	Website
Authentication/LoginThrottling/IpAddressTimeoutTimeSpan	00:10:00	Expense Power Pages - expensepowerpages
Authentication/LoginThrottling/MaxAttemptsTimeLimitTimeSpan	00:03:00	Expense Power Pages - expensepowerpages
Authentication/LoginThrottling/MaxInvalidAttemptsFromIpAddress	1000	Expense Power Pages - expensepowerpages
Authentication/LoginTracking/Enabled	False	Expense Power Pages - expensepowerpages
Authentication/Registration/AzureADLoginEnabled	true	Expense Power Pages - expensepowerpages
Authentication/Registration/DenyMinors	false	Expense Power Pages - expensepowerpages
Authentication/Registration/DenyMinorsWithoutParentalConsent	false	Expense Power Pages - expensepowerpages
Authentication/Registration/EmailConfirmationEnabled	true	Expense Power Pages - expensepowerpages
Authentication/Registration/Enabled	true	Expense Power Pages - expensepowerpages
Authentication/Registration/ExternalLoginEnabled	true	Expense Power Pages - expensepowerpages

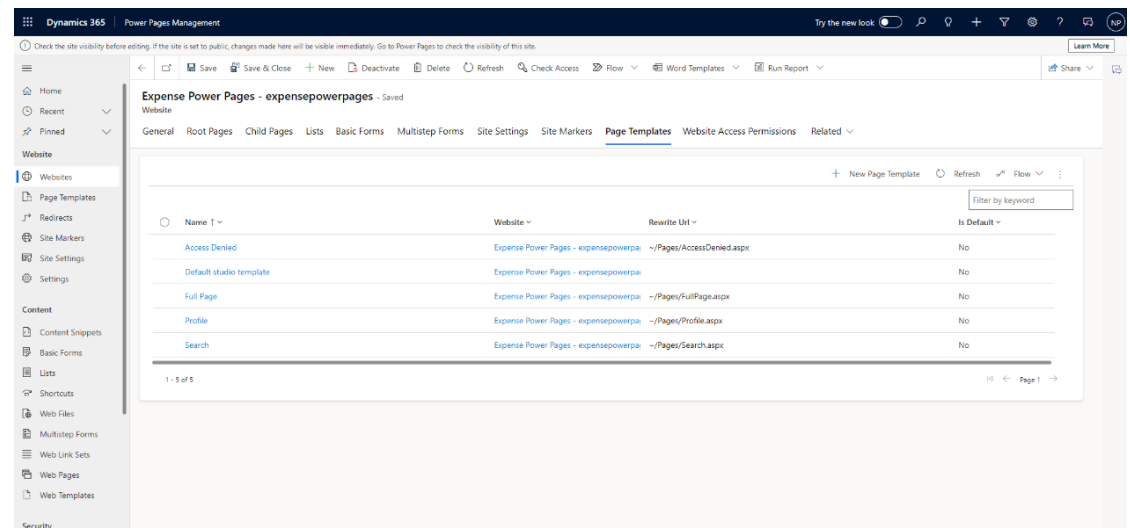
Site Markers are shown below.



The screenshot shows the 'Site Markers' tab for the same website. The main area displays a table of site markers.

Name	Page	Website
Access Denied	Access Denied	Expense Power Pages - expensepowerpages
Default Offline Page	Default Offline Page	Expense Power Pages - expensepowerpages
Home	Home	Expense Power Pages - expensepowerpages
Page Not Found	Page Not Found	Expense Power Pages - expensepowerpages
Profile	Profile	Expense Power Pages - expensepowerpages
Search	Search	Expense Power Pages - expensepowerpages

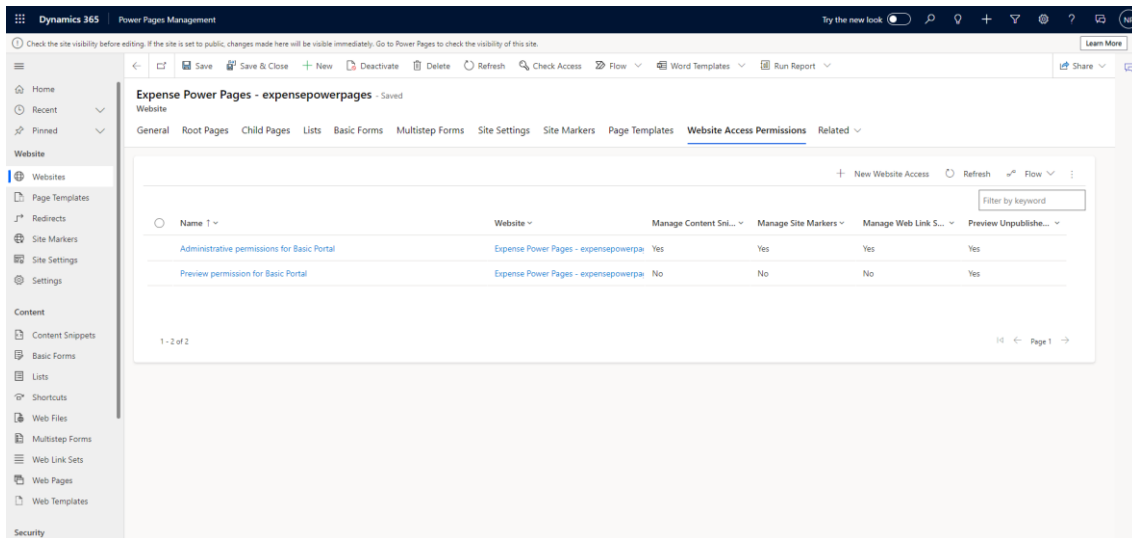
Page Templates contains the information about how a particular page should be displayed.



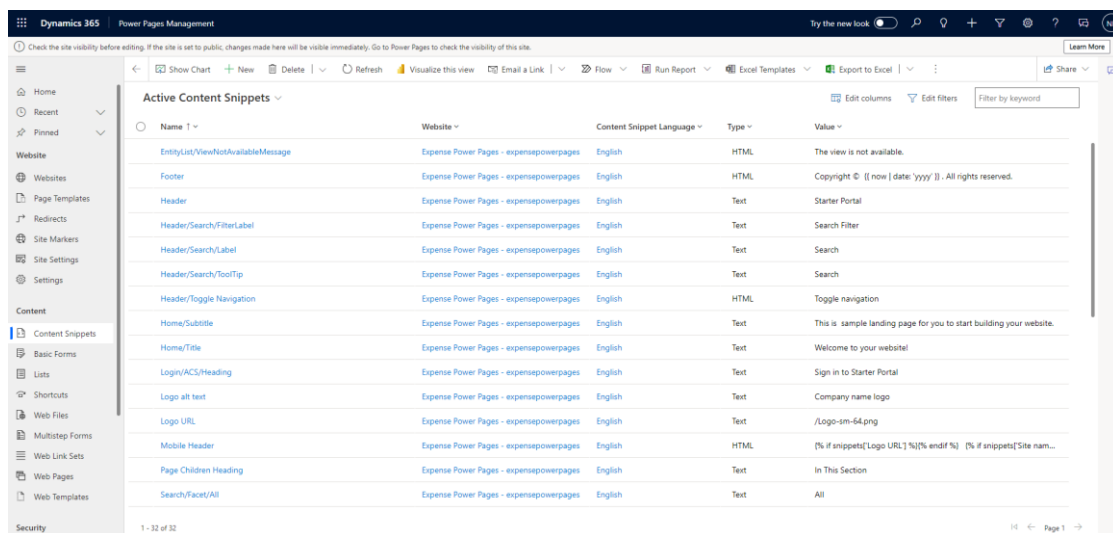
The screenshot shows the 'Page Templates' tab for the same website. The main area displays a table of page templates.

Name	Website	Relative Url	Is Default
Access Denied	Expense Power Pages - expensepowerpages	~/Pages/AccessDenied.aspx	No
Default studio template	Expense Power Pages - expensepowerpages	~/Pages/Default.aspx	No
Full Page	Expense Power Pages - expensepowerpages	~/Pages/FullPage.aspx	No
Profile	Expense Power Pages - expensepowerpages	~/Pages/Profile.aspx	No
Search	Expense Power Pages - expensepowerpages	~/Pages/Search.aspx	No

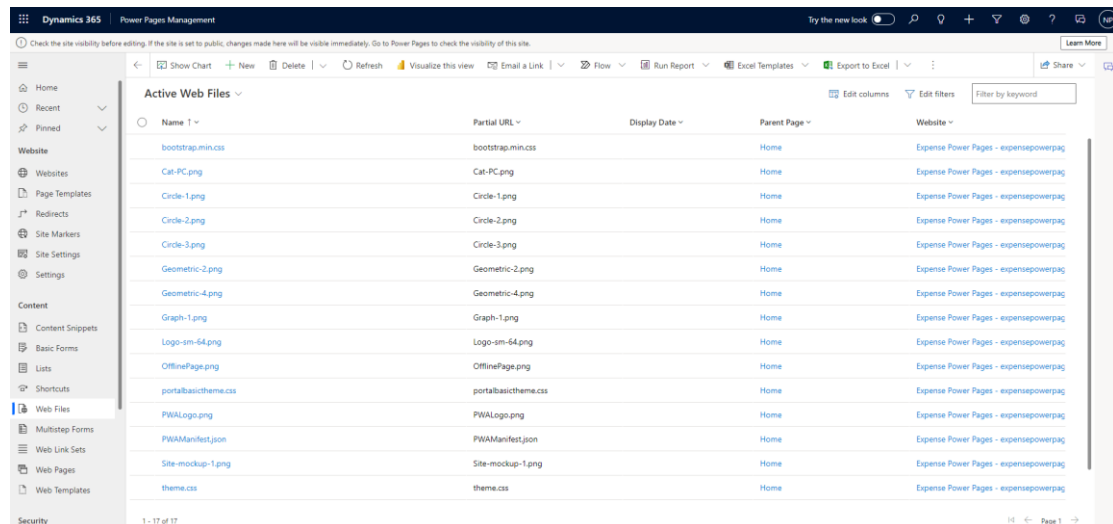
Website Access Permissions shows who can manage the site, its content, web links, and other stuff.



Content Snippets shows a list of snippets which are created once and then can be used any number of times in our site.



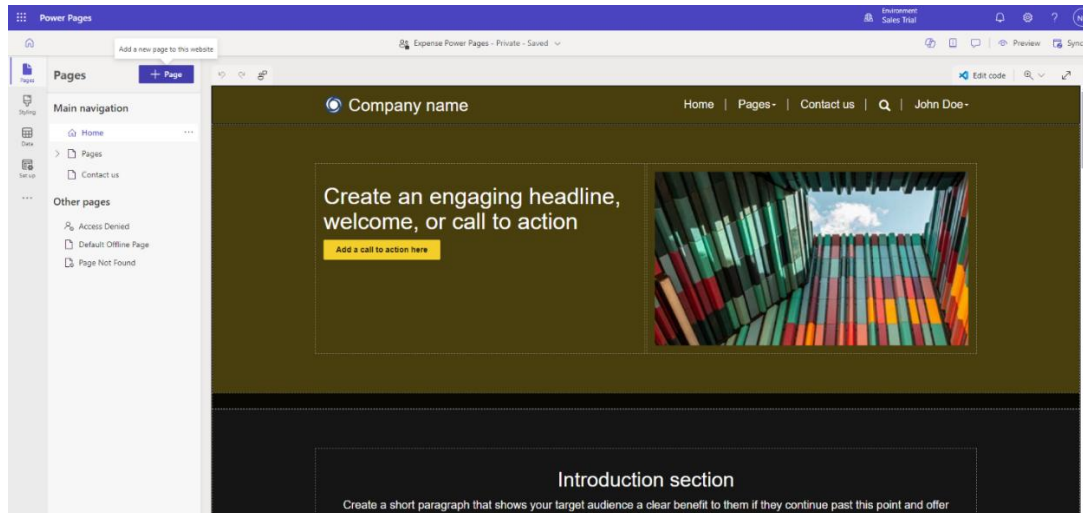
Web Files are files which are used in our website. These can be images, videos, html code, or text files.



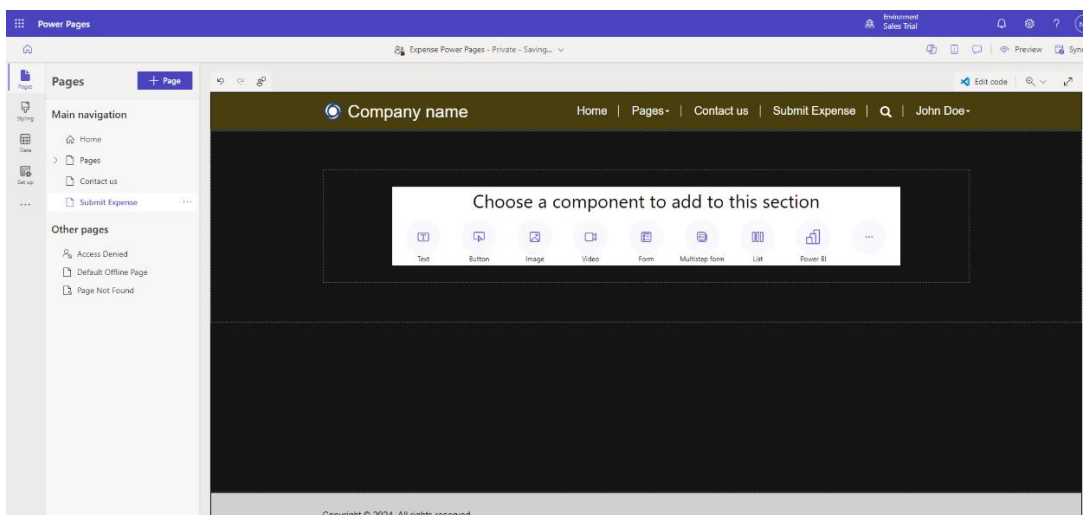
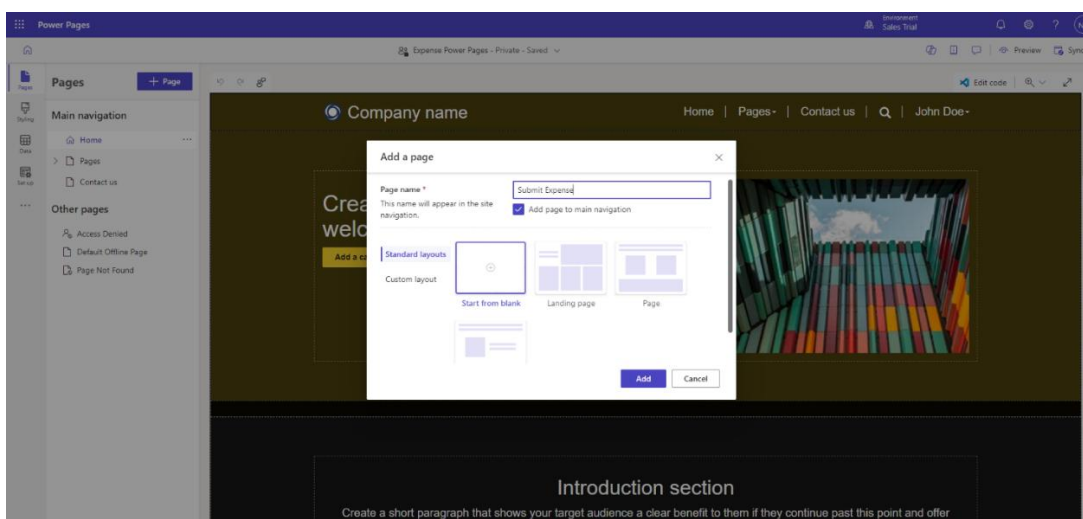
Web links, web pages, web templates, are all things that we can create and manage in Dynamics 365 as well as in Power Pages.

3. Adding a new page for expense requests

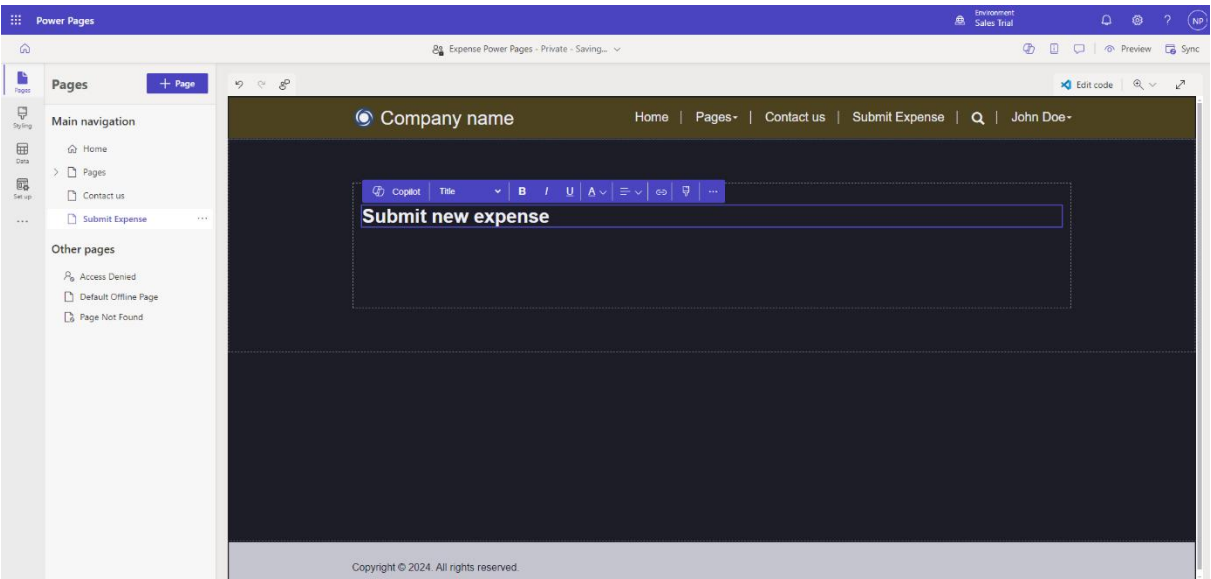
Let us add a new page to our website.



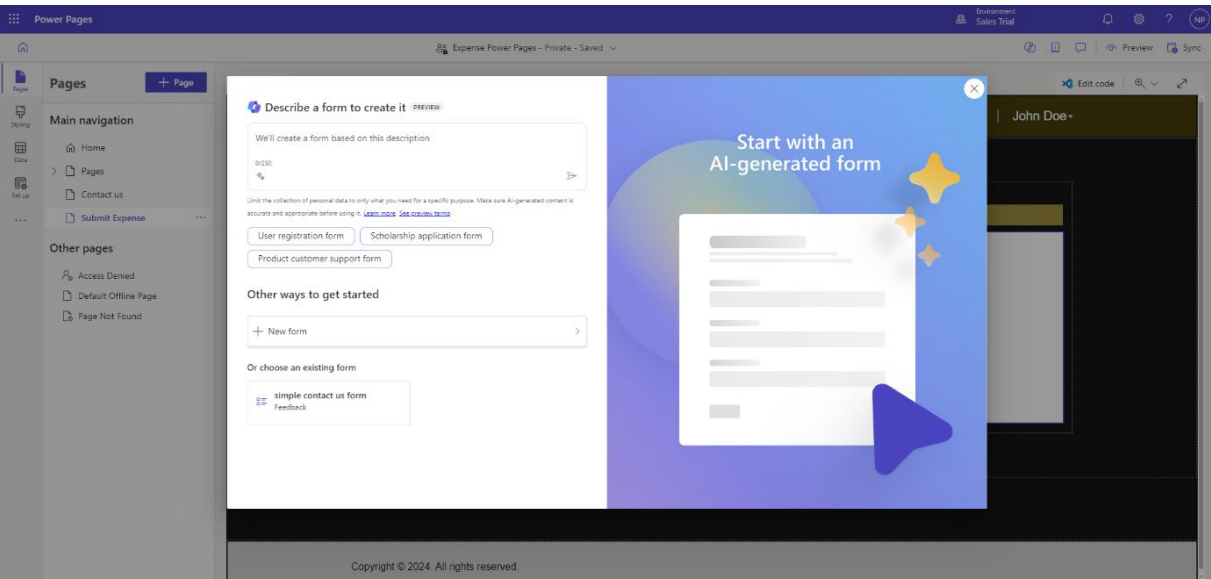
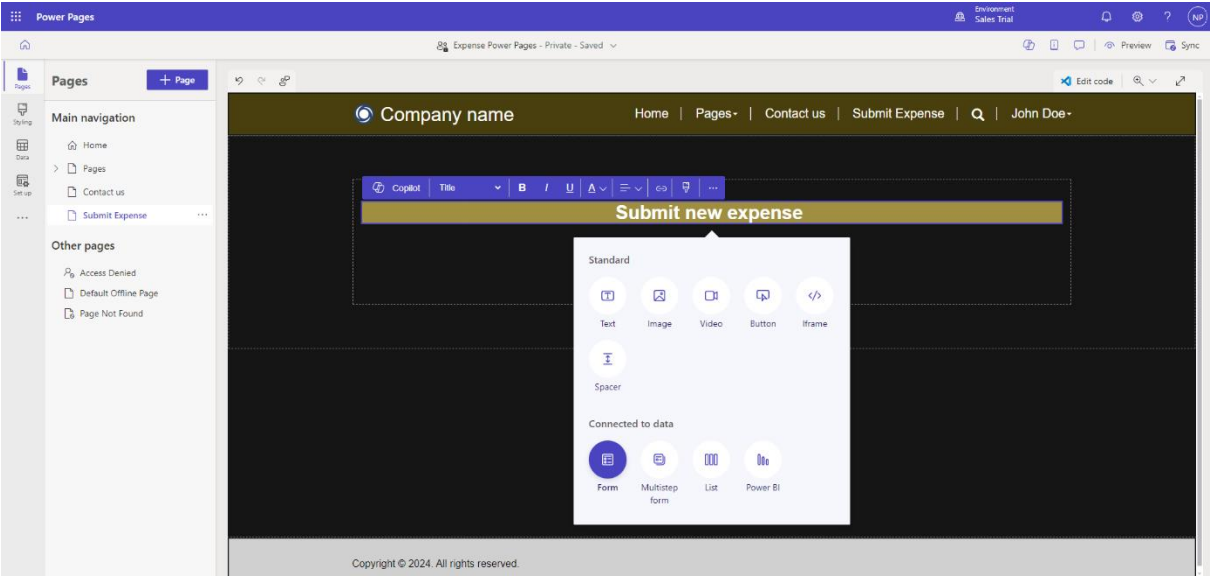
The purpose of this new page will be to allow a contact to submit their expenses.



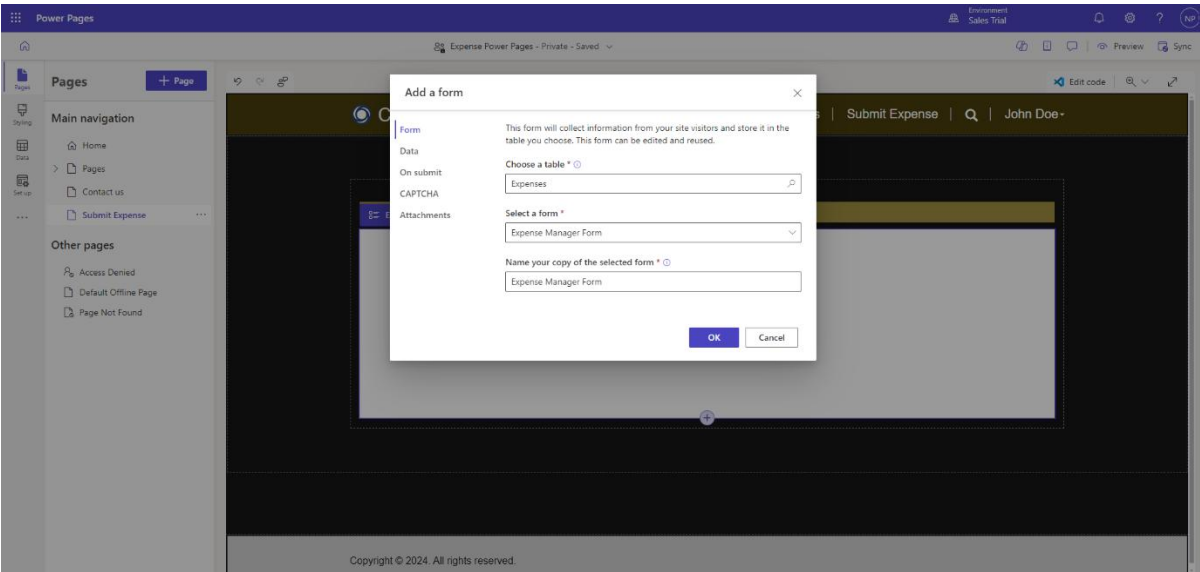
Let us display a title.



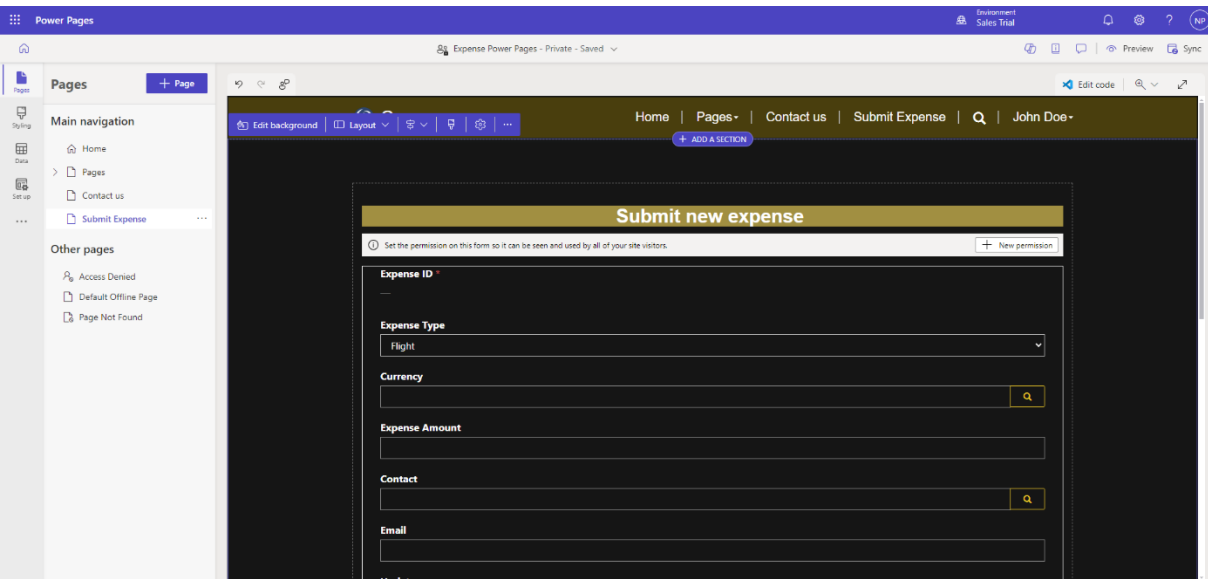
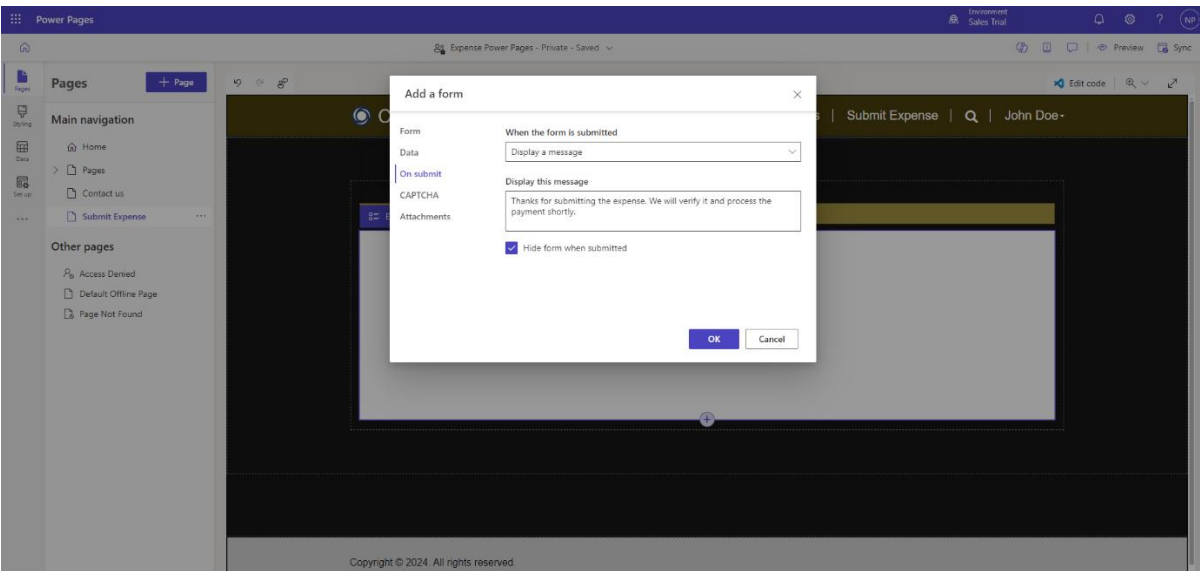
Next, we add a form for a user to submit an expense request.



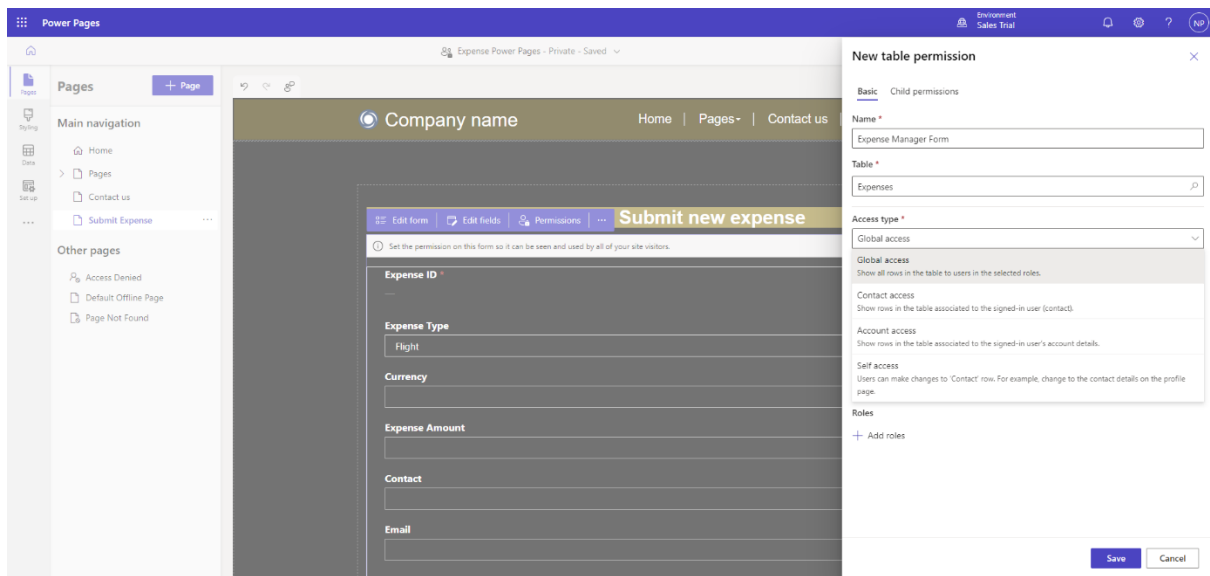
Click on + New form.



Here we have chosen Expenses as table, and Expense Manager Form as form.

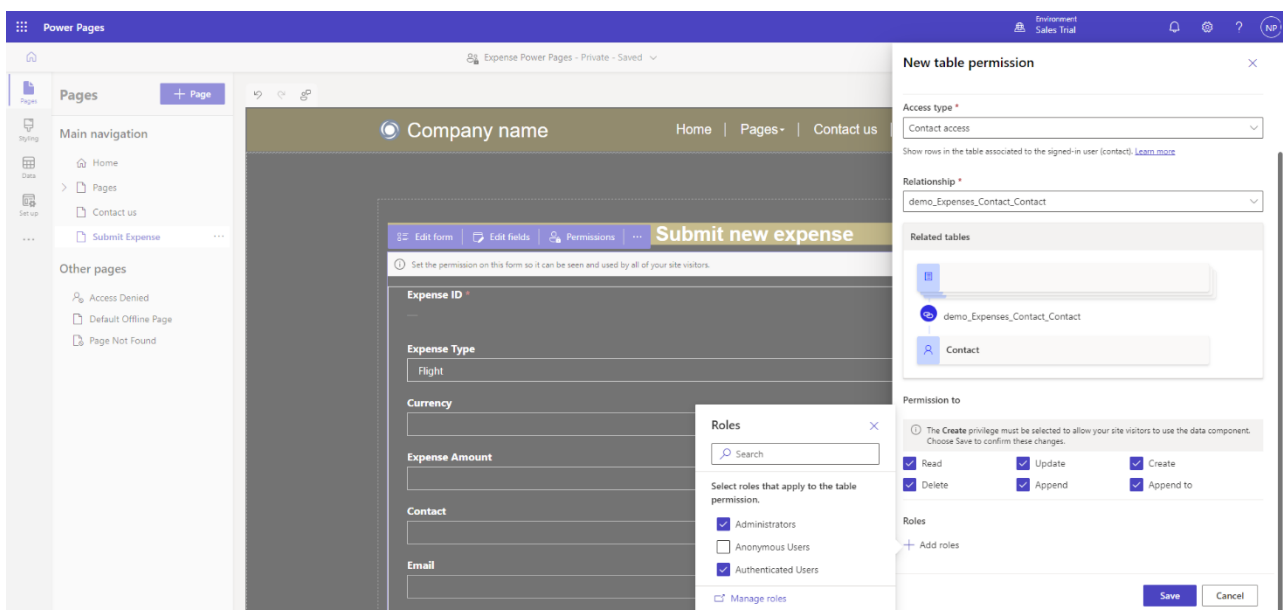


Once the form is displayed, we click on + New permission.



Global access means that everybody can see everyone's expenses. Contact access means that a contact can see only its own expenses. Account access is for accounts with multiple contacts; any contact belonging to a specific account will have the possibility to see the expenses of all the other contacts belonging to the same account. In our case we choose Contact access.

Subsequently, we will be asked for the relationship existing between Contact and Expenses.



The checked options define the operations that Administrators and Authenticated Users (in this case) are entitled to perform.

This is how our website will look like.

The screenshot shows a web form titled "Submit new expense" on a dark background. The form is contained within a light-colored box. It includes the following fields and controls:

- Expense ID**: A text input field.
- Expense Type**: A dropdown menu.
- Currency**: A text input field with a search icon (magnifying glass) to its right.
- Expense Amount**: A text input field.
- Contact**: A text input field with a search icon (magnifying glass) to its right.
- Email**: A text input field.
- Update expense**: Radio buttons for "No" (selected) and "Yes".
- Exchange Rate**: A text input field.
- Status Reason**: A text input field.

Obviously, our form will be now available under Basic Forms in Dynamics 365.

The screenshot shows the Dynamics 365 Power Pages Management interface. The left sidebar contains navigation options: Home, Recent, Pinned, Website, Websites, Page Templates, Redirects, Site Markers, Site Settings, Settings, Content, Content Snippets, Basic Forms (selected), Lists, Shortcuts, Web Files, Multistep Forms, Web Link Sets, Web Pages, and Web Templates. The main area displays a table of Active Basic Forms.

Name	Website	Table Name	Mode	Form Name	Tab Name	Table Source Ty...	Record ID Para...	On Success	Redirect Web P...	Redirect URL
Expense Manager Form	Expense Powe	demo_expenses	Insert	Expense Manager...				Display Succes...		
simple contact us form	Expense Powe	feedback	Insert	simple contact us...				Display Succes...		

If we use the form to create a new expense of type Flight and amount 830, our custom message will appear.

The screenshot shows the website after a successful submission. The "Submit new expense" form is still visible, but a green message box has appeared below it with the text: "Thanks for submitting the expense. We will verify it and process the payment shortly." The footer of the page reads "Copyright © 2024. All rights reserved."

The new expense record is present in Dynamics 365 as well.

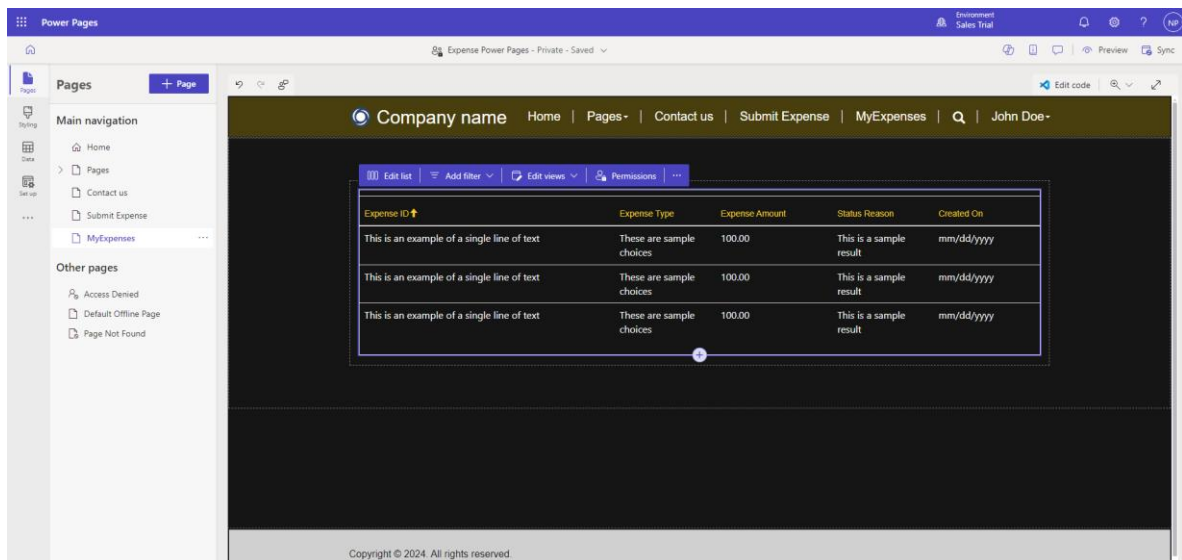
The screenshot shows the Dynamics 365 Expense System interface. The top navigation bar includes 'Dynamics 365' and 'Expense System'. The left sidebar shows a navigation menu with 'Home', 'Recent', 'Pinned', 'Expense Group', 'Contacts', 'Expenses', and 'Open Google'. The main content area displays the 'EXPENSE-100040 - Saved' form. The 'General' tab is active, showing fields for 'Expense ID' (EXPENSE-100040), 'Expense Type' (Flight), 'Currency' (Euro), 'Expense Amount' (€630.00), 'Contact' (---), 'Email' (---), 'Update expense' (No), 'Exchange Rate' (1.000000000), 'Status Reason' (Active), 'Created By' (# Portals-Expense Power Pages), 'Modified By' (# Portals-Expense Power Pages), and 'Owner' (# Portals-Expense Power Pages). A 'Description' field is visible at the bottom.

At this point we add a new list to show Active Expenses.

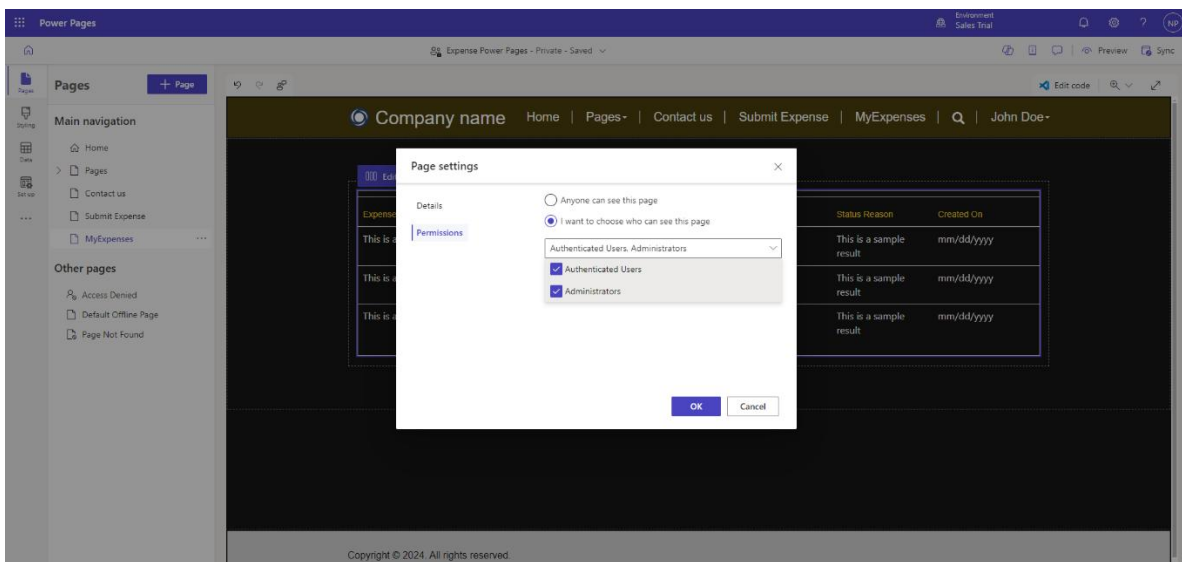
The screenshot shows the Dynamics 365 Power Pages Management interface. The top navigation bar includes 'Dynamics 365' and 'Power Pages Management'. The left sidebar shows a navigation menu with 'Home', 'Recent', 'Pinned', 'Website', 'Content', and 'Security'. The main content area displays the 'New List - Unsaved' form. The 'General' tab is active, showing fields for 'Name' (Expense), 'Table Name' (demo_expenses), and 'Website' (Expense Power Pages - expenspowerpages). The 'Views' section shows a dropdown menu with options: 'Active Expenses', 'Expenses Associated View', 'Expenses Greater than 500', 'Expenses Lookup View', and 'Inactive Expenses'.

Since we just added a new item to our Power Pages site, first we need to sync. Next, we create a new page called MyExpenses and add to it the list we have created.

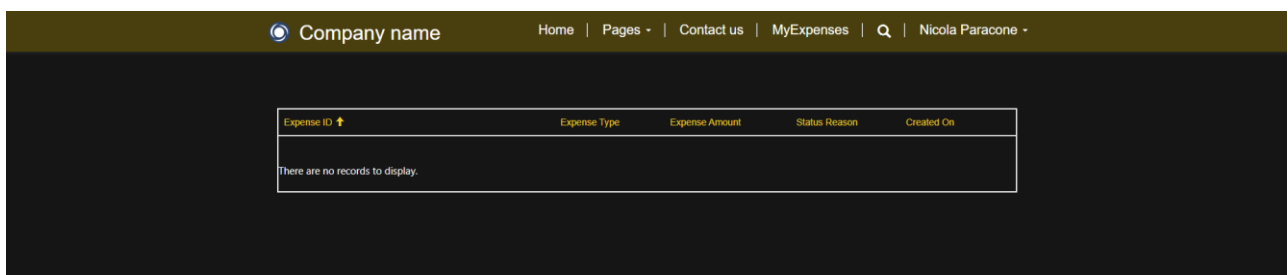
The screenshot shows the Dynamics 365 Power Pages interface. The top navigation bar includes 'Power Pages' and 'Expense Power Pages - Private - Saved'. The left sidebar shows a navigation menu with 'Pages', 'Main navigation', and 'Other pages'. The main content area displays the 'MyExpenses' page. The page has a header with 'Company name' and navigation links: 'Home', 'Pages', 'Contact us', 'Submit Expense', 'MyExpenses', and 'John Doe'. The main content area shows a 'Choose a component to add to this section' dialog with options: 'Text', 'Button', 'Image', 'Video', 'Form', 'Multistep form', 'List', and 'Power BI'.



Hence, we set some user permissions.



Before visualizing the preview, we move Submit Expense to Other pages, so that it will not be visible in our site.



Even though we have expense records in CRM, they are not shown here because the permission we have set up is for contact access. This means that the user currently logged doesn't own any expense request.

Let us then create a new form in Dynamics 365 for editing,

The screenshot shows the Dynamics 365 Power Pages Management console. The left sidebar contains navigation options: Home, Recent, Pinned, Website, Content, and Security. The 'Website' section is expanded, showing 'Basic Forms' as the selected option. The main area displays the 'New Basic Form' configuration for 'Edit Expense'. The 'General' tab is active, showing fields for Name, Table Name, Form Name, Tab Name, Mode, Record Source Type, Record ID Parameter Name, and Website. The 'Enable Table Permissions' checkbox is checked.

Field	Value
Name	Edit Expense
Table Name	demo_expenses
Form Name	Expense Manager Form
Tab Name	---
Mode	Edit
Record Source Type	Query String
Record ID Parameter Name	id
Website	Expense Power Pages - expensepowerpages
Enable Table Permissions	☒

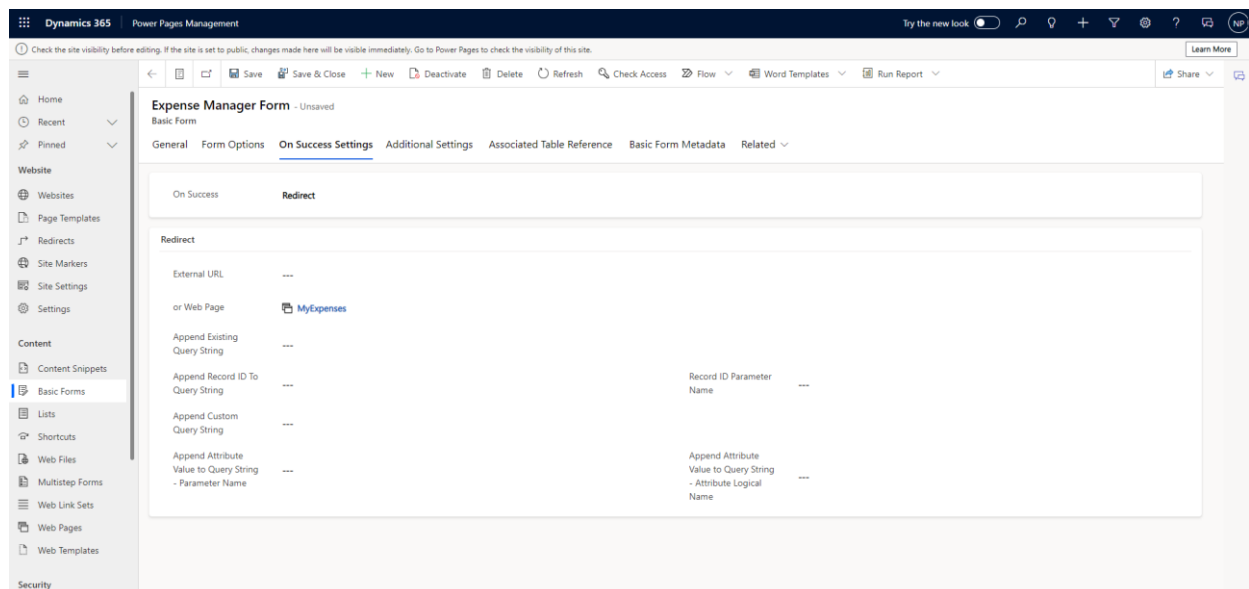
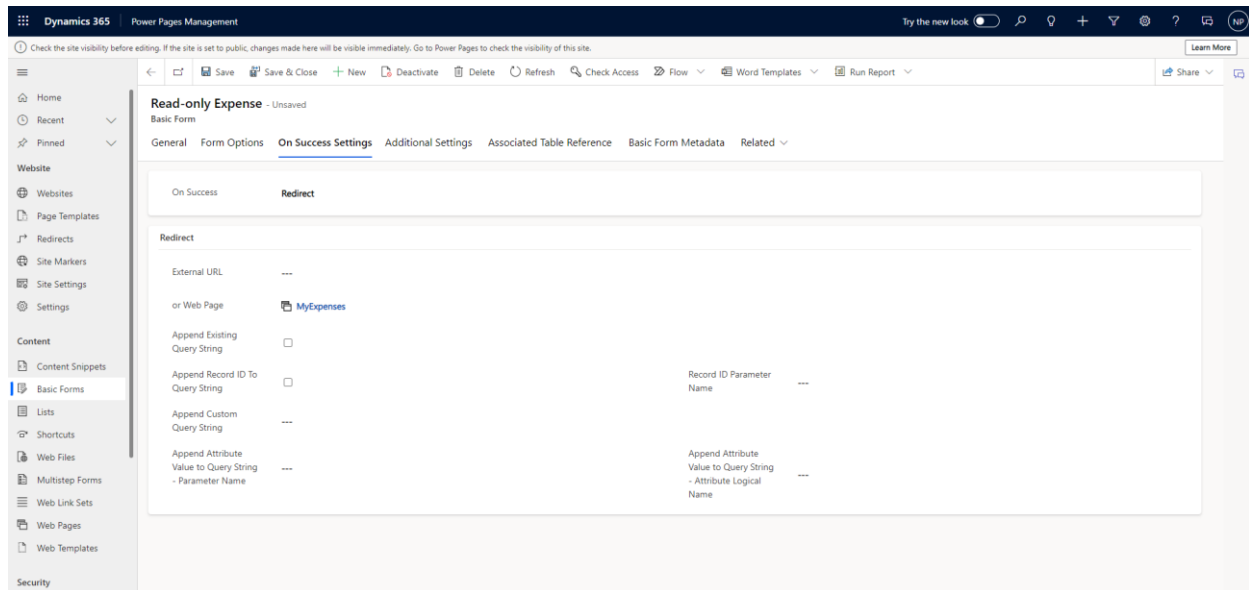
The screenshot shows the Dynamics 365 Power Pages Management console. The left sidebar is the same as the previous screenshot. The main area displays the 'Edit Expense' form configuration. The 'On Success Settings' tab is active, showing the 'Redirect' section. The 'Redirect' section contains fields for External URL, or Web Page, Append Existing Query String, Append Record ID To Query String, Append Custom Query String, and Append Attribute Value to Query String - Parameter Name. The 'or Web Page' field is set to 'MyExpenses'.

Field	Value
External URL	---
or Web Page	MyExpenses
Append Existing Query String	☐
Append Record ID To Query String	☐
Append Custom Query String	---
Append Attribute Value to Query String - Parameter Name	---

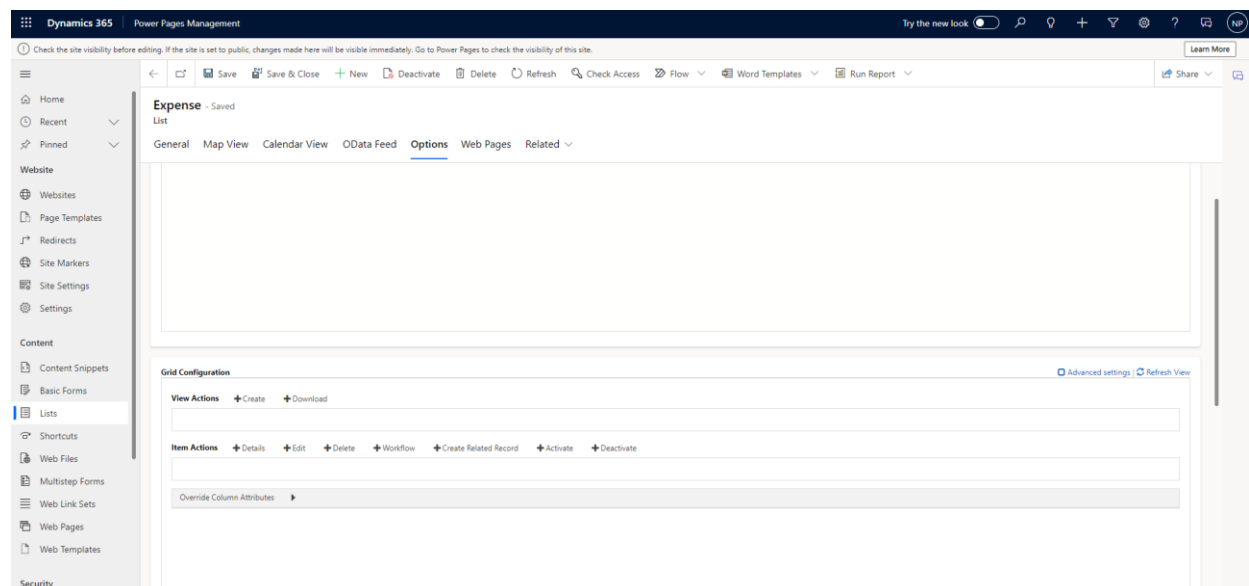
and a new form for reading.

The screenshot shows the Dynamics 365 Power Pages Management console. The left sidebar is the same as the previous screenshots. The main area displays the 'New Basic Form' configuration for 'Read-only Expense'. The 'General' tab is active, showing fields for Name, Table Name, Form Name, Tab Name, Mode, Record Source Type, Record ID Parameter Name, and Website. The 'Enable Table Permissions' checkbox is checked.

Field	Value
Name	Read-only Expense
Table Name	demo_expenses
Form Name	Expense Manager Form
Tab Name	---
Mode	ReadOnly
Record Source Type	Query String
Record ID Parameter Name	id
Website	Expense Power Pages - expensepowerpages
Enable Table Permissions	☒



At this point we need to go back to our list and define the create and edit functionalities.



Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close New Deactivate Delete Refresh Check Access Flow Word Templates Run Report Share

Expense - Saved
List

General Map View Calendar View OData Feed **Options** Web Pages Related

Grid Configuration [Advanced settings](#) [Refresh View](#)

Create	Webpage
Target Type	Submit Expense (Expense Power Pages - expenspowerpages)
Redirect to Webpage	English (United States) (1033) The record has been saved.
Success Message	English (United States) (1033)
Button Label	English (United States) (1033)
Button Tooltip	English (United States) (1033)
Filter Criteria	

Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close New Deactivate Delete Refresh Check Access Flow Word Templates Run Report Share

Expense - Unsaved
List

General Map View Calendar View OData Feed **Options** Web Pages Related

Button Tooltip English (United States) (1033)

Filter Criteria

Item Actions +Details +Edit +Delete +Workflow +Create Related Record +Activate +Deactivate

Details

Target Type	Basic Form
Basic Form	Read-only Expense
Record ID Query String Parameter Name	id
Success Message	English (United States) (1033) The record has been saved.
Button Label	English (United States) (1033)
Button Tooltip	English (United States) (1033)
Filter Criteria	

Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close New Deactivate Delete Refresh Check Access Flow Word Templates Run Report Share

Expense - Saved
List

General Map View Calendar View OData Feed **Options** Web Pages Related

Grid Configuration [Advanced settings](#) [Refresh View](#)

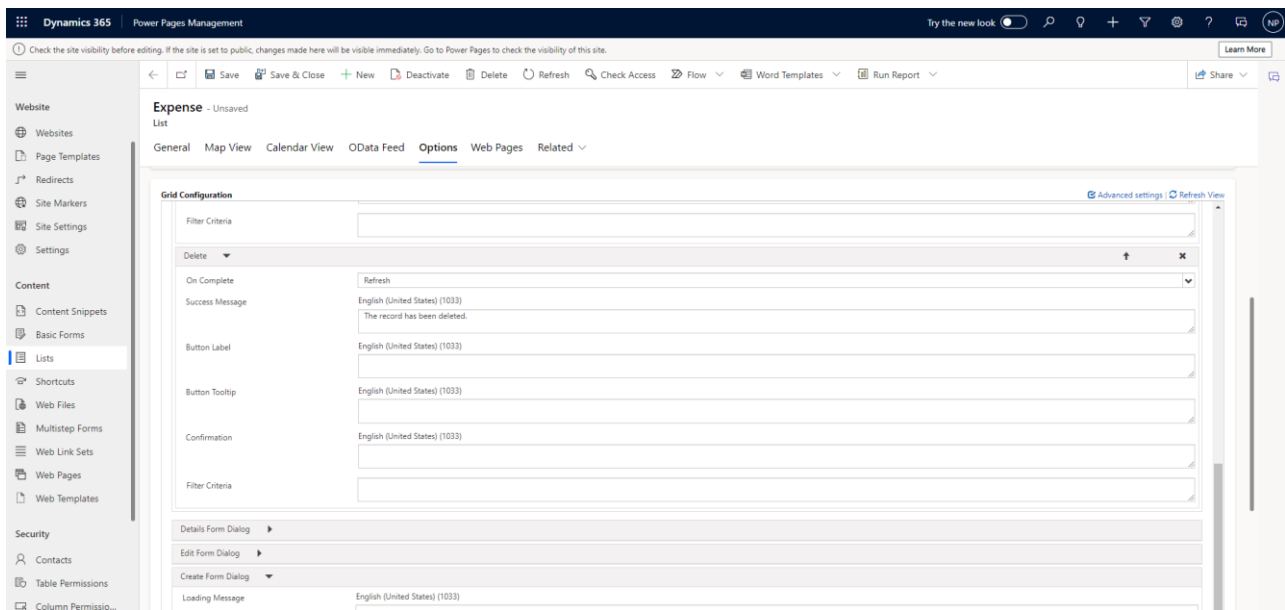
Filter Criteria	
Edit	
Target Type	Basic Form
Basic Form	Edit Expense
Record ID Query String Parameter Name	id
Success Message	English (United States) (1033) The record has been saved.
Button Label	English (United States) (1033)
Button Tooltip	English (United States) (1033)
Filter Criteria	

Details Form Dialog

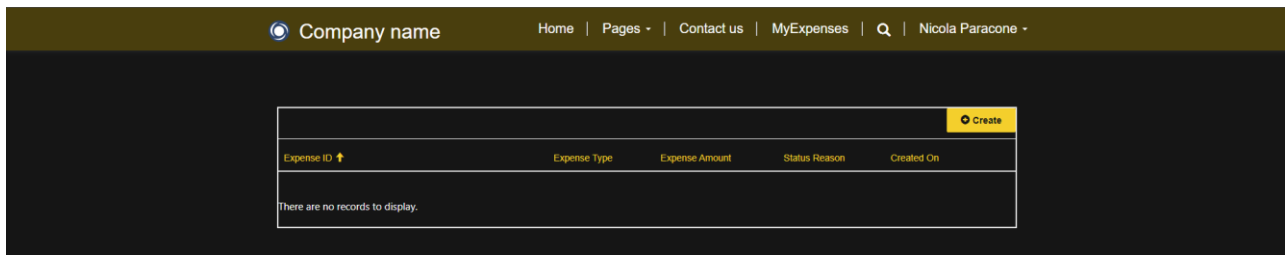
Edit Form Dialog

Create Form Dialog

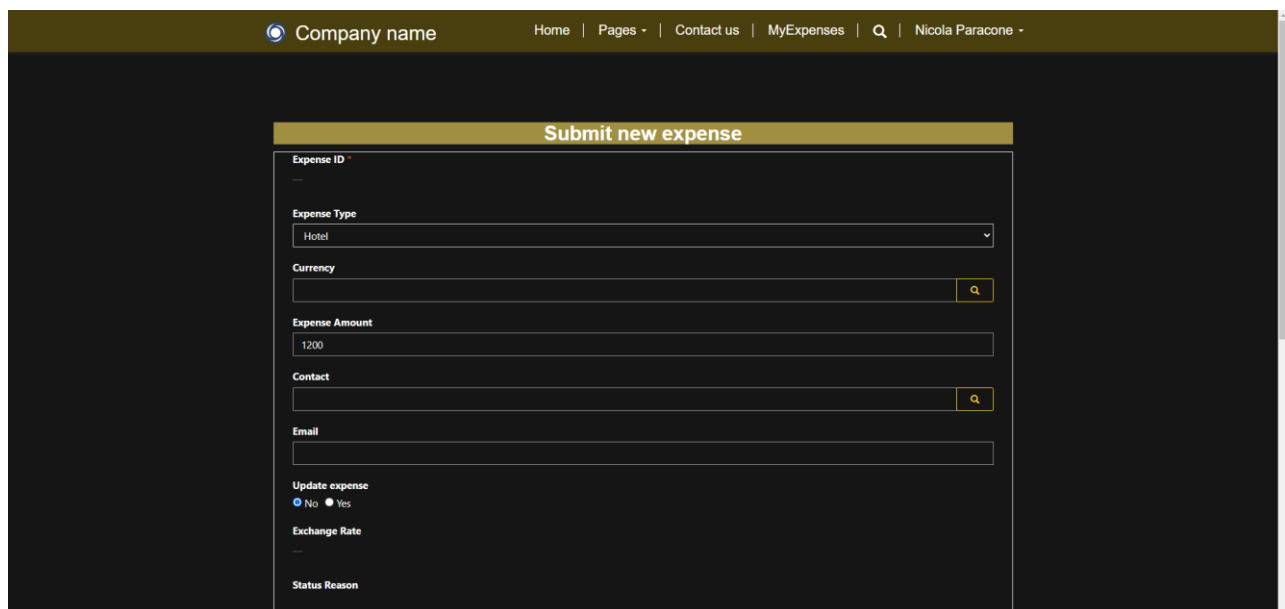
Loading Message English (United States) (1033)



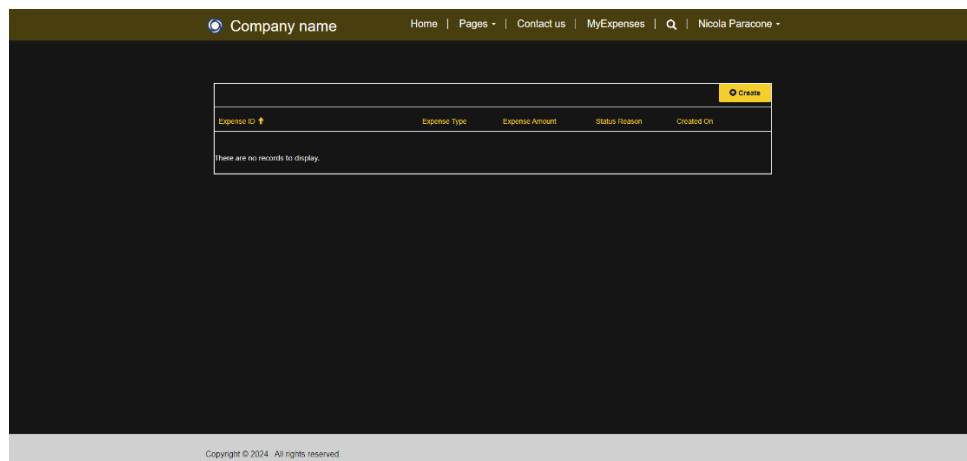
Back to Power Pages, we click on Sync to import the changes. The Create button will now appear.



Clicking on it brings us to the Submit new expense form we previously have created.

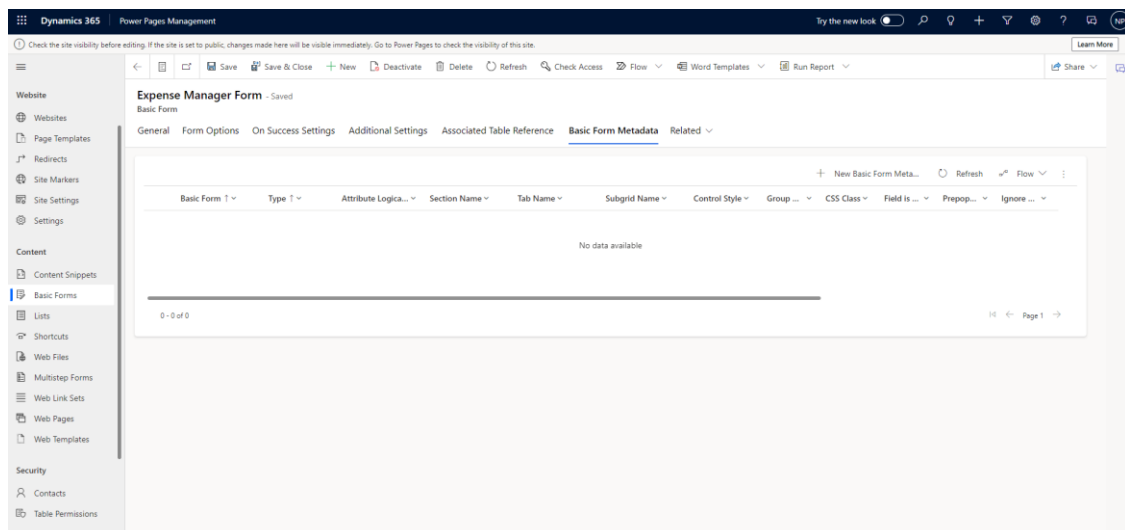


When the new expense is submitted, we are redirected to the grid containing all the expenses.

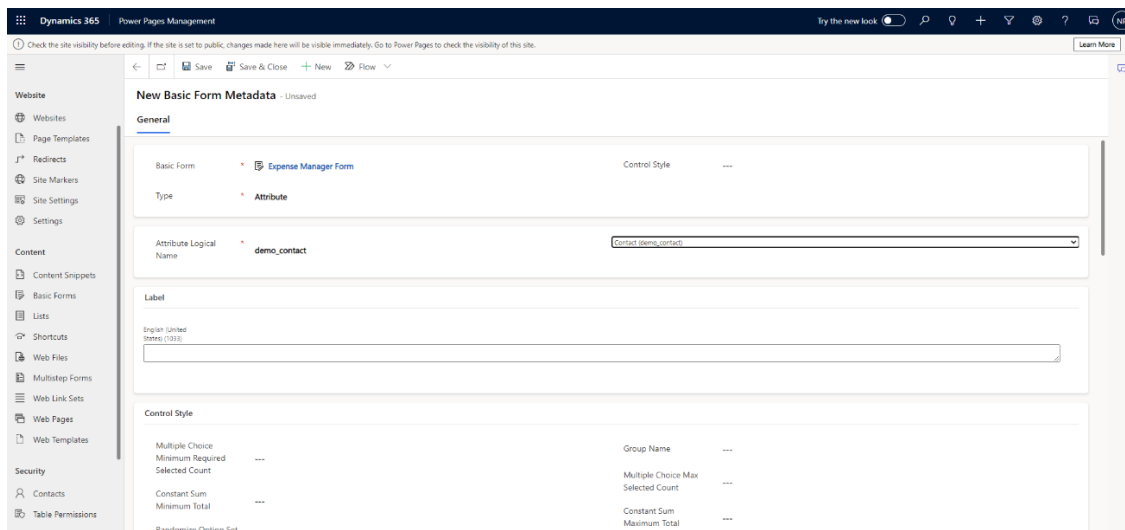


Notice that according to our permissions we don't see any expense. This is because we do not result as the contact owners.

Let us go to Basic Forms in Dynamics 365 and open Expense Manager Form. We then navigate to Basic Form Metadata.



Click on + New Basic Form Metadata.



Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close + New Flow

Website

- Web Sites
- Page Templates
- Redirects
- Site Markers
- Site Settings
- Settings

Content

- Content Snippets
- Basic Forms
- Lists
- Shortcuts
- Web Files
- Multistep Forms
- Web Link Sets
- Web Pages
- Web Templates

Security

- Contacts
- Table Permissions
- Column Permissions

New Basic Form Metadata - Unsaved

General

Prepopulate Field

Ignore Default Value ☒

Type **Current Portal User**

Value ---

From Attribute *** contactid** Contact (contactid)

Set Value On Save

Set Value On Save ☐

Type ---

Value ---

From Attribute ---

Lookup Settings

Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close + New Deactivate Delete Refresh Check Access View Hierarchy Flow Word Templates Run Report Share

Website

- Web Sites
- Page Templates
- Redirects
- Site Markers
- Site Settings
- Settings

Content

- Content Snippets
- Basic Forms
- Lists
- Shortcuts
- Web Files
- Multistep Forms
- Web Link Sets
- Web Pages
- Web Templates

Security

- Contacts
- Table Permissions
- Column Permissions

Expense Manager Form demo_contact - Unsaved

Basic Form Metadata

General Related

Set Value On Save

Set Value On Save ☒

Type **Current Portal User**

Value ---

From Attribute *** contactid** Contact (contactid)

Lookup Settings

Basic Form for Create **Expense Manager Form**

Description/Instructions

Add Description **No**

Validation

Dynamics 365 Power Pages Management Try the new look

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site.

Save Save & Close + New Deactivate Delete Refresh Check Access View Hierarchy Flow Word Templates Run Report Share

Website

- Web Sites
- Page Templates
- Redirects
- Site Markers
- Site Settings
- Settings

Content

- Content Snippets
- Basic Forms
- Lists
- Shortcuts
- Web Files
- Multistep Forms
- Web Link Sets
- Web Pages
- Web Templates

Security

- Contacts
- Table Permissions
- Column Permissions

Expense Manager Form demo_contact - Unsaved

Basic Form Metadata

General Related

Description/Instructions

Add Description **Yes**

Position **Above the field**

Use Attribute's Description Property **No**

Description

English United States (US)

This is a simple description.

Validation

Validation Error Message

English United States (US)

Regular Expression ---

Sync the configurations.

Note that the contact which is creating the new expense is now automatically added.

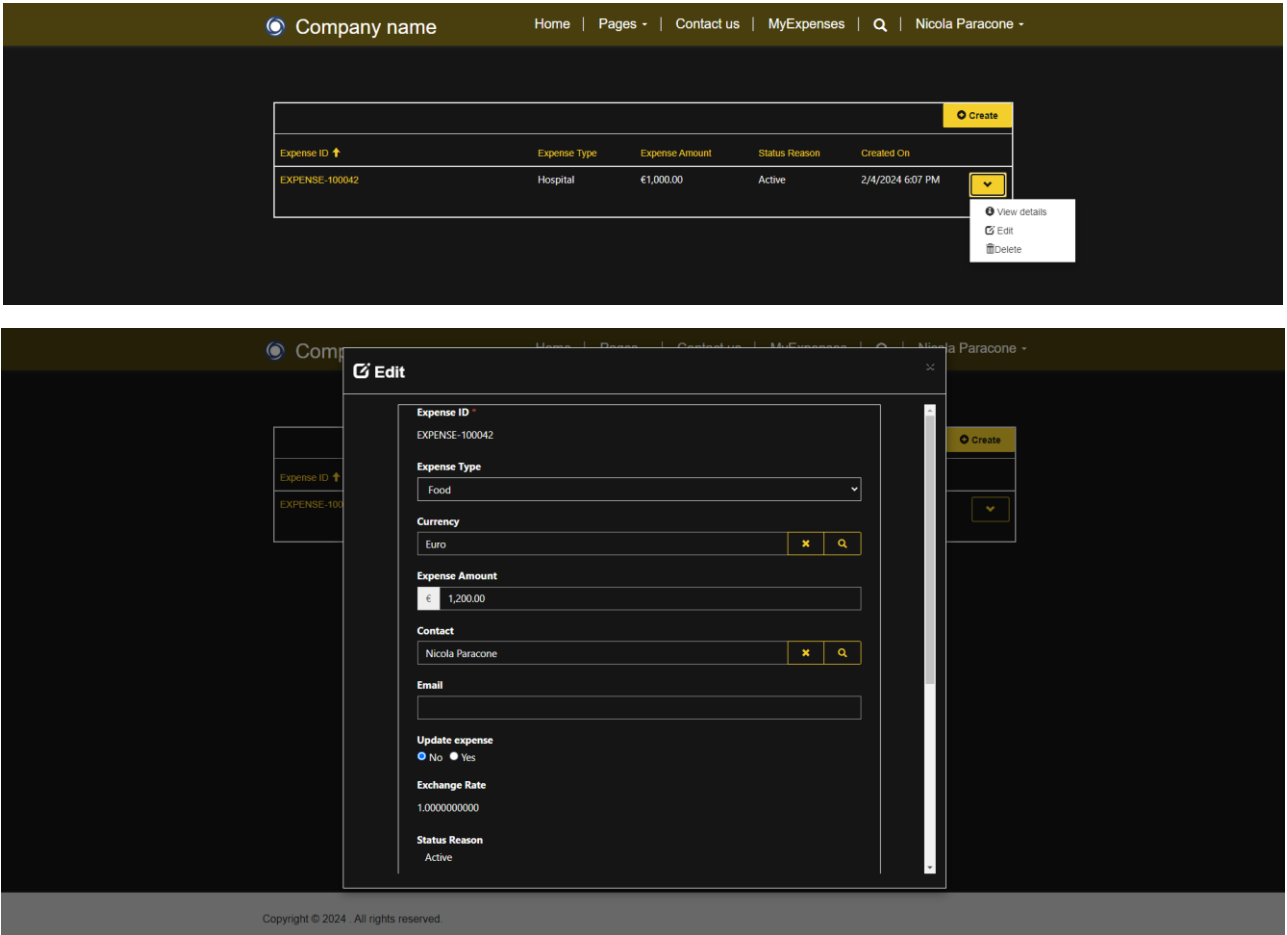
The screenshot shows a web application interface with a dark header bar containing the company name and navigation links. The main content area features a form titled 'Submit new expense'. The form includes fields for 'Expense ID', 'Expense Type' (a dropdown), 'Currency' (with a search icon), 'Expense Amount', 'Contact' (with a description 'This is a simple description.' and a search icon), 'Email', 'Update expense' (a radio button group with 'No' selected), 'Exchange Rate', and 'Status Reason'.

As a last step, we need to give the correct table permissions also to Contact.

The screenshot shows the 'New Table Permission' form in the Dynamics 365 Power Pages Management interface. The form is titled 'New Table Permission - Unsaved'. It has a 'General' tab. The 'Name' field is set to 'Contact'. The 'Table Name' field is set to 'contact'. The 'Website' field is set to 'Expense Power Pages - expensepowerpages'. The 'Access Type' field is set to 'Self'. The 'Privileges' section shows a table with columns for 'Read', 'Write', 'Create', 'Delete', 'Append', and 'Append To'. The 'Read', 'Write', 'Append', and 'Append To' checkboxes are checked. The 'Create' checkbox is unchecked. The 'Delete' checkbox is unchecked. The 'Child Table Permissions' section is empty.

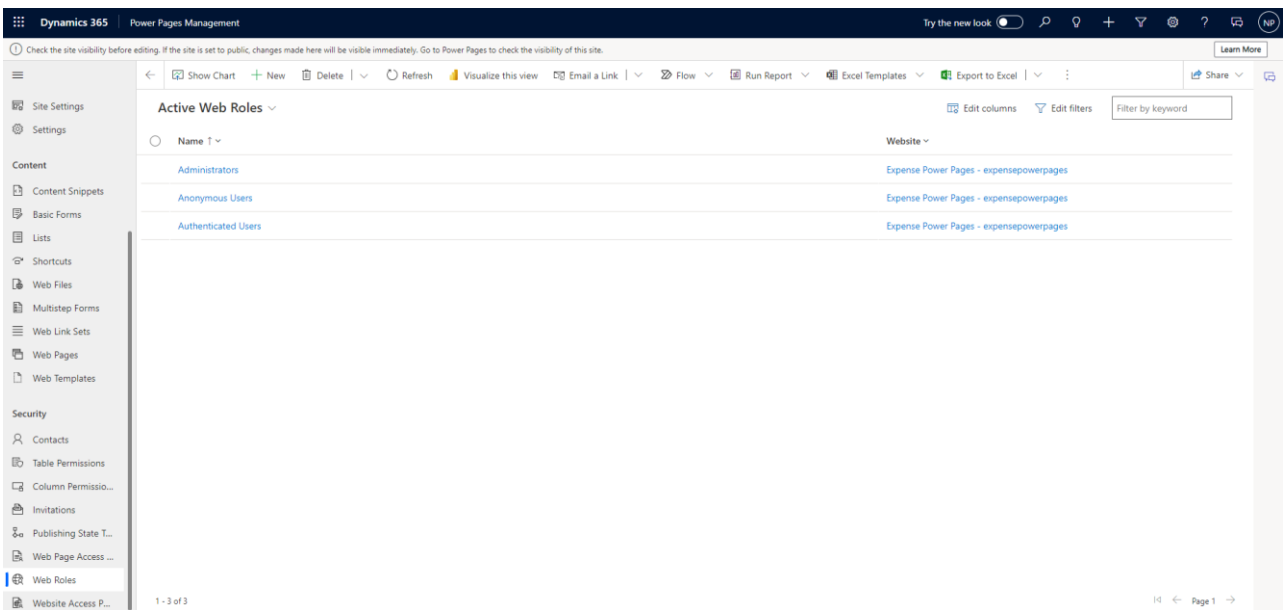
The screenshot shows the 'Contact' table permission configuration in the Dynamics 365 Power Pages Management interface. The form is titled 'Contact - Saved Table Permission'. It has a 'General' tab. The 'Name' field is set to 'Contact'. The 'Table Name' field is set to 'contact'. The 'Parent Relationship' field is set to 'Contact (contact)'. The 'Read', 'Write', 'Append', and 'Append To' checkboxes are checked. The 'Create' checkbox is unchecked. The 'Delete' checkbox is unchecked. The 'Child Table Permissions' section is empty. A 'Web Roles' section is visible at the bottom, showing a table with columns for 'Name' and 'Website'. The 'Add Existing Web Role' button is visible. A 'Lookup Records' dialog is open on the right, showing a search bar and a list of records. The 'Add' button is visible at the bottom right.

After syncing again the configurations, a newly created expense will be displayed as expected. There we also have the possibility to view details, edit, or delete the record.



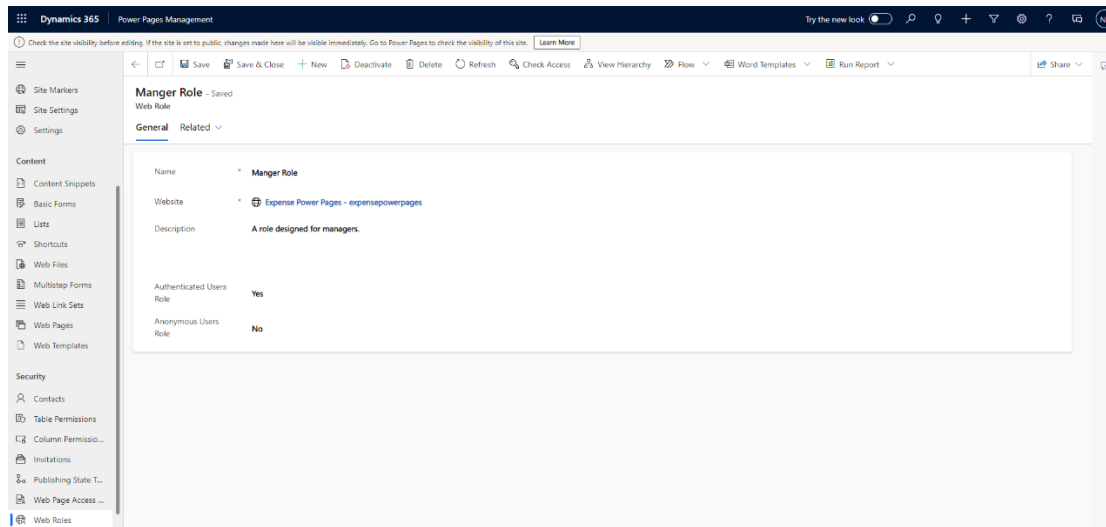
4. Custom web roles

So far, we have added some web roles to our table permissions.

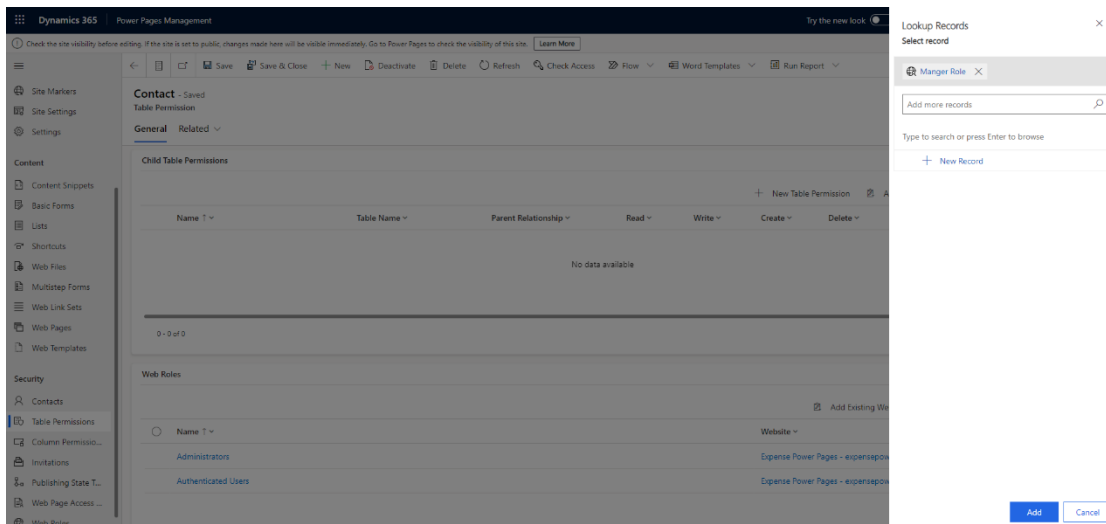


In most scenarios, we will create custom web roles.

For example, we can create a Manger Role.

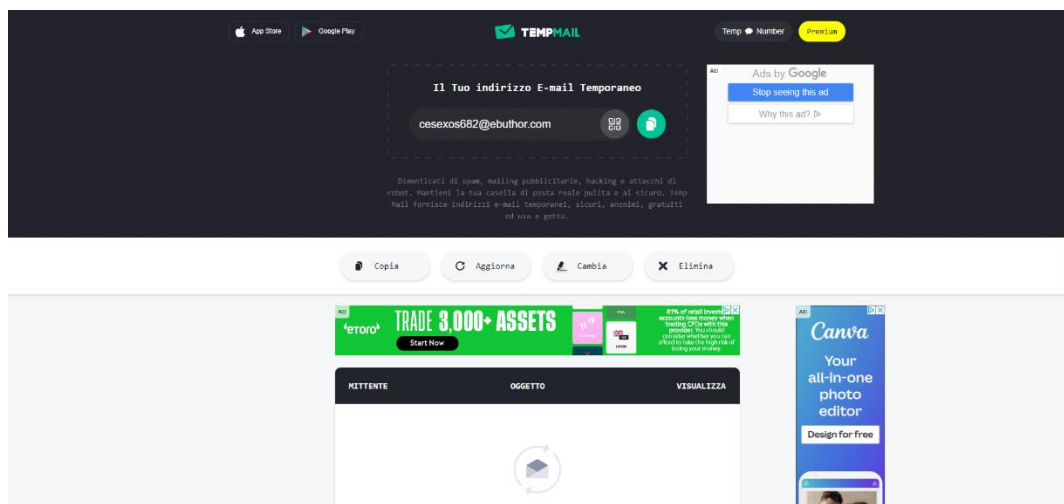


Next, we add the newly created web role to the table permissions of Contact.



5. Redeeming an invitation

Let us create a new contact having an email generated as a temporary one (<https://temp-mail.org/it>).



Dynamics 365 Power Pages Management

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site. [Learn More](#)

Alastor Moody - Saved
Contact - Contact

Summary Details Files Expenses Related

CONTACT INFORMATION

First Name	Alastor
Last Name	Moody
Job Title	---
Account Name	---
Email	cesexos62@ebuthor.com
Business Phone	---
Mobile Phone	---
Fax	---
Preferred Method of Contact	Any
Address 1: Street 1	---
Address 1: Street 2	---
Address 1: Street 3	---

Timeline

Search timeline
Enter a note...

Who Knows Whom

No connections found
Add first name or last name to find the common connections. [Learn More](#)

Assistant

Notifications

No notifications or suggestions
Check back later to see what's new and stay up to date.

Company

RECENT OPPORTUNITIES

No data available

0 - 0 of 0 Page 1

We want this contact to access our website. So we click on Create Invitation.

Dynamics 365 Power Pages Management

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site. [Learn More](#)

New Invitation - Unsaved

General Advanced Activities & Notes

Name Alastor Moody Expiry Date ---

Type Single Inviter Alex Baker

Owner/Sender Nicola Paracone

Invitee

Invited Contact Alastor Moody

When Redeemed

Assign To Account ---

Execute Workflow on Redeeming Contact ---

Redemption

Redeemed Contact ---

This will create a new invitation record with a unique identifier. In the Advanced tab we can choose the Invitation Code, or CRM will create it for us after saving the record.

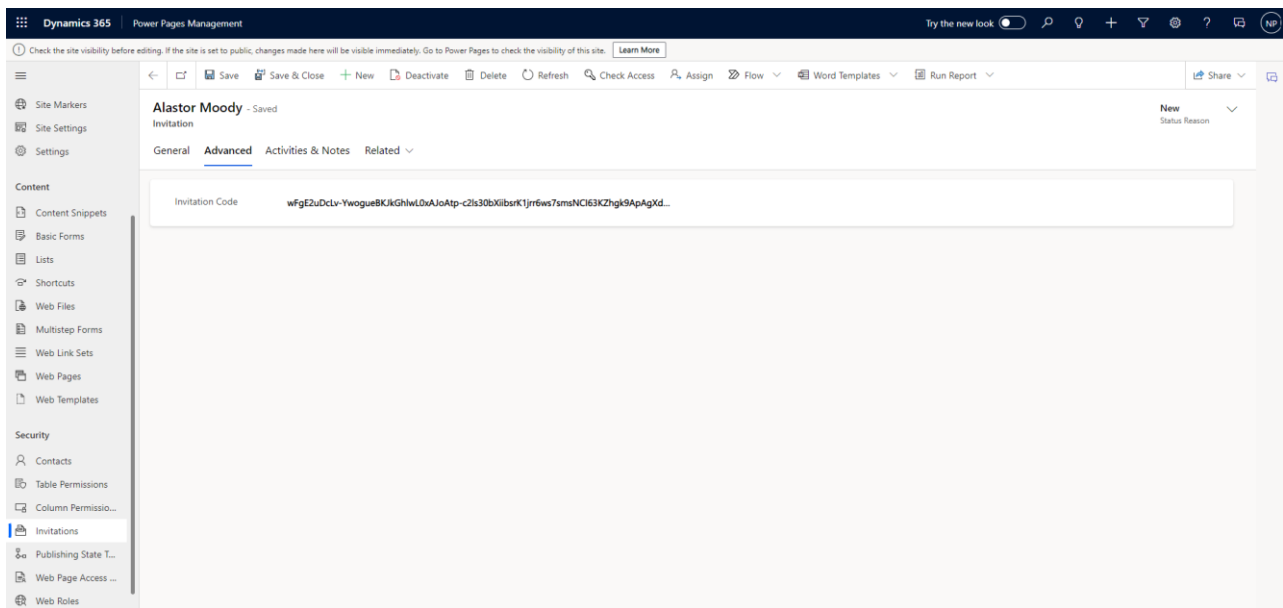
Dynamics 365 Power Pages Management

Check the site visibility before editing. If the site is set to public, changes made here will be visible immediately. Go to Power Pages to check the visibility of this site. [Learn More](#)

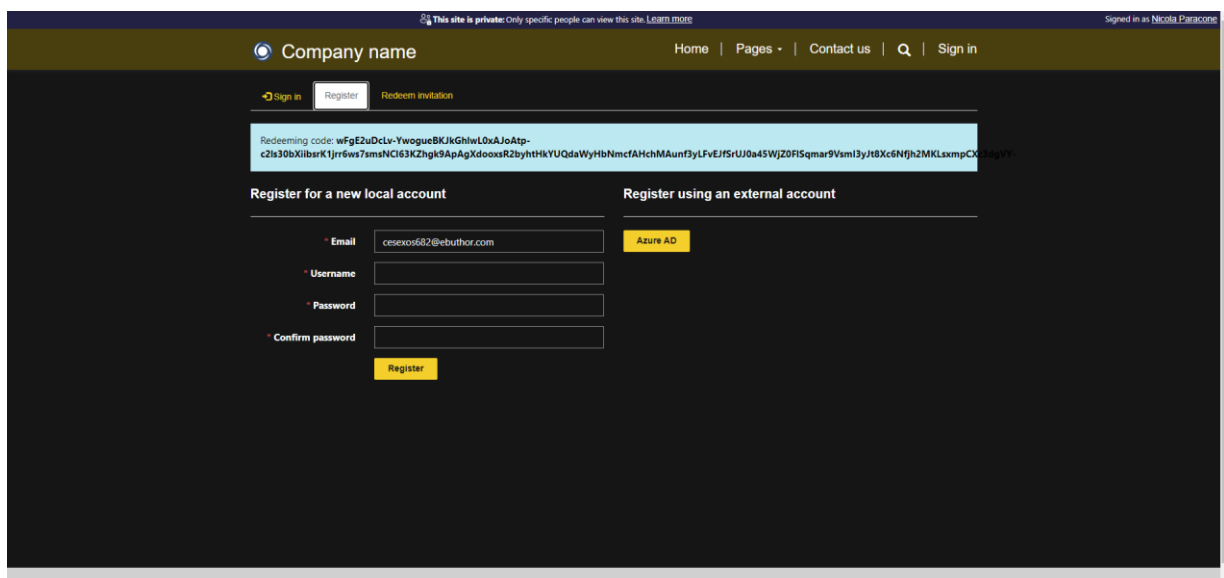
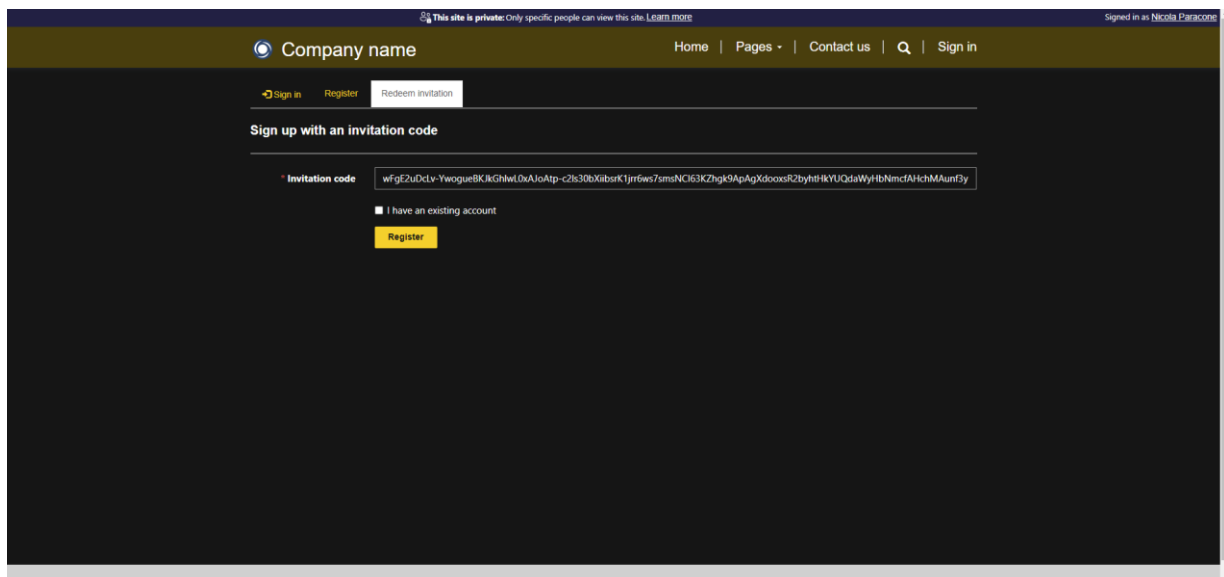
New Invitation - Unsaved

General Advanced Activities & Notes

Invitation Code ---



Copy the invitation code. In our website we go to Redeem invitation and paste the code.



This site is private: Only specific people can view this site. [Learn more](#) Signed in as Nicola Paracore

Company name Home | Pages | Contact us | Search | Sign in

Sign in Register Redeem invitation

Redeeming code: wFgE2uDcLv-YwoqueBKJkGhlwL0xJoAtp-c2Is30bXilbsrKtJrr6ws7smstNCI63KZhgk9ApAgXdoosR2byhtHkYUQdaWyHbNmcfAHchMAunf3yLFvEJf5rUJ0a45WjZ0FISqmar9VsmI3yJt8Xc6Nfjh2MKLxmpCXongVY-

Register for a new local account

Email

Username

Password

Confirm password

Register

Register using an external account

Azure AD

Company name Home | Pages | Contact us | MyExpenses | Search | Alastor Moody

Create

Expense ID ↑	Expense Type	Expense Amount	Status Reason	Created On
There are no records to display.				

6. Adding Liquid code

Liquid is an open-source template language that is integrated natively into Power Pages. It acts as a bridge between Microsoft Dataverse and the HTML or text output which is sent to the browser. We can use Liquid to add dynamic content to pages and to create various custom templates. Navigate to the following two links for a detailed introduction to Liquid:

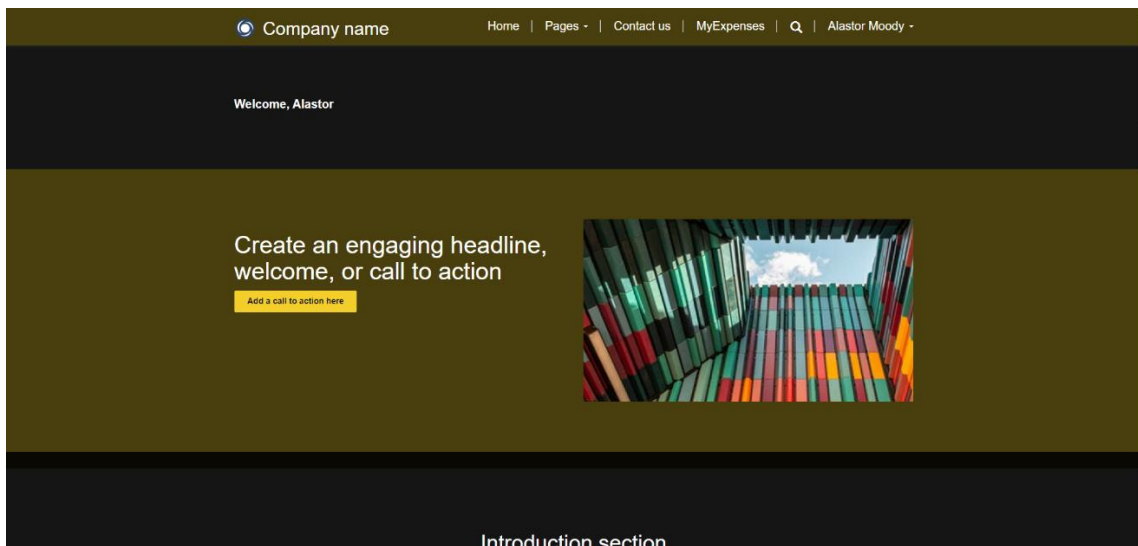
<https://learn.microsoft.com/en-us/training/modules/power-pages-liquid>

<https://learn.microsoft.com/en-us/power-pages/configure/liquid/liquid-overview>

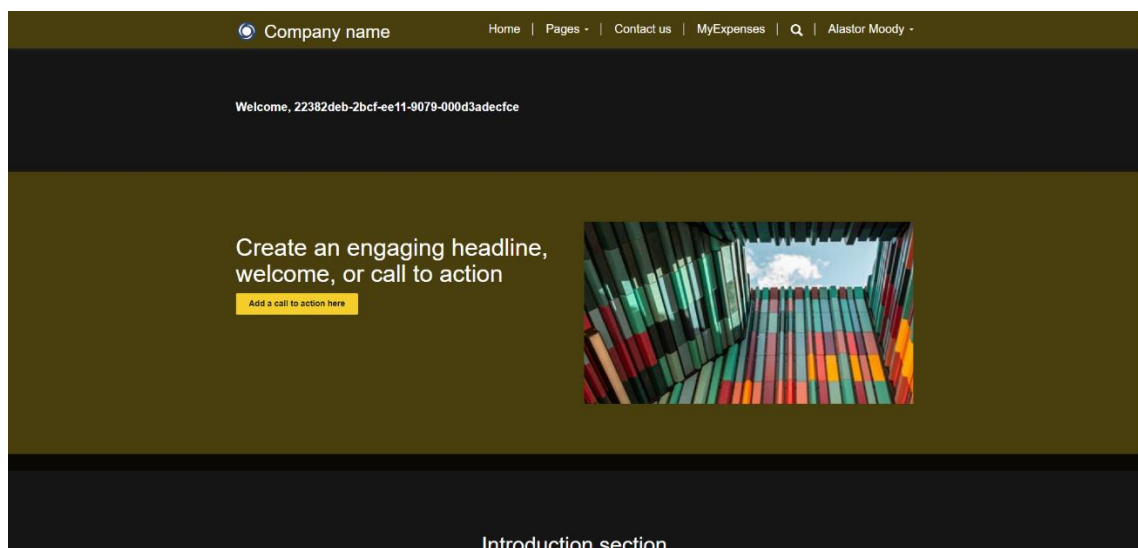
For example, the following code displays a welcome message for the user currently logged.

The screenshot displays a web browser window with a live preview of a landing page. The page has a dark header with a navigation menu, a large hero section with a blue background and white text, and a grid of service cards. The browser's developer tools are open, showing the HTML structure of the page. The HTML includes various Bootstrap classes and custom CSS for styling. The page is titled "MyServices" and features a "Services" section with a grid of service cards. The cards are styled with Bootstrap classes and custom CSS for a modern, clean look.

This message can be displayed, for instance, in a new section of the home page.



The line `{{user.contactid}}` will display, instead, the Guid of the contact.



```
EXPLODER
  MyExpenses.en-US.webpage copy.html
  Home.en-US.webpage copy.html X
  MyExpenses.en-US.costumcs.css

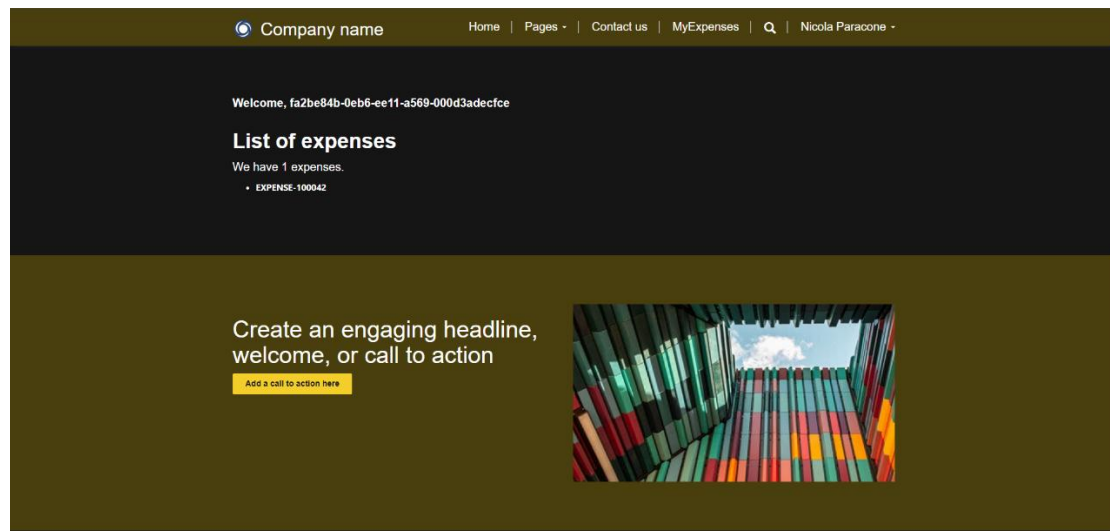
OPEN EDITORS
  MyExpenses.en-US.webpage copy.html...
  Home.en-US.webpage copy.html web...
  MyExpenses.en-US.costumcs.css web-p...

POWERPAGE (WORKING)
  Expense Power Pages
  advanced-forms
  basic-forms
  content-snippets
  data
  web-files
  web-pages
  Access Denied
  Contact us
  Default Office Page
  Home
  Home.en-US.costumcs.css
  Home.en-US.costumjs.js
  Home.en-US.webpage copy.html
  MyExpenses
  Page Not Found
  Pages
  Profile
  Search
  Submit Expense
  Subpage 1
  Subpage 2
  web-templates

OUTLINE
TIMELINE
POWER PAGES ACTIONS
  Extension Power Platform Tools added 1 folder to the workspace
```

The image shows the top portion of a website. The header is a dark blue bar with a logo on the left and navigation links on the right. The logo consists of a blue circle with a white dot inside, followed by the text "Company name". The navigation links are "Home", "Pages -", "Contact us", a magnifying glass icon, and "Sign in". Below the header is a dark grey hero section. On the left, it says "Welcome," followed by "Please log in." in a smaller font. On the right, there is a large, vibrant image of a colorful, geometric structure made of many small, rectangular blocks in various colors (red, orange, yellow, green, blue, purple) arranged in a way that creates a sense of depth and perspective, resembling a modern architectural installation or a large-scale art piece.

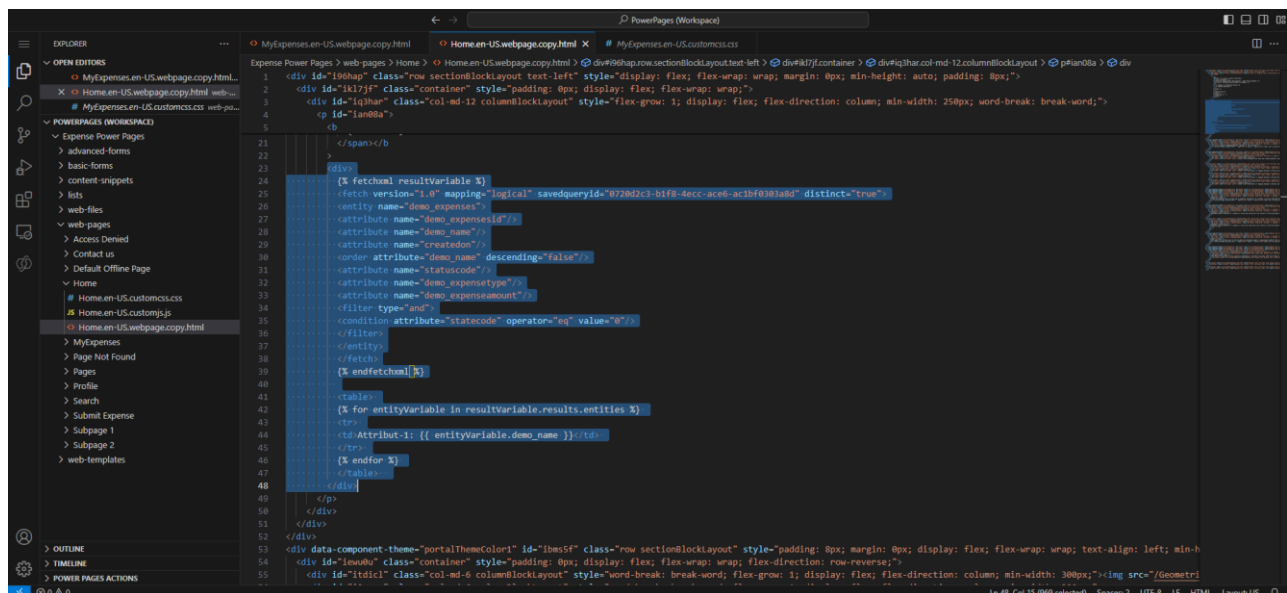
```
Visual Studio Code (Preview). Anyone, anytime, entirely on your browser. Read the Announcement Privacy & Cookies Terms of Use Download VS Code
EXPLORE
OPEN EDITORS
MyExpenses.en-US.webpage.cop...
Home.en-US.webpage.cop...
MyExpenses.en-US.customcss.c...
POWERPAGES (WORKSPACE)
Expense Power Pages
advanced forms
basic forms
current snippets
libs
web-files
web-pages
Access Denied
Contact us
Default Offline Page
Home
Home.en-US.customcss.css
Home.en-US.customjs.js
Home.en-US.webpage.cop...
MyExpenses
Page Not Found
Pages
Profile
Search
Subtotal Expense
Snippet 1
Snippet 2
web-templates
EXPLORE
TIMELINE
POWER PAGES ACTIONS
a o a o
```



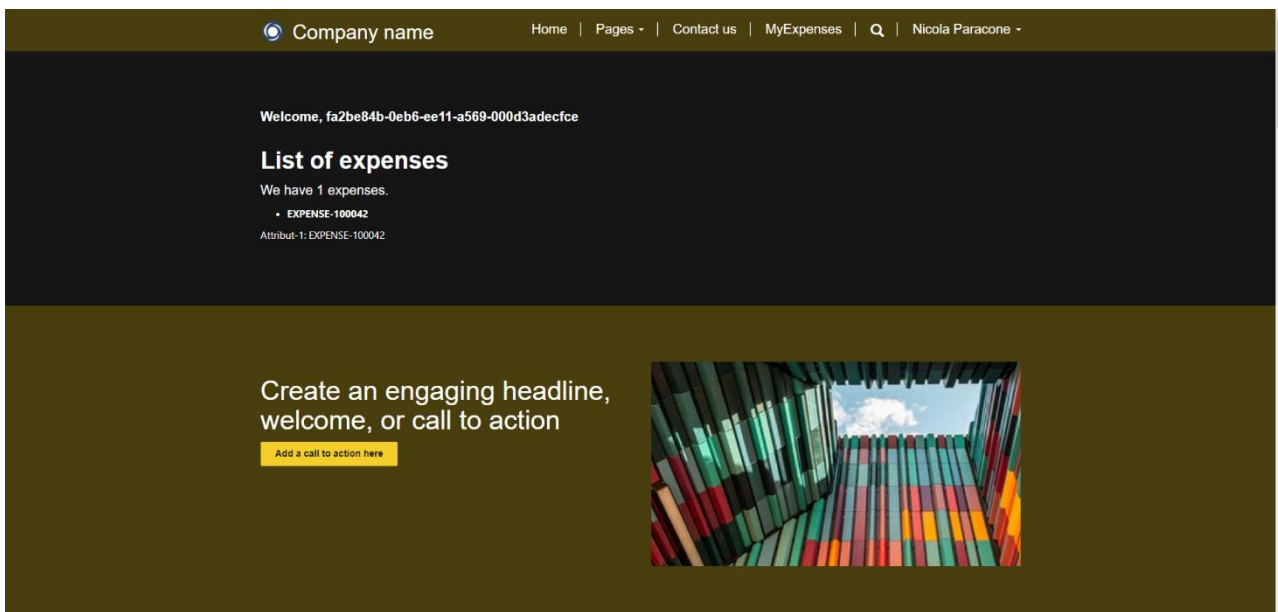
Notice that security works by default.

Inside our code snippet we can also insert a FetchXML query.

```
<fetch version="1.0" mapping="logical" savedqueryid="0720d2c3-b1f8-4ecc-ace6-ac1bf0303a8d" distinct="true">
  <entity name="demo_expenses">
    <attribute name="demo_expensesid"/>
    <attribute name="demo_name"/>
    <attribute name="createdon"/>
    <order attribute="demo_name" descending="false"/>
    <attribute name="statuscode"/>
    <attribute name="demo_expensetype"/>
    <attribute name="demo_expenseamount"/>
    <filter type="and">
      <condition attribute="statecode" operator="eq" value="0"/>
    </filter>
  </entity>
</fetch>
```

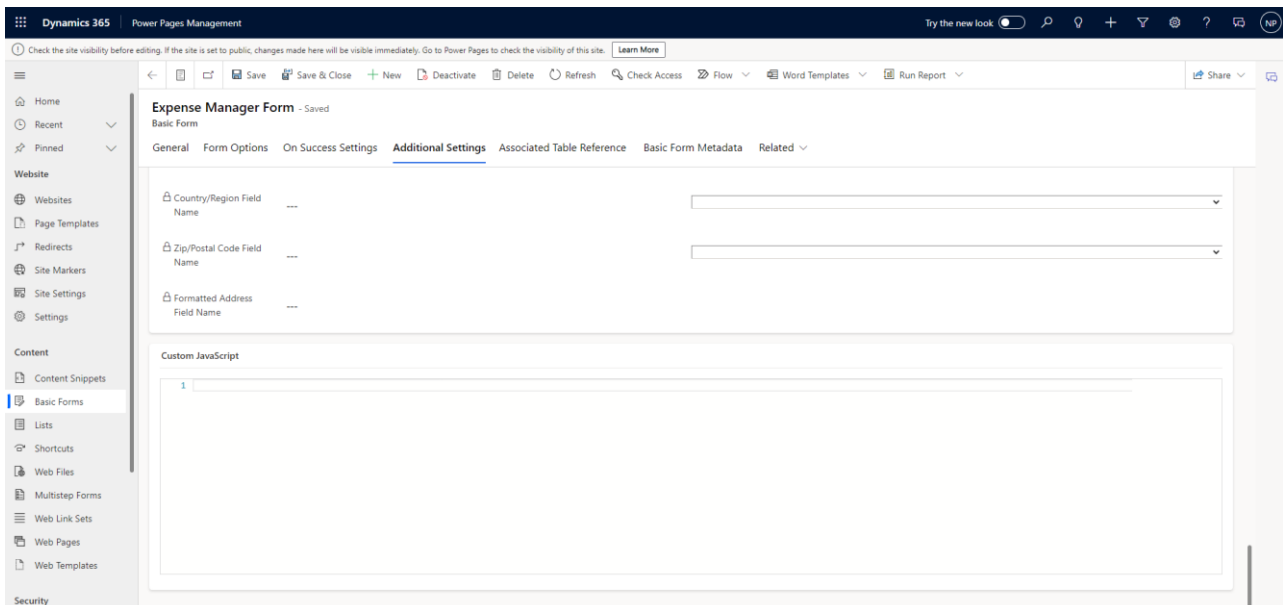


This what we will see in the website.



7. Custom JavaScript code

Under Additional Settings, in the Expense Manger Form basic form, we have a section called Custom JavaScript.



There we can type custom JavaScript codes. In fact, in some scenario, we may want to hide/show or lock some fields in a page of our website. Business rules will work in Dynamics 365, but not on a Power Pages site. The only way to achieve similar functionalities, therefore, is by means of JavaScript code and JQuery.

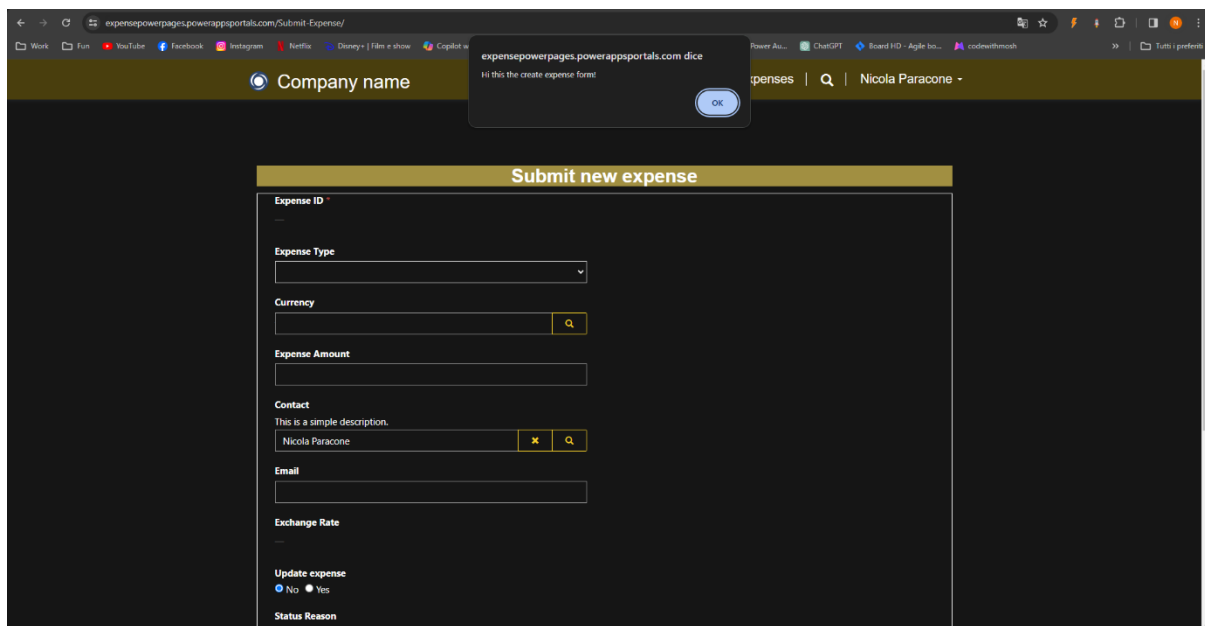
Navigate to <https://learn.microsoft.com/en-us/power-pages/configure/add-custom-javascript> for more details.

Consider the following example.

Custom JavaScript

```
1 $(document).ready(function() {  
2   alert("Hi this the create expense form!");  
3 });
```

This will produce the effect shown in the figure below on load.



The following code, instead, can be used for hiding a field and its label,

Custom JavaScript

```
1 $(document).ready(function() {  
2   debugger;  
3   alert("Hi this the create expense form!");  
4   $("#demo_name").hide();  
5   $("#demo_name_label").hide();  
6 });
```

and this to show them:

Custom JavaScript

```
1 $(document).ready(function() {  
2   debugger;  
3   alert("Hi this the create expense form!");  
4   $("#demo_name").show();  
5   $("#demo_name_label").show();  
6 });
```

Navigate to <https://softchief.com/2024/01/09/hide-form-field-in-power-pages-by-using-javascript/> for more about hiding fields in a page.

At <https://shafiuddinmicrosoftdynamics365crm.wordpress.com/2021/04/20/how-to-make-field-mandatory-and-non-mandatory-using-jquery-in-ms-power-apps-portal/> some instructions to make fields mandatory and non-mandatory can be found.

By means of the following code we can hide the Expense Amount field when Expense Type changes.

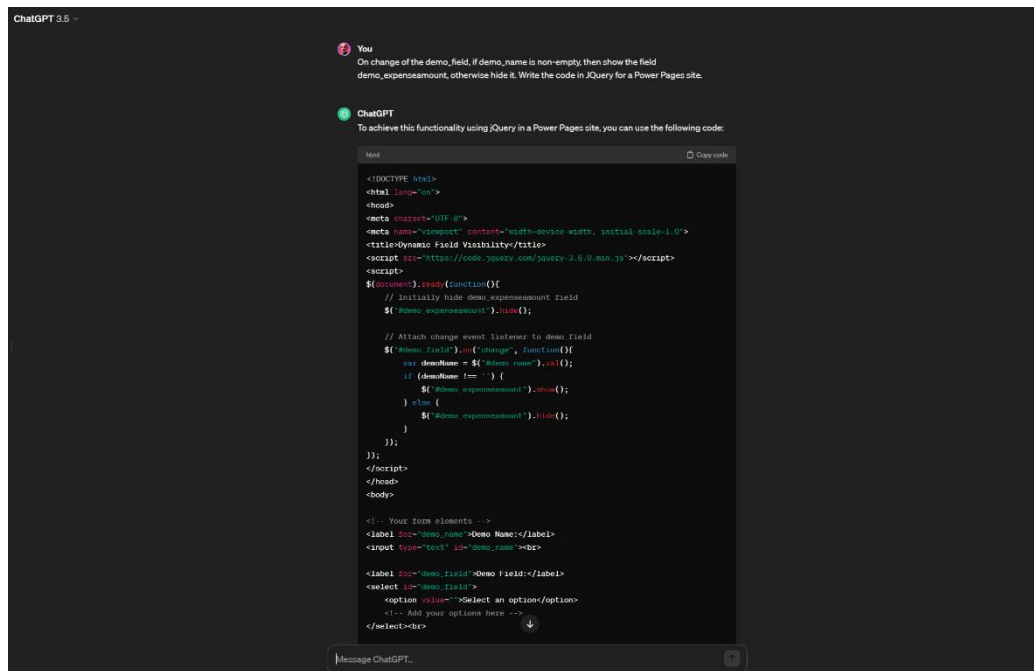
Custom JavaScript

```
1 $(document).ready(function() {  
2     debugger;  
3     $("#demo_expenseamount").show();  
4     $("#demo_expenseamount_label").show();  
5     $("#demo_expensetype").change(function(){  
6         alert("Hi this the create expense form!");  
7         $("#demo_expenseamount").hide();  
8         $("#demo_expenseamount_label").hide();  
9     }  
10 }  
11 });
```

The screenshot shows a web portal with a dark theme. At the top is a navigation bar with the company name and links to Home, Pages, Contact us, MyExpenses, and a user profile for Nicola Paracone. Below the navigation bar is a form titled "Submit new expense". The form contains several sections: "Expense ID" with a text input, "Expense Type" with a dropdown menu, "Currency" with a text input and a search icon, "Expense Amount" with a text input, "Contact" with a text input and a search icon, "Email" with a text input, and "Exchange Rate" with a text input. At the bottom of the form are two radio buttons for "Update expense" (No and Yes) and a "Status Reason" section.

This screenshot shows the same "Submit new expense" form, but with the "Expense Type" dropdown menu set to "Hotel". The "Expense Amount" field is now hidden, as indicated by the JavaScript code above. The rest of the form, including the "Currency", "Contact", "Email", "Exchange Rate", and "Update expense" sections, remains visible and unchanged.

Very often, using ChatGPT is the easiest way to recover the code needed for some desired functionality.



The screenshot shows a ChatGPT 3.5 interface. The user asks: "On change of the demo_field, if demo_name is non-empty, then show the field demo_expenseamount, otherwise hide it. Write the code in JQuery for a Power Pages site." The ChatGPT response provides a jQuery code snippet to achieve this functionality. The code includes a document ready function that initially hides the 'demo_expenseamount' field and then attaches a change event listener to the 'demo_field' select element. When the field value changes, it checks if 'demo_name' is not empty. If it is, it shows the 'demo_expenseamount' field; otherwise, it hides it. The code also includes HTML snippets for the form elements: a label and input for 'demo_name' and a label and select for 'demo_field'.

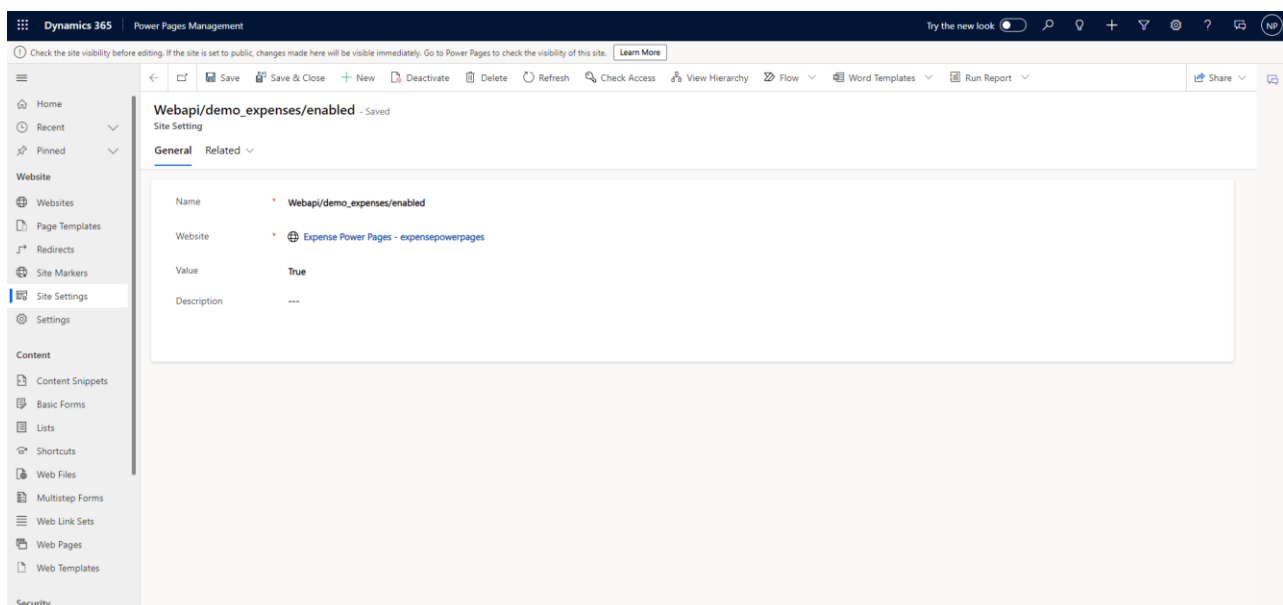
```
<!-- Your form elements -->
<label for="demo_name">Demo Name:</label>
<input type="text" id="demo_name">
<br>
<label for="demo_field">Demo field:</label>
<select id="demo_field">
  <option value="">Select an option/>
  <!-- Add your options here -->
</select>
</div>
```

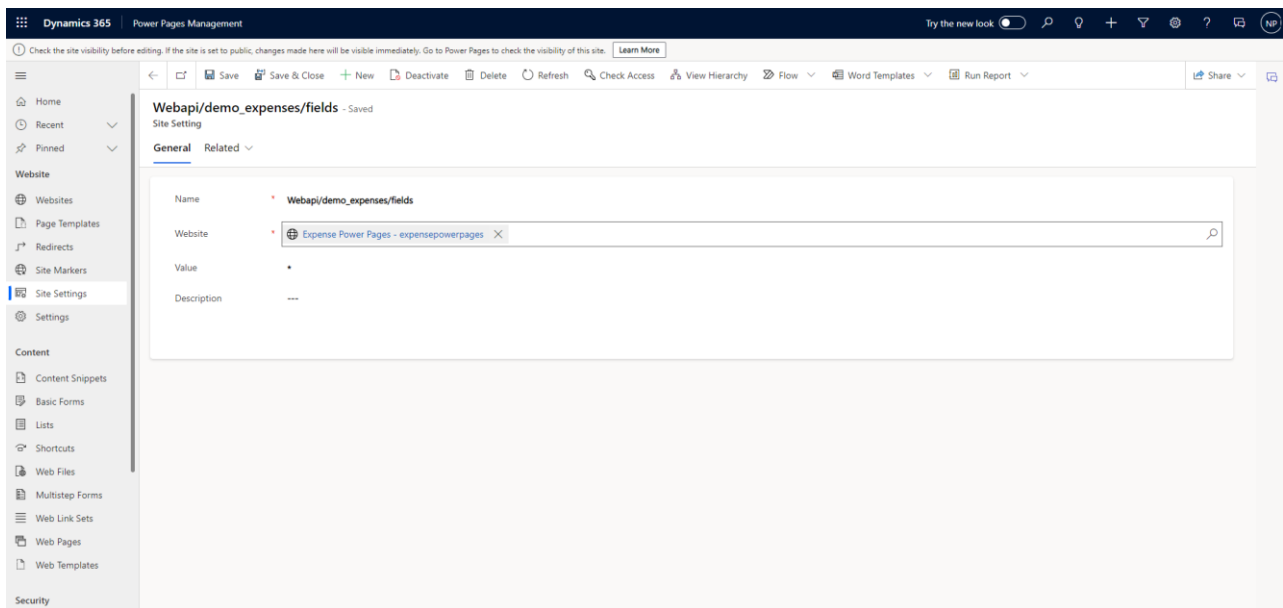
Similarly, tabs and sections can also be hidden or shown depending on certain conditions.

8. CRUD operations

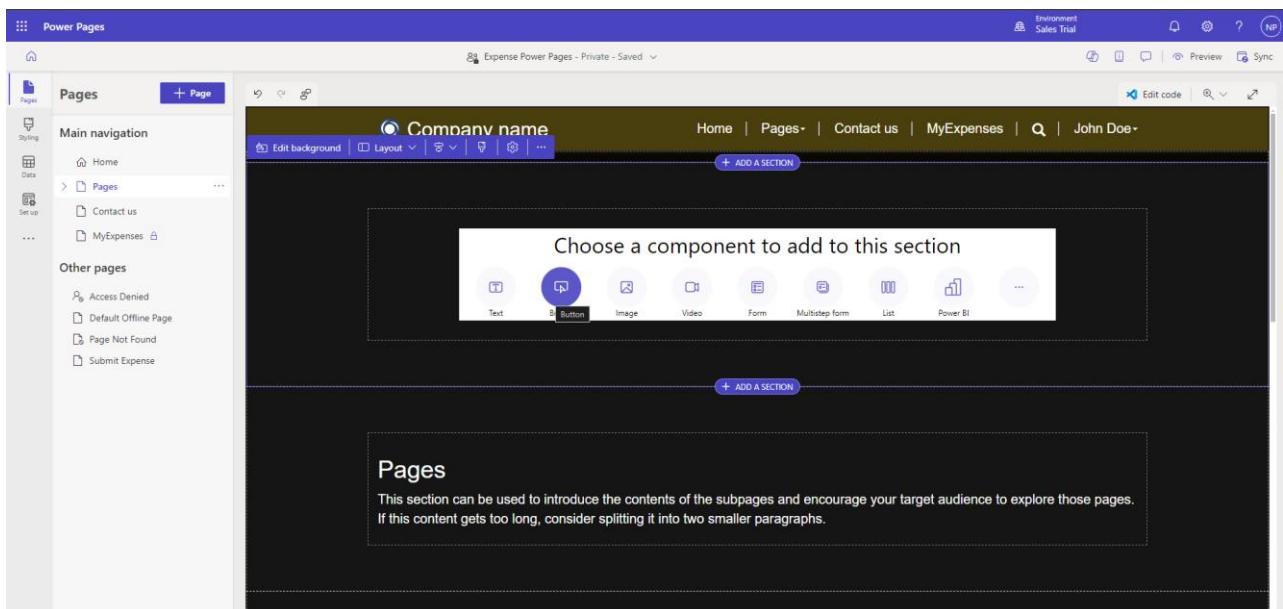
At <https://learn.microsoft.com/en-us/power-pages/configure/write-update-delete-operations> we find a rich description about CRUD using Web API. Basically, we can use the Web API to perform create, read, update, and delete operations across all Microsoft Dataverse tables from our webpages. For example, we can create a new account, update a contact, without using a form or multistep form by using the portals Web API.

Let us navigate to Site Settings in the Power Pages Management app and create the following site settings.



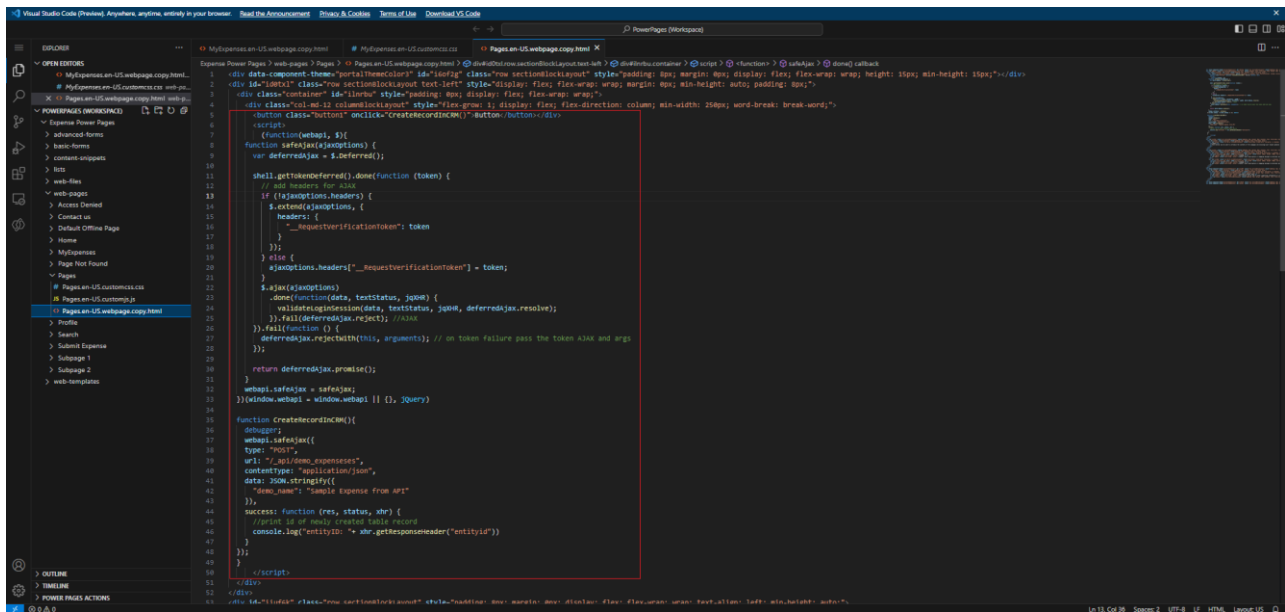


We add a new section and create a button.



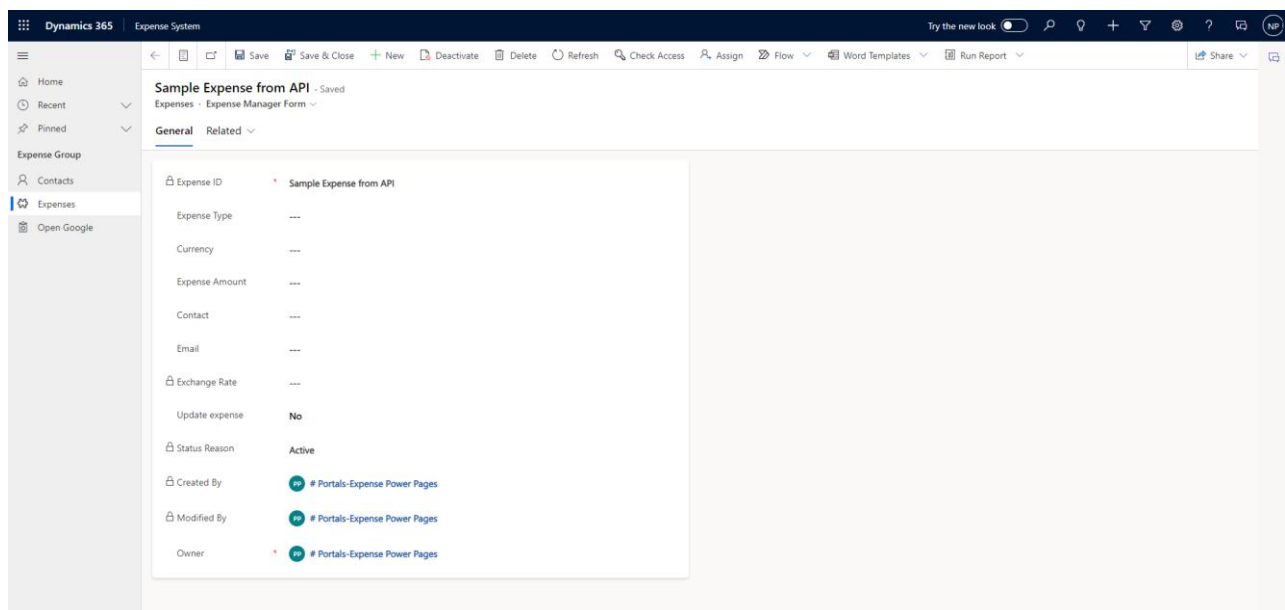
Then click on  Edit Code.

To the button we add the functionality of creating a new expense record in CRM when it is hit.



```
1 <div data-component-theme="portalThemeColor" id="ioefg" class="row sectionblocklayout" style="padding: 5px; margin: 0px; display: flex; flex-wrap: wrap; height: 15px; min-height: 15px;"/></div>  
2 <div id="ioefg" class="row sectionblocklayout text-left" style="display: flex; flex-wrap: wrap; margin: 0px; min-height: auto; padding: 5px;"/>  
3 <div class="container" id="ioefg" style="padding: 5px; display: flex; flex-wrap: wrap;"/>  
4 <div class="col-md-12 columnblocklayout" style="flex-grow: 1; display: flex; flex-direction: column; min-width: 200px; word-break: break-word;"/>  
5 <div class="button" onclick="CreateRecordInCRM()" style="display: flex; flex-direction: column; min-width: 200px; word-break: break-word;"/></div>  
6 </script>  
7 (function(webapi, $){  
8   function safeAjax($ajaxOptions) {  
9     var deferredAjax = $.Deferred();  
10  
11     $.ajax({  
12       url: $ajaxOptions.url,  
13       type: $ajaxOptions.type,  
14       data: $ajaxOptions.data,  
15       headers: {  
16         'RequestVerificationToken': token  
17       },  
18       success: function(data, textStatus, jqXHR) {  
19         deferredAjax.resolve(data);  
20       },  
21       error: function(jqXHR, textStatus, errorThrown) {  
22         deferredAjax.reject(jqXHR, textStatus, errorThrown);  
23       }  
24     });  
25     return deferredAjax.promise();  
26   }  
27   function CreateRecordInCRM() {  
28     $.ajax({  
29       url: "/api/demo_expenses",  
30       type: "POST",  
31       data: {  
32         "demo_name": "Sample Expense from API"  
33       },  
34       success: function(res, status, xhr) {  
35         console.log("entityId: " + xhr.getResponseHeader("entityid"));  
36       }  
37     });  
38   }  
39 })(window.webapi, window.$);
```

This is will be a newly created expense.



Field	Value
Expense ID	Sample Expense from API
Expense Type	---
Currency	---
Expense Amount	---
Contact	---
Email	---
Exchange Rate	---
Update expense	No
Status Reason	Active
Created By	# Portals-Expense Power Pages
Modified By	# Portals-Expense Power Pages
Owner	# Portals-Expense Power Pages

Following the official guide by Microsoft, the other CRUD operations can be implemented in a similar way. Only the structure of the second function (inside the red box in the figure above) is to be changed in dependence of the functionality that we want to achieve.